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(54) **INTEGRATED CIRCUIT AND METHOD OF
MANUFACTURING SAME**

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H10D 84/83 (2025.01)
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(2020.01); **G06F 30/392** (2020.01); **H01L**
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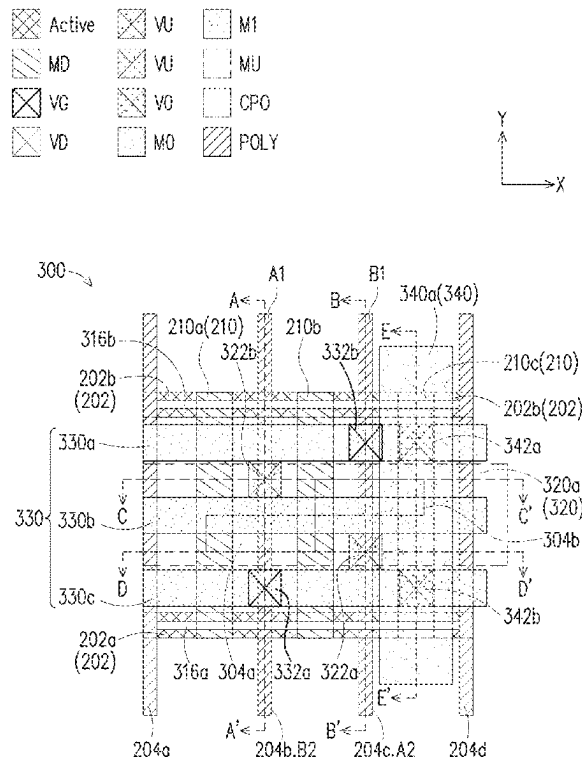
Related U.S. Application Data

(60) Continuation of application No. 18/053,012, filed on Nov. 7, 2022, now Pat. No. 12,341,103, which is a continuation of application No. 16/936,249, filed on Jul. 22, 2020, now Pat. No. 11,508,661, which is a division of application No. 16/135,684, filed on Sep. 19, 2018, now Pat. No. 10,734,321.

(60) Provisional application No. 62/564,663, filed on Sep. 28, 2017.

(57) **ABSTRACT**

An integrated circuit includes a first and second active region, a first conductor, a first and second contact, and a set of gates. The first and second active region extend in a first direction, and is located on a first level. The first conductor extends in the first direction and a second direction, is located on a lower portion of the first level, and is between the first and second active region. The first contact extends in the second direction, overlaps the first conductor and is located on a second level. The second contact extends in the second direction, overlaps the first conductor, is located on the second level, and is separated from the first contact in the first direction. The first conductor electrically couples the first contact and the second contact together.



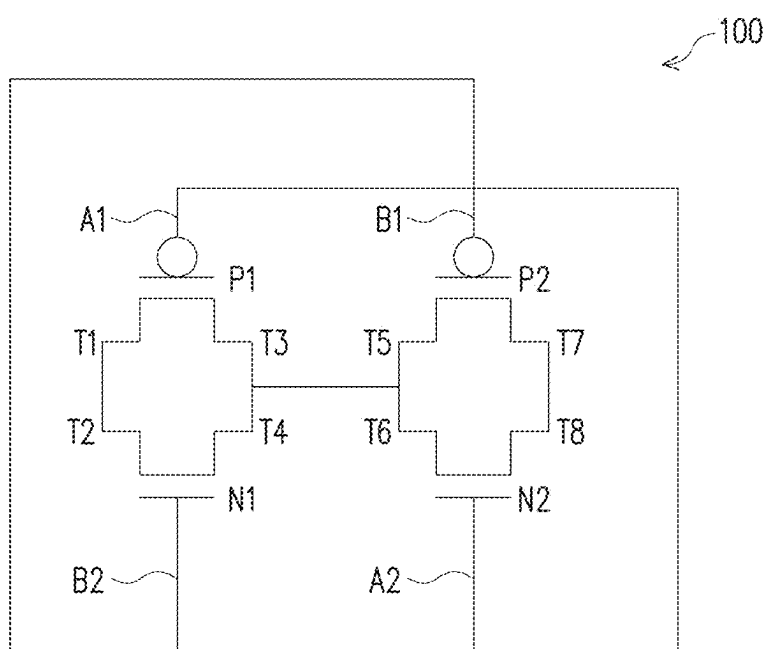


FIG. 1

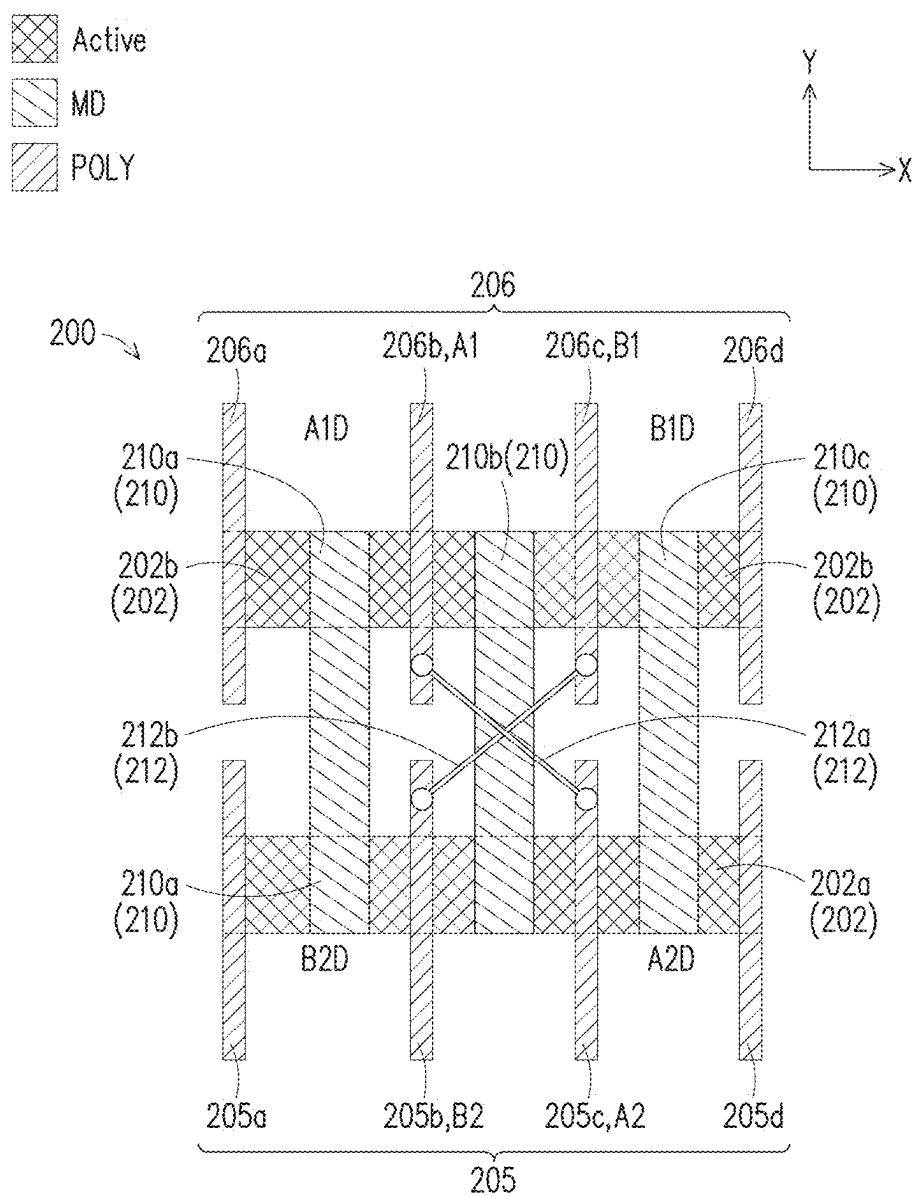


FIG. 2

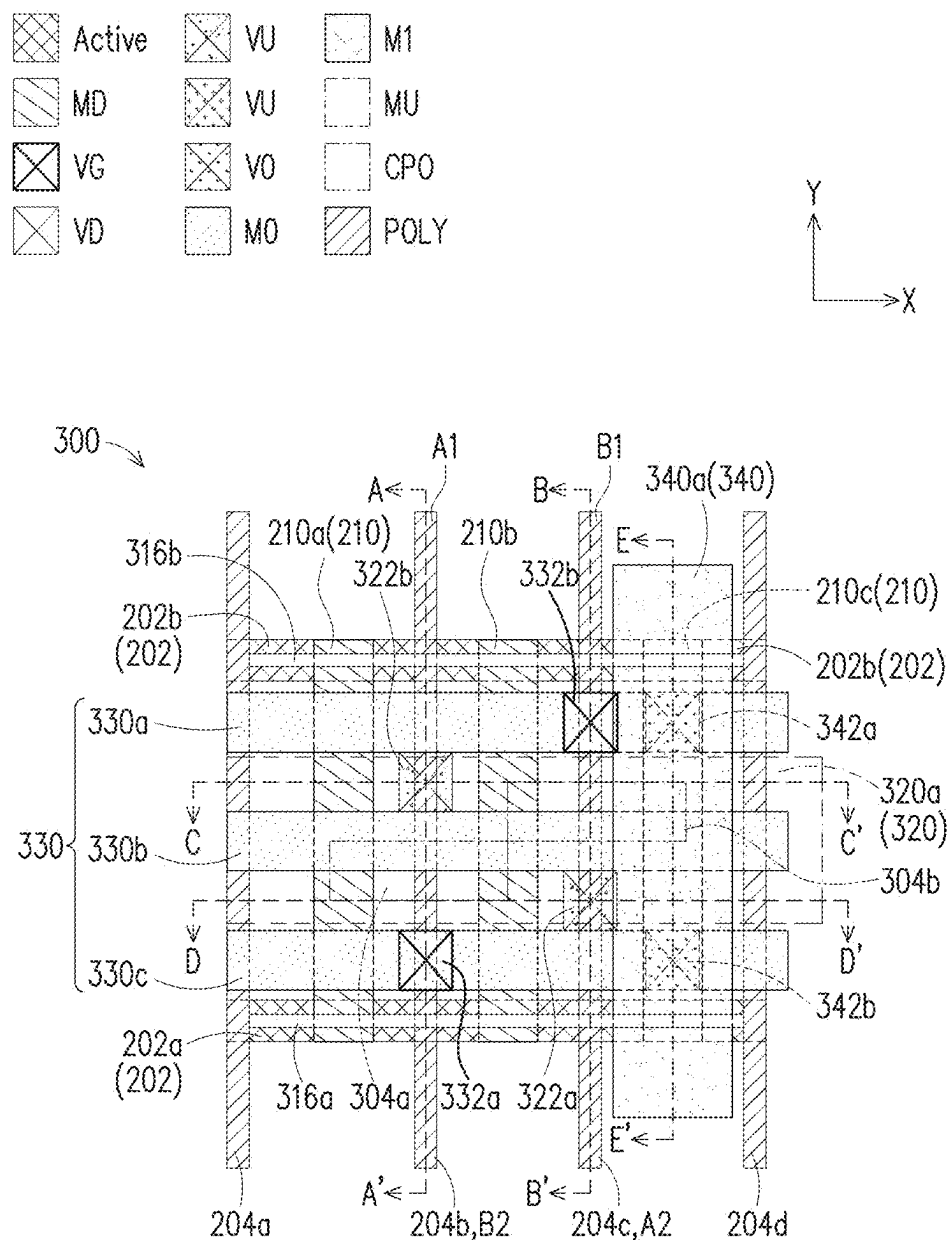


FIG. 3

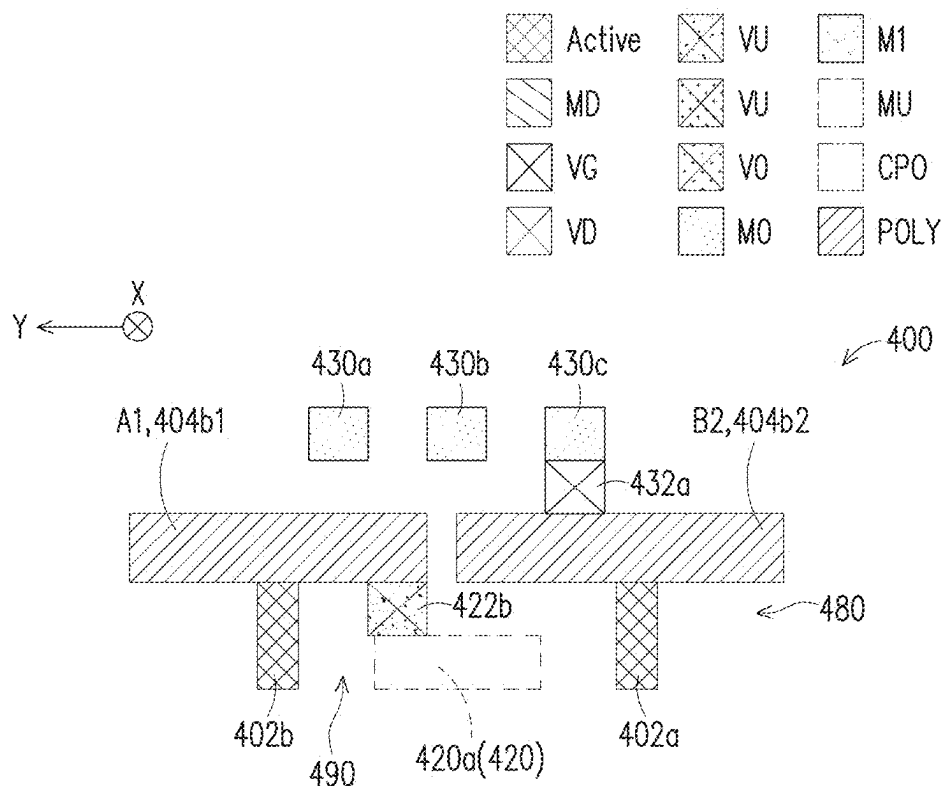


FIG. 4A

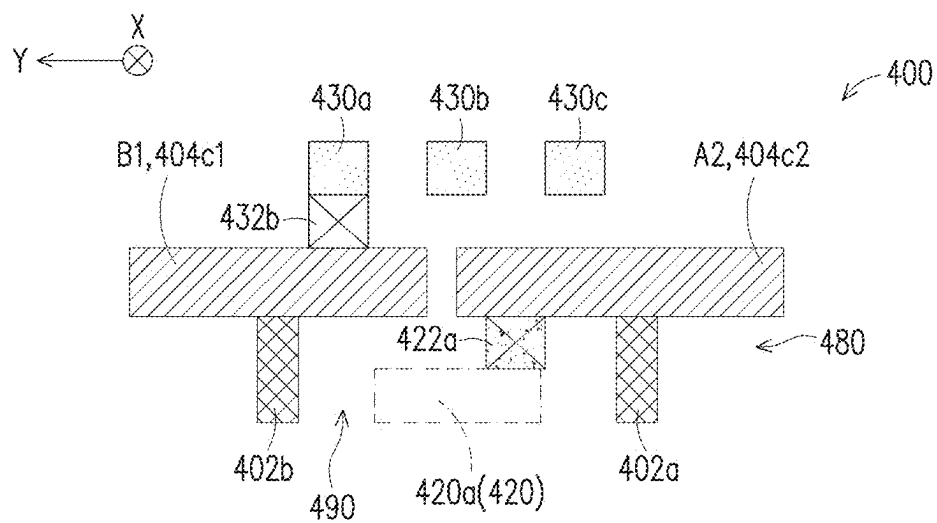


FIG. 4B

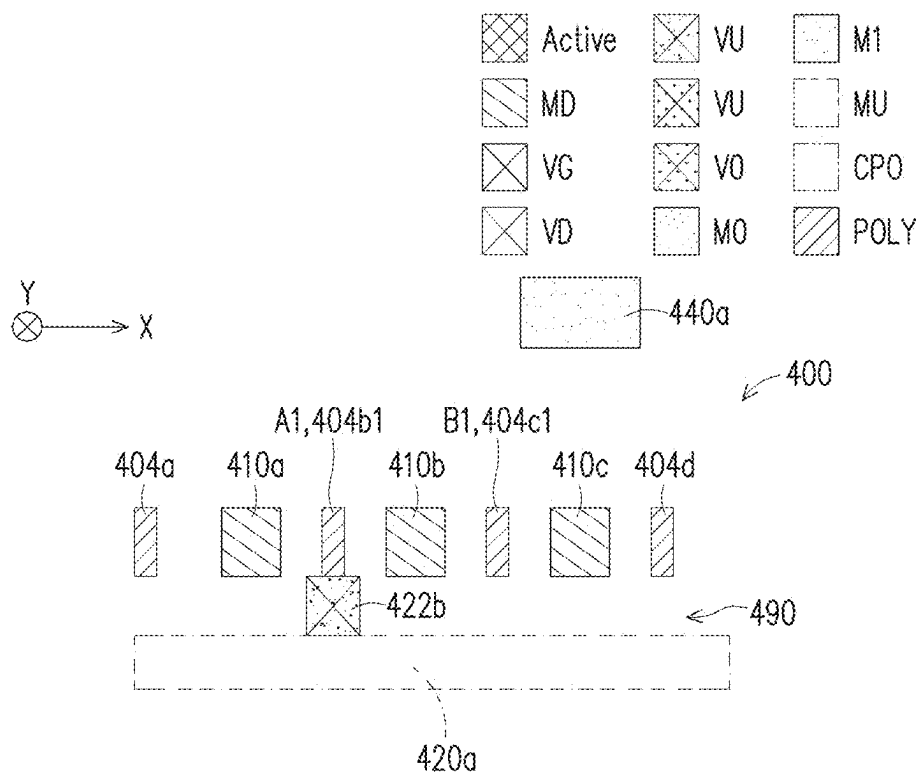


FIG. 4C

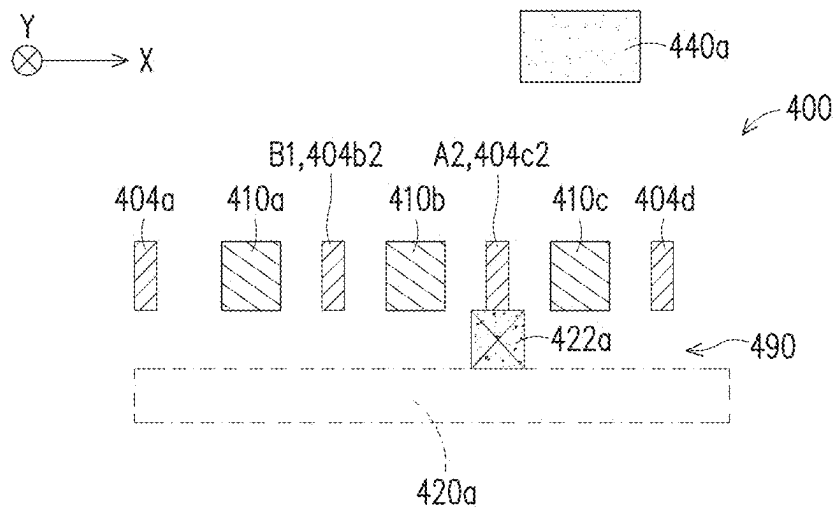


FIG. 4D

 Active	 VU	 M1
 MD	 VU	 MU
 VG	 VO	 CP0
 VD	 M0	 POLY

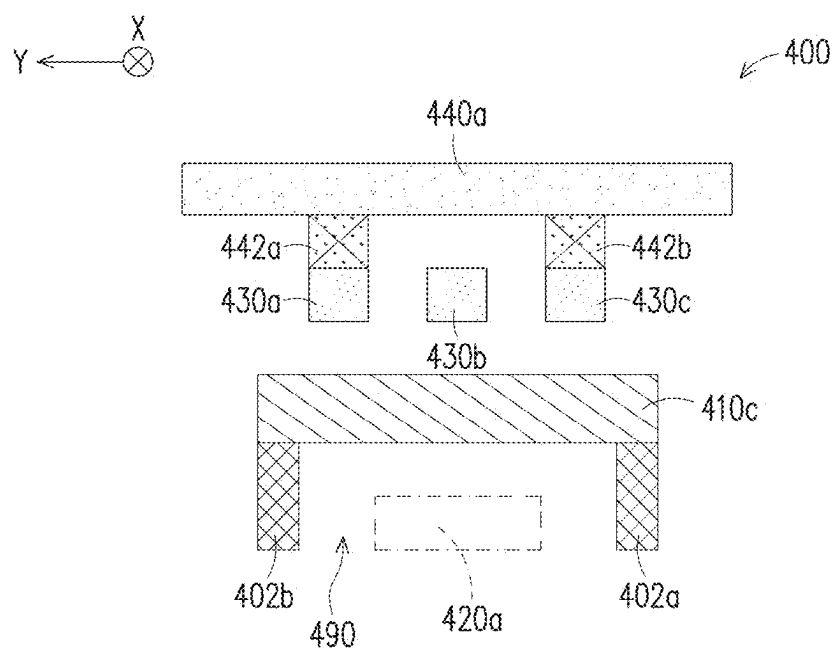


FIG. 4E

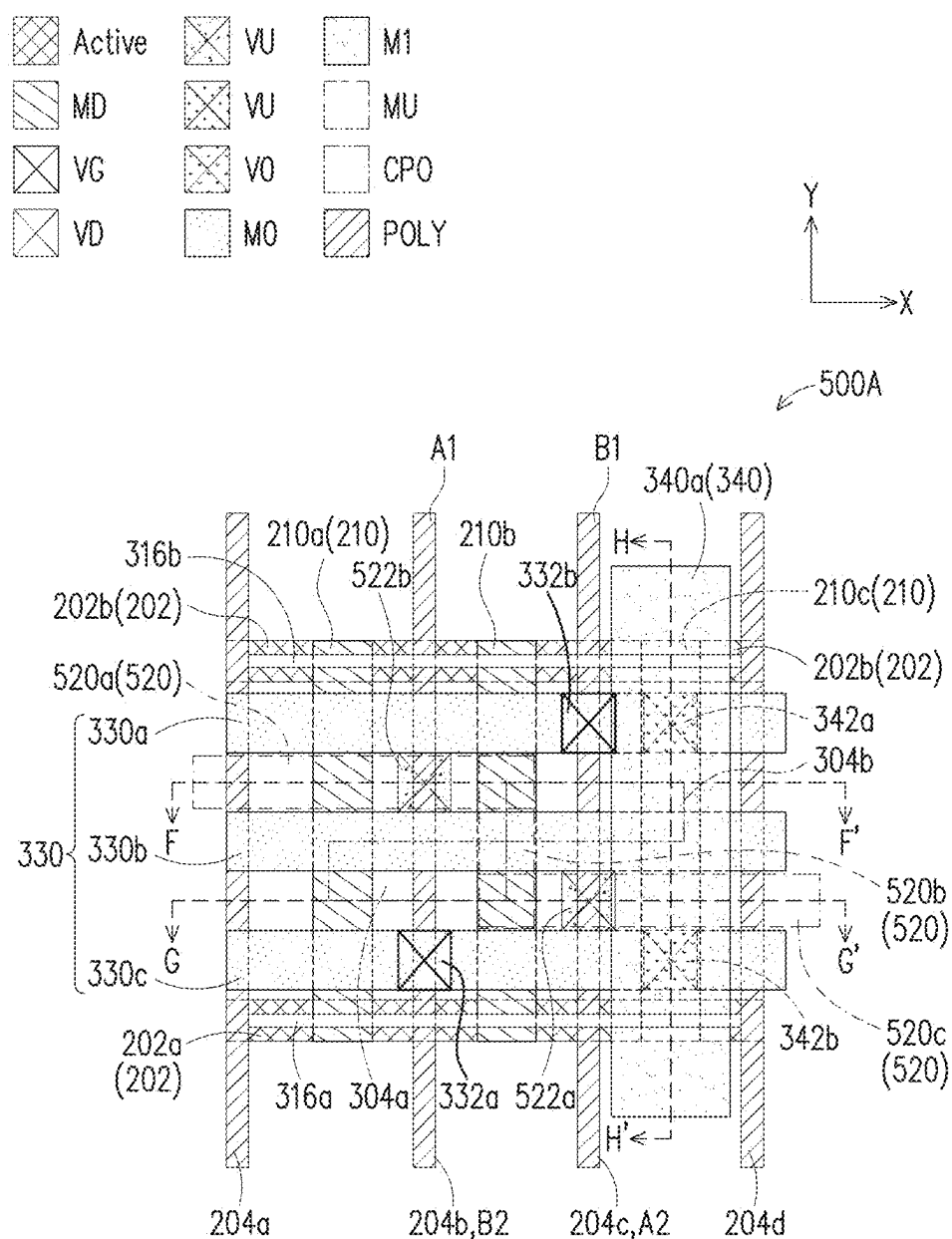


FIG. 5A

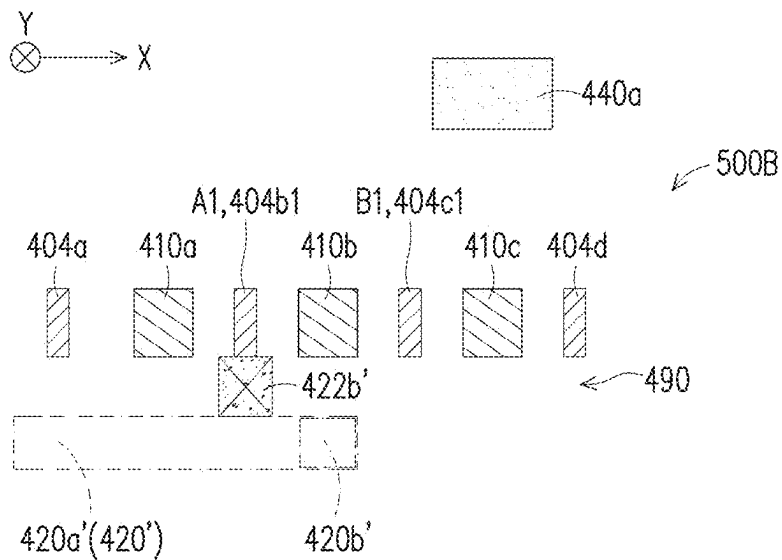
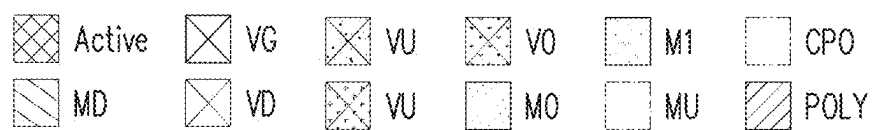


FIG. 5B

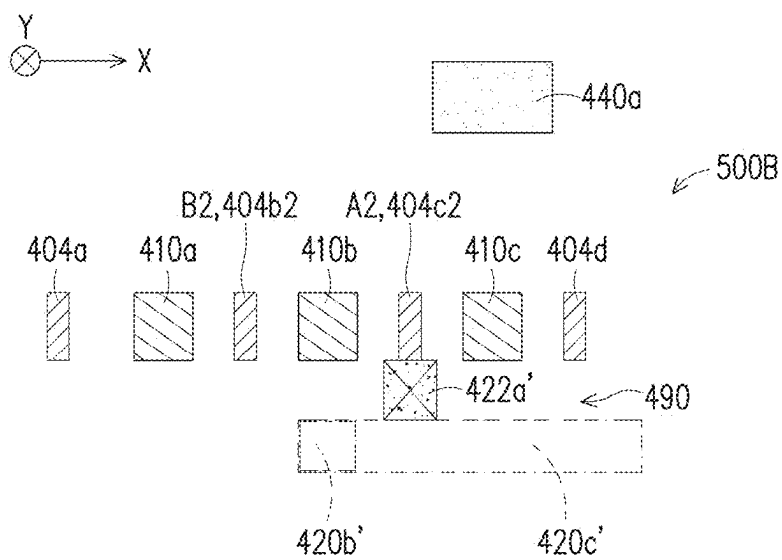


FIG. 5C

 Active	 VU	 M1
 MD	 VU	 MU
 VG	 VO	 CPO
 VD	 M0	 POLY

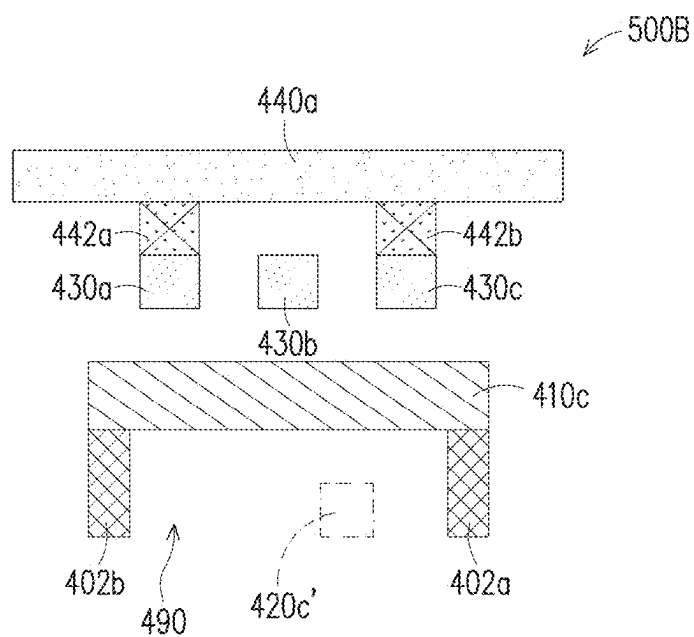
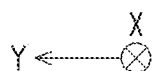


FIG. 5D

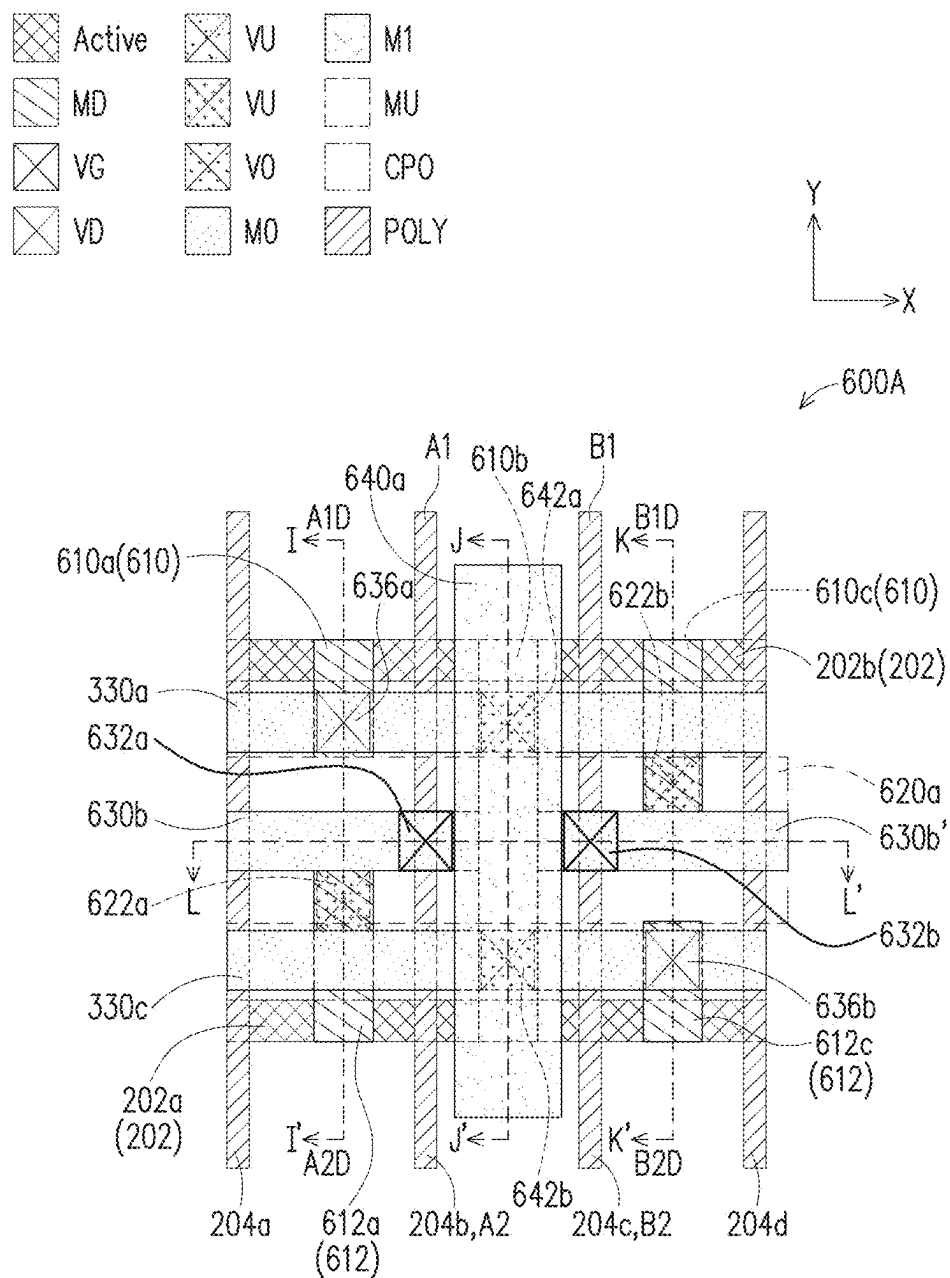


FIG. 6A

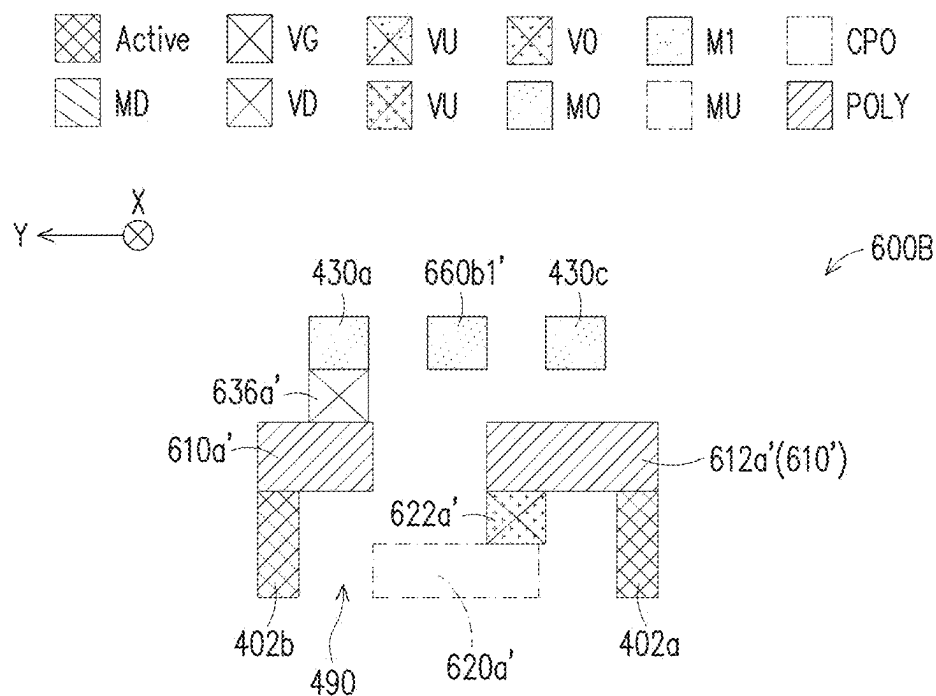


FIG. 6B

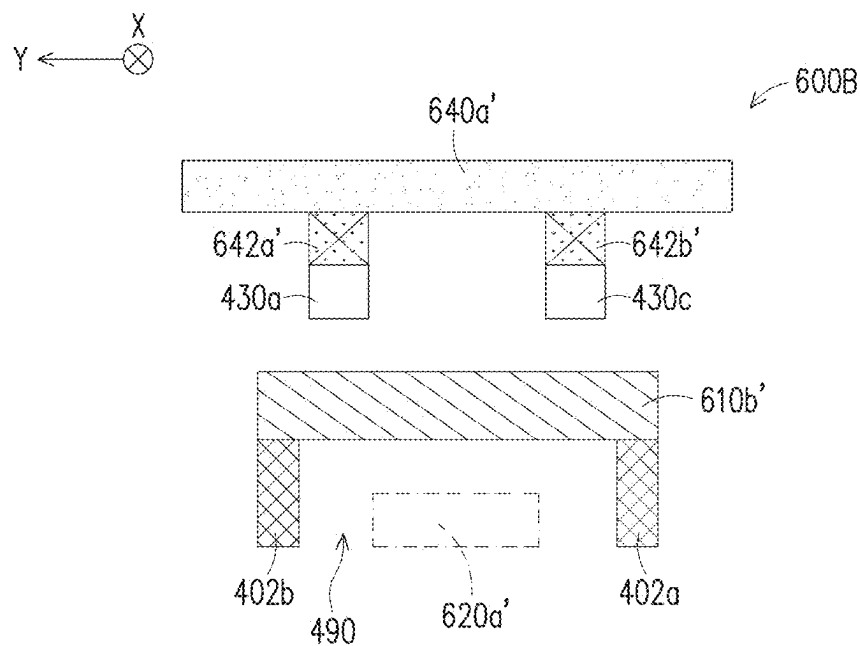


FIG. 6C

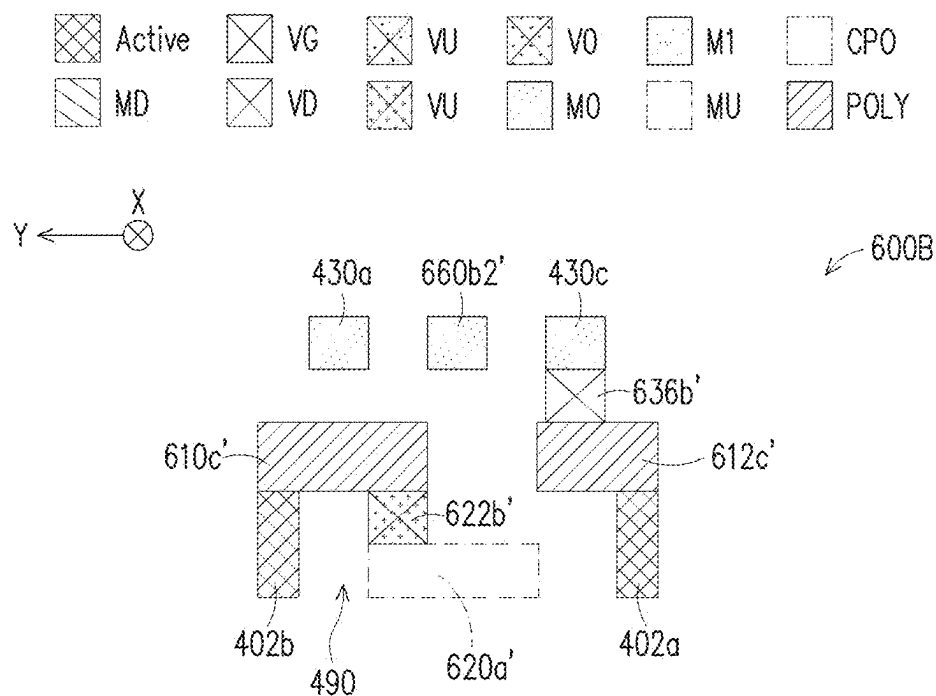


FIG. 6D

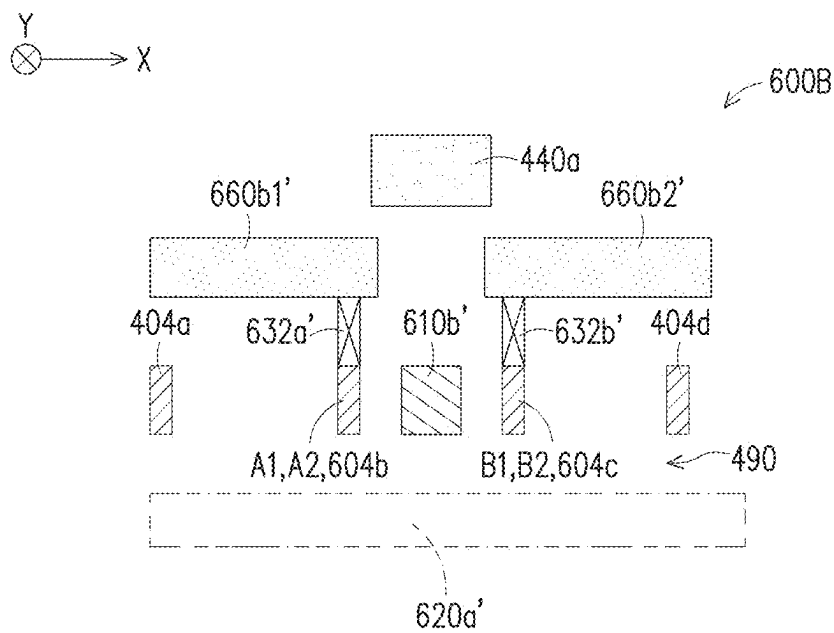


FIG. 6E

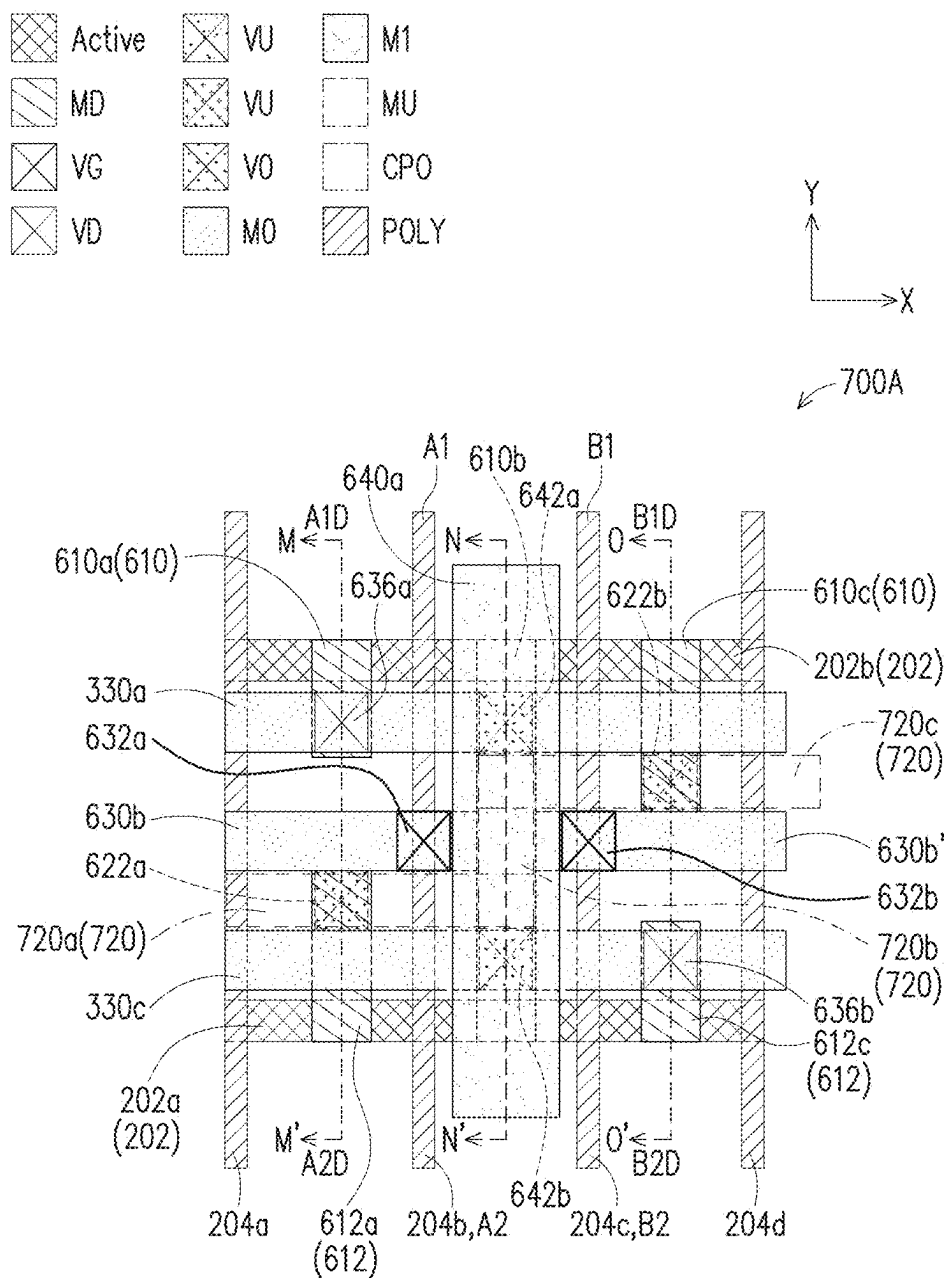


FIG. 7A

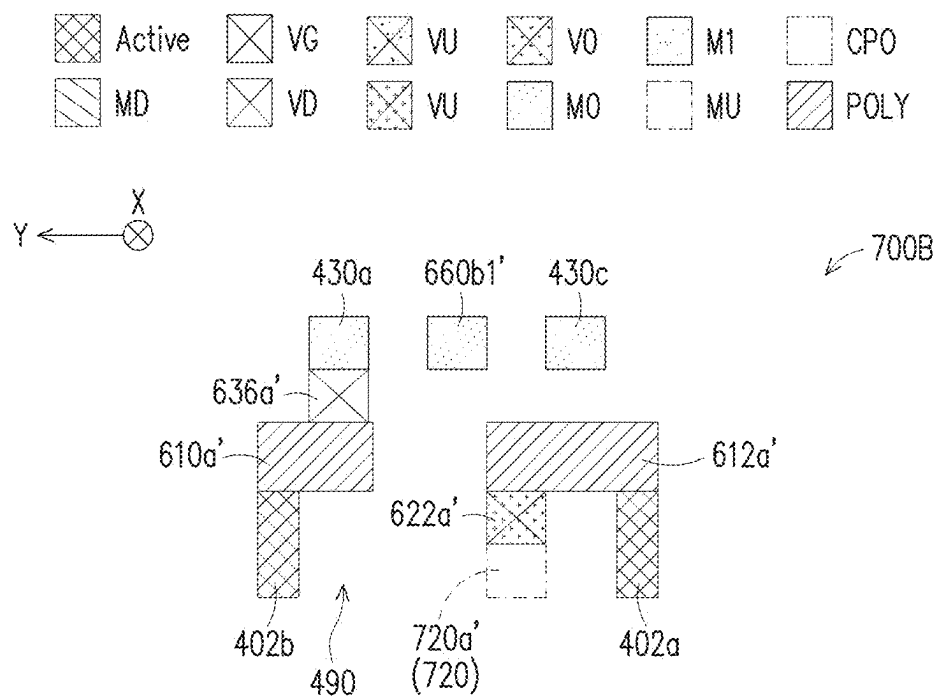


FIG. 7B

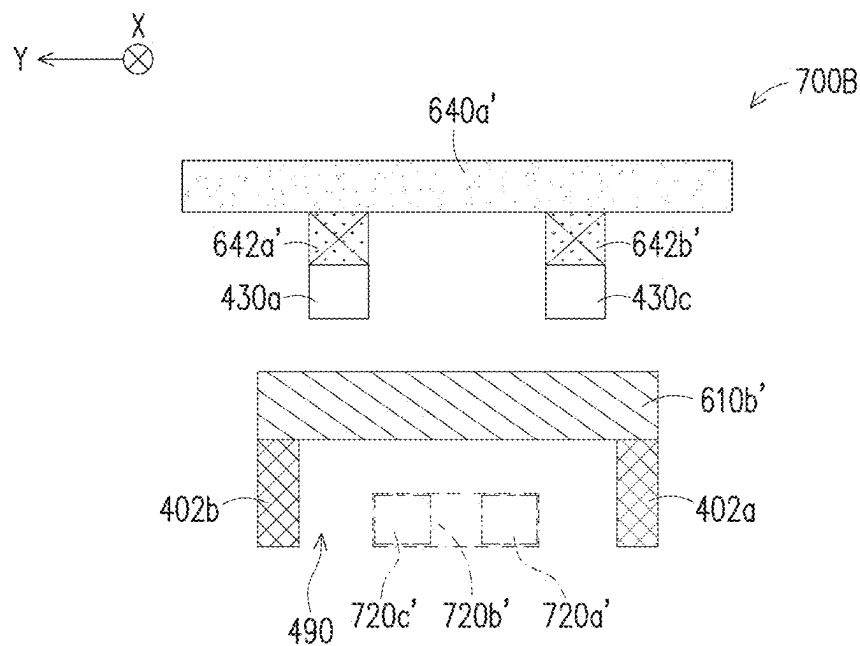


FIG. 7C

 Active	 VU	 M1
 MD	 VU	 MU
 VG	 VO	 CPO
 VD	 M0	 POLY

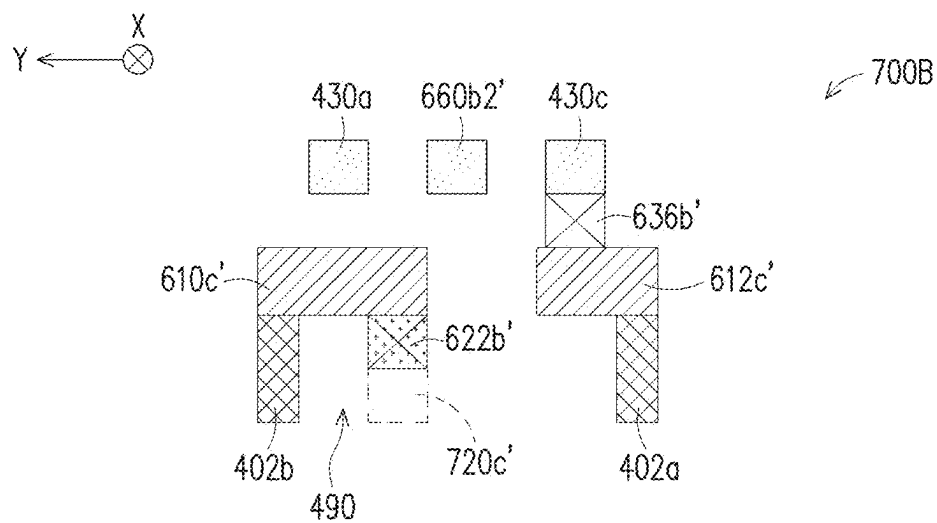


FIG. 7D

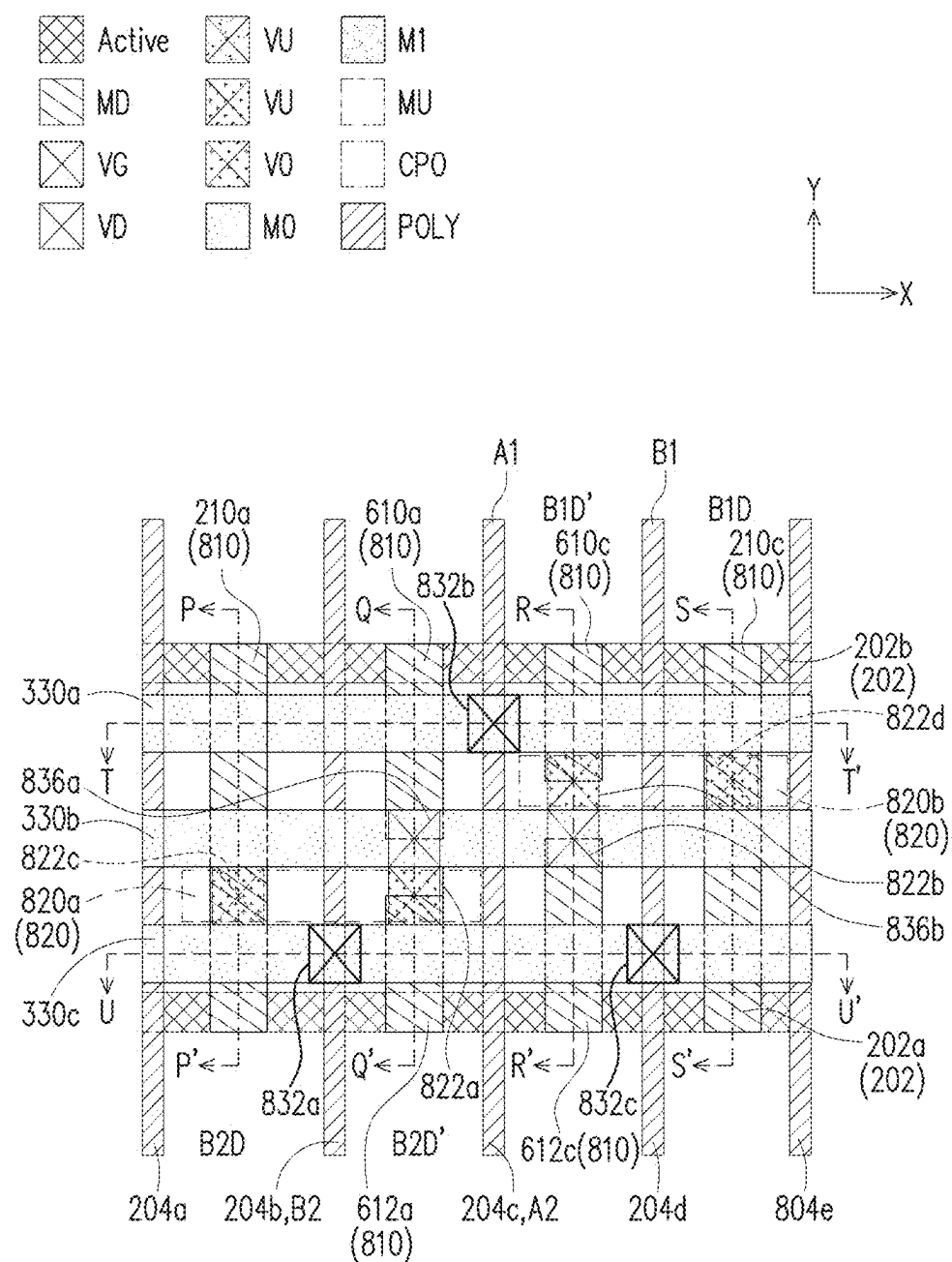


FIG. 8A

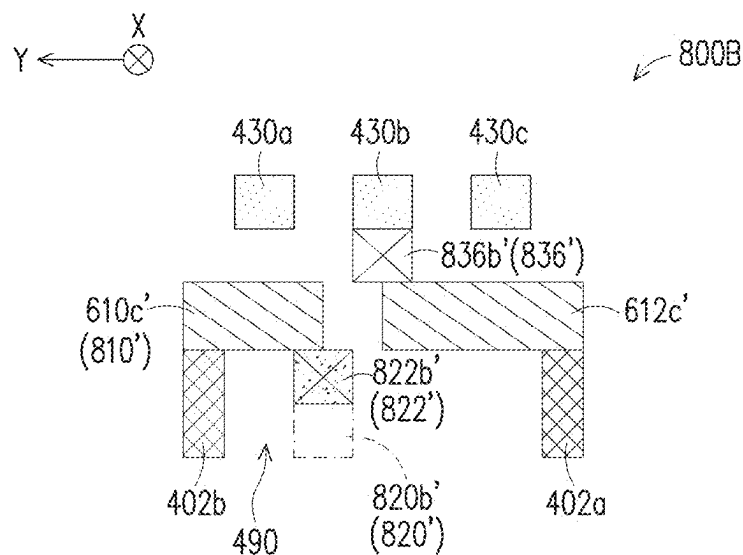
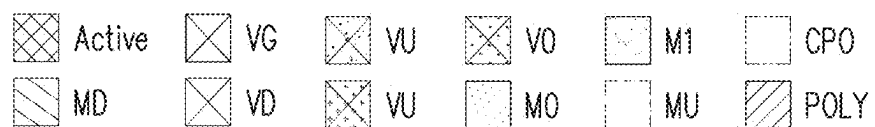


FIG. 8D

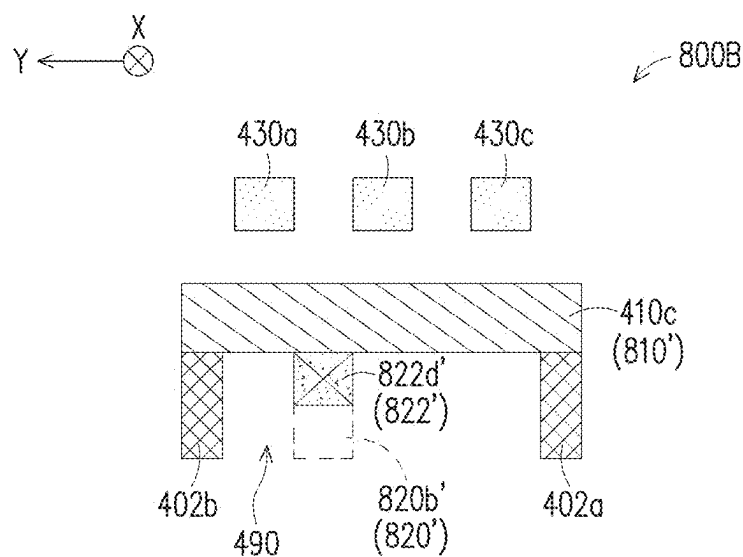


FIG. 8E

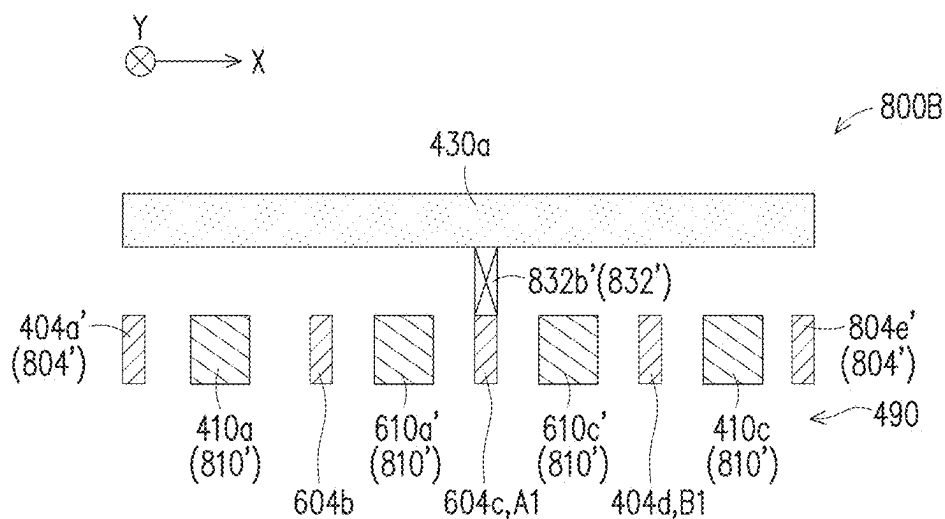
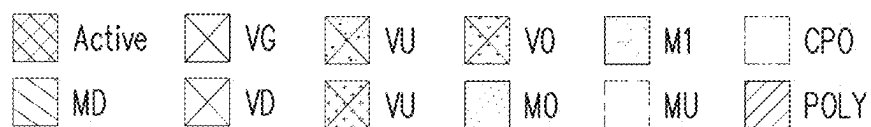


FIG. 8F

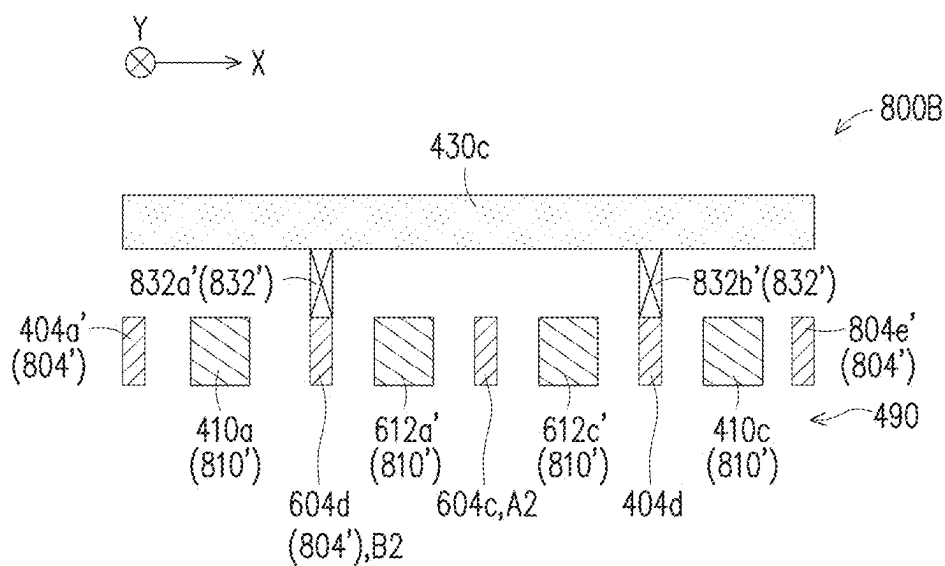


FIG. 8G

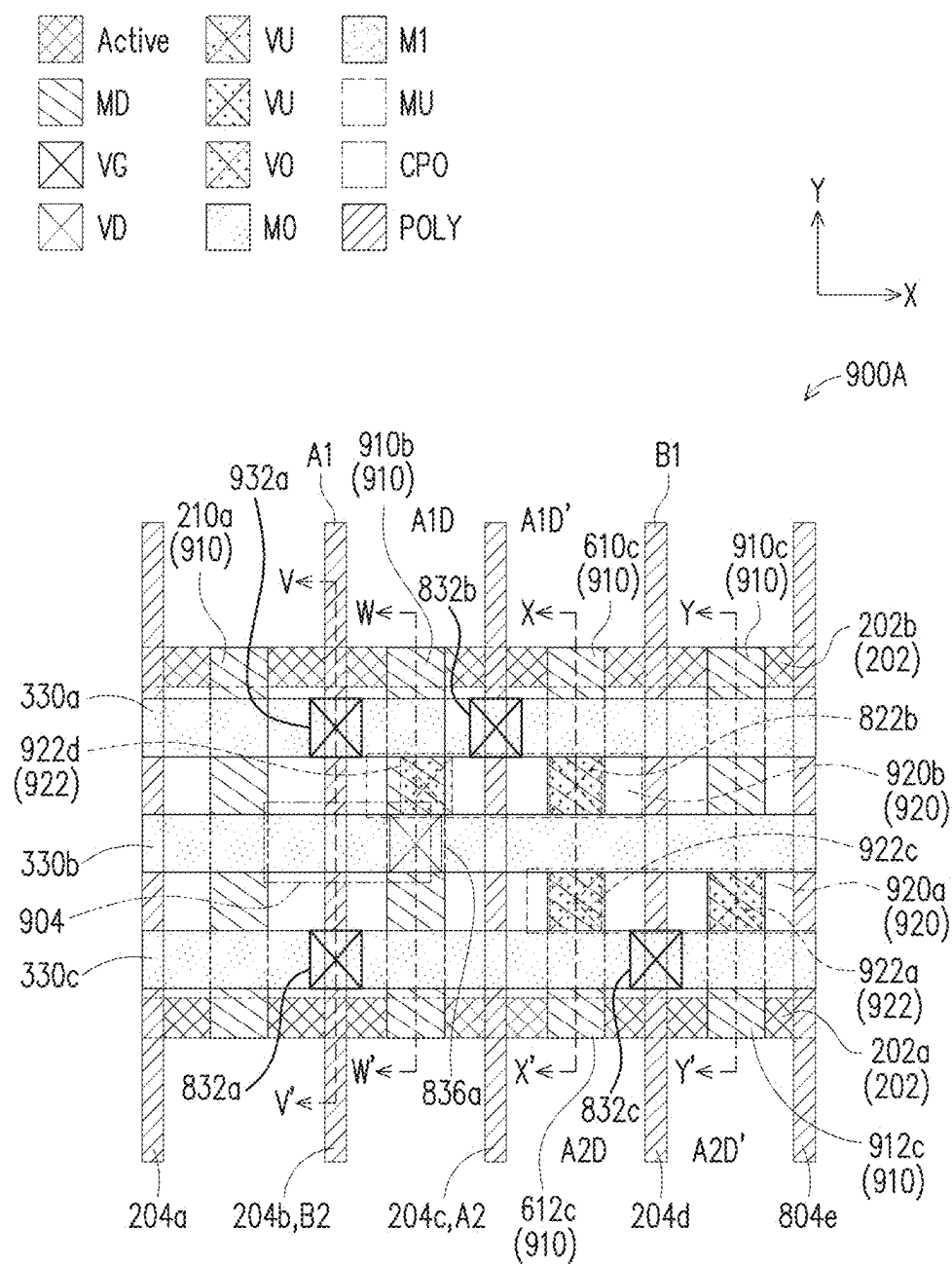


FIG. 9A

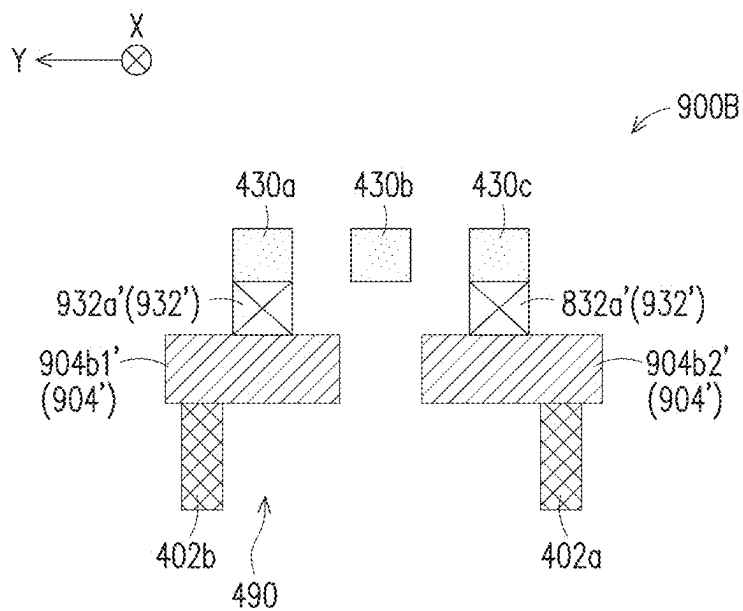


FIG. 9B

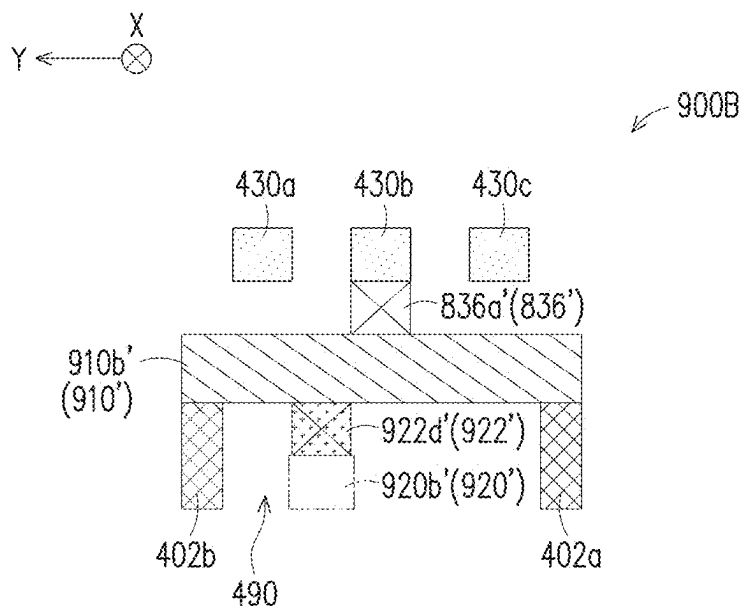


FIG. 9C

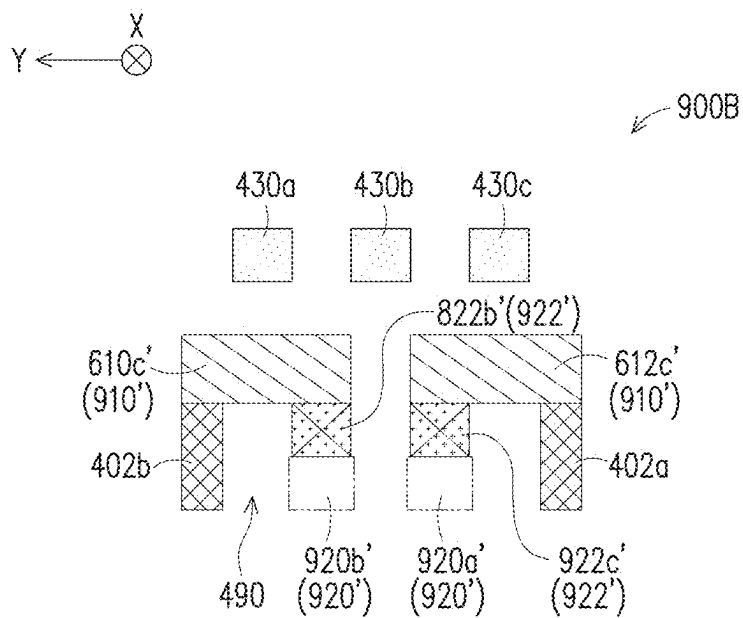


FIG. 9D

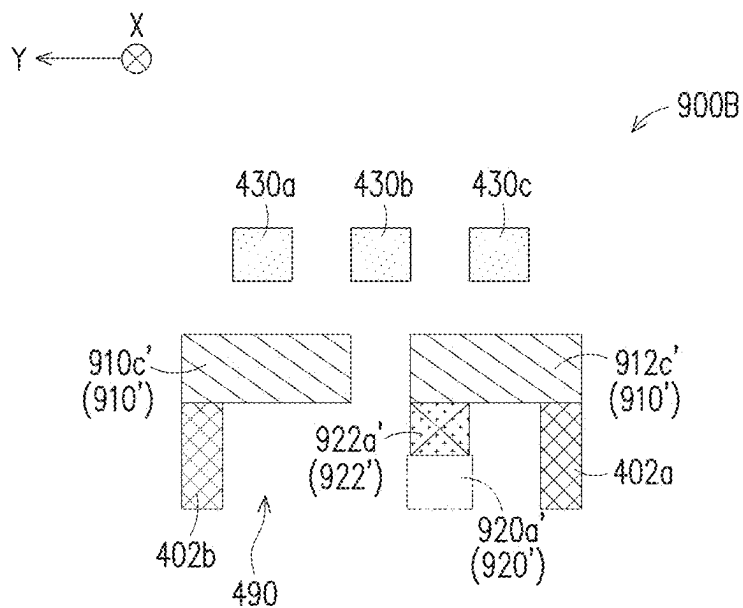


FIG. 9E

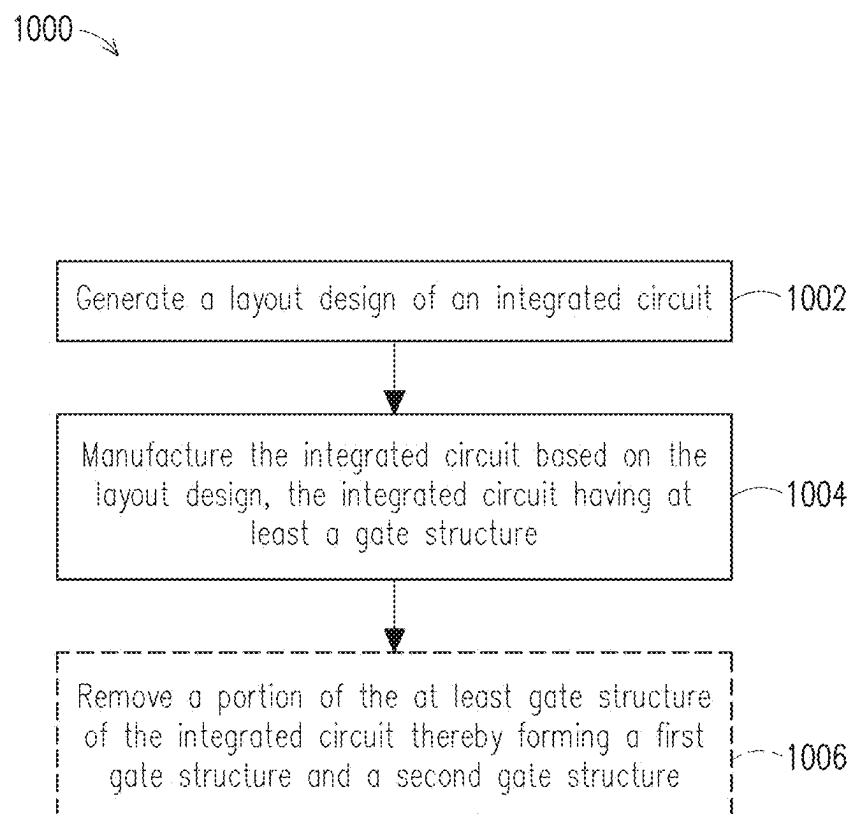


FIG. 10

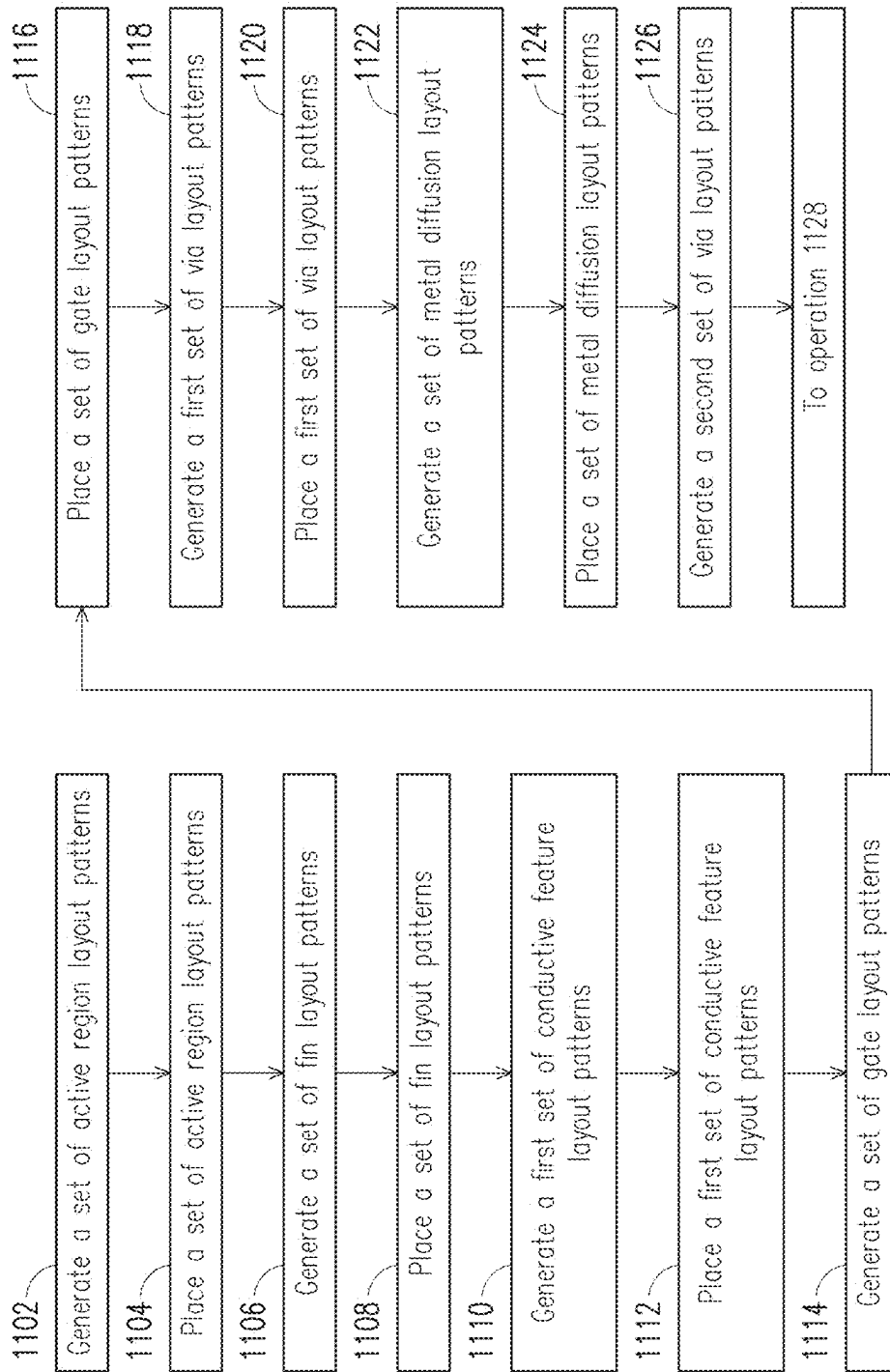


FIG. 11A

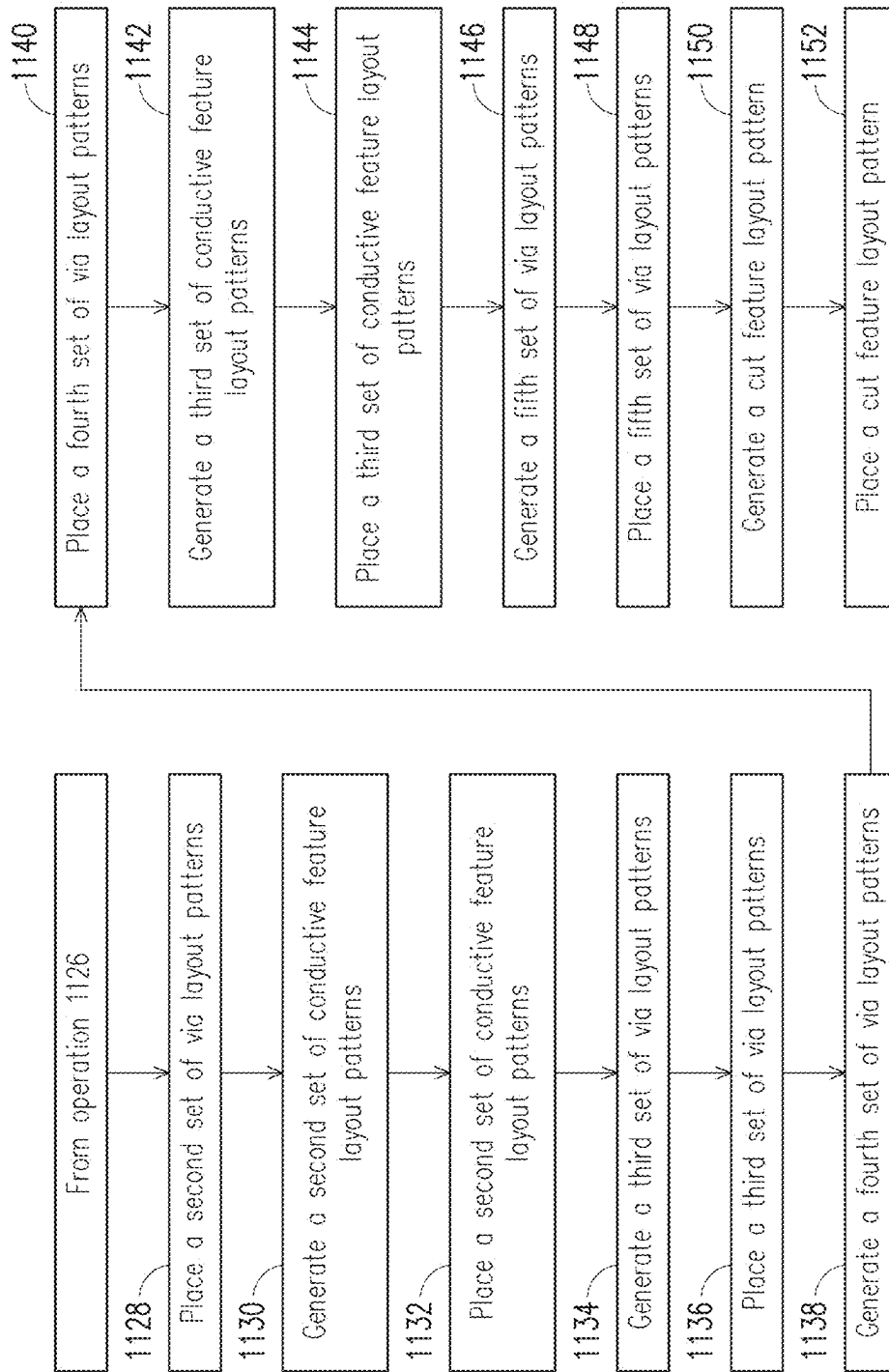


FIG. 11B

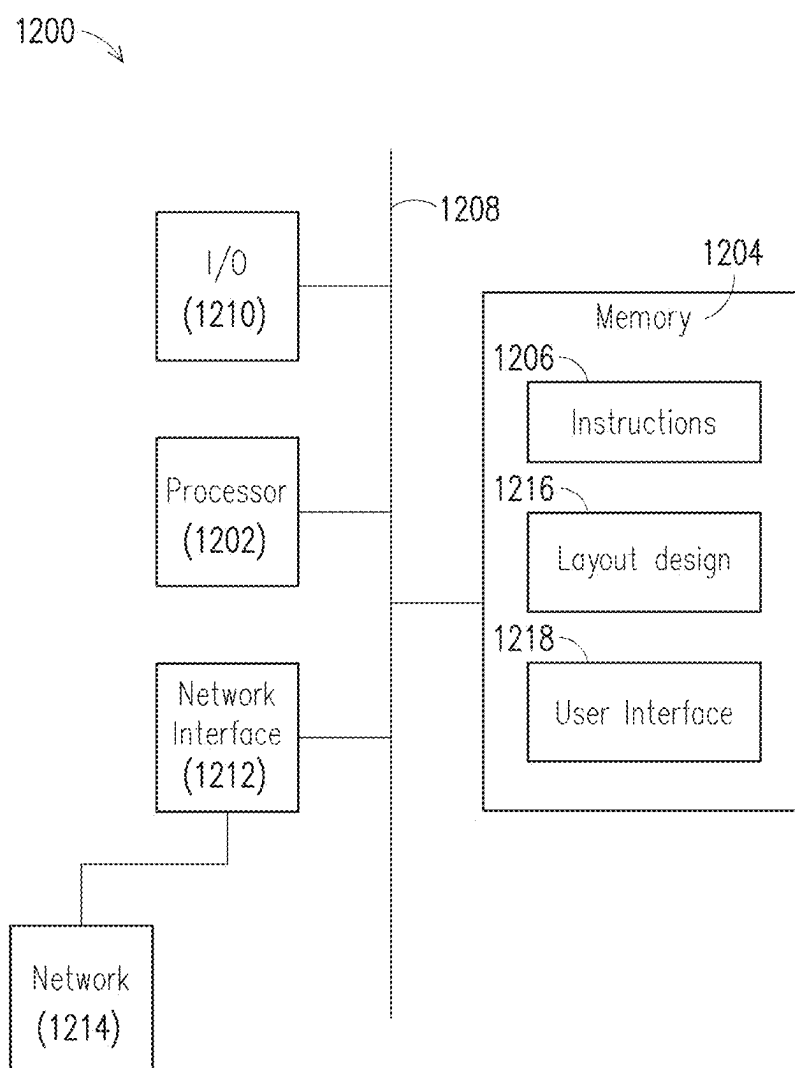


FIG. 12

INTEGRATED CIRCUIT AND METHOD OF MANUFACTURING SAME

PRIORITY CLAIM

[0001] The present application is a continuation of U.S. application Ser. No. 18/053,012, filed Nov. 7, 2022, which is a continuation of U.S. application Ser. No. 16/936,249, filed Jul. 22, 2020, now U.S. Pat. No. 11,508,661, issued Nov. 22, 2022, which is a divisional of U.S. application Ser. No. 16/135,684, filed Sep. 19, 2018, now U.S. Pat. No. 10,734,321, issued Aug. 4, 2020, which claims the benefit of U.S. Provisional Application No. 62/564,663, filed Sep. 28, 2017, which are herein incorporated by reference in their entireties.

BACKGROUND

[0002] The recent trend in miniaturizing integrated circuits (ICs) has resulted in smaller devices which consume less power yet provide more functionality at higher speeds. The miniaturization process has also resulted in stricter design and manufacturing specifications as well as reliability challenges. Various electronic design automation (EDA) tools generate, optimize and verify standard cell layout designs for integrated circuits while ensuring that the standard cell layout design and manufacturing specifications are met.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 is a circuit diagram of an integrated circuit, in accordance with some embodiments.

[0005] FIG. 2 is a diagram of a layout design of an integrated circuit, in accordance with some embodiments.

[0006] FIG. 3 is a diagram of a layout design of an integrated circuit, in accordance with some embodiments.

[0007] FIGS. 4A, 4B, 4C, 4D and 4E are cross-sectional views of an integrated circuit, in accordance with some embodiments.

[0008] FIG. 5A is a diagram of a layout design of an integrated circuit, in accordance with some embodiments.

[0009] FIGS. 5B, 5C and 5D are cross-sectional views of an integrated circuit, in accordance with some embodiments.

[0010] FIG. 6A is a diagram of a layout design of an integrated circuit, in accordance with some embodiments.

[0011] FIGS. 6B, 6C, 6D and 6E are cross-sectional views of an integrated circuit, in accordance with some embodiments.

[0012] FIG. 7A is a diagram of a layout design of an integrated circuit, in accordance with some embodiments.

[0013] FIGS. 7B, 7C and 7D are cross-sectional views of an integrated circuit, in accordance with some embodiments.

[0014] FIG. 8A is a diagram of a layout design of an integrated circuit, in accordance with some embodiments.

[0015] FIGS. 8B, 8C, 8D, 8E, 8F and 8G are cross-sectional views of an integrated circuit, in accordance with some embodiments.

[0016] FIG. 9A is a diagram of a layout design of an integrated circuit, in accordance with some embodiments.

[0017] FIGS. 9B, 9C, 9D and 9E are cross-sectional views of an integrated circuit, in accordance with some embodiments.

[0018] FIG. 10 is a flowchart of a method of manufacturing an IC, in accordance with some embodiments.

[0019] FIGS. 11A and 11B are a flowchart of a method of generating a layout design of an integrated circuit, in accordance with some embodiments.

[0020] FIG. 12 is a block diagram of a system of designing an IC layout design, in accordance with some embodiments.

[0021] The patent or application file contains drawings/photographs executed in color. Copies of this patent with color drawing(s)/photograph(s) will be provided by the Office upon request and payment of the necessary fee.

DETAILED DESCRIPTION

[0022] The following disclosure provides different embodiments, or examples, for implementing features of the provided subject matter. Specific examples of components, materials, values, steps, arrangements, or the like, are described below to simplify the present disclosure. These are, of course, merely examples and are not limiting. Other components, materials, values, steps, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0023] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0024] In accordance with some embodiments, an integrated circuit includes a set of active regions in a substrate, a first set of conductive structures, a shallow trench isolation (STI) region, a set of gates and a first set of vias. In some embodiments, the first set of conductive structures are located below the set of gates and the metal-0 (M0) layer of the integrated circuit. In some embodiments, the first set of conductive structures are referred to as “a buried track,” as the conductive structures of the first set of conductive structures are buried in the substrate, and beneath at least the M0-layer, the MD layer or the POLY layer. In some embodiments, by using the buried track beneath the M0-layer, one M0 track is reduced yielding a standard cell with a smaller height and smaller area compared with other approaches. In

some embodiments, the height of a standard cell of the present disclosure is reduced by up to 50% compared with other approaches.

[0025] In some embodiments, the first set of conductive structures includes a single conductive structure that extends in a first direction or a second direction. In some embodiments, the single conductive structure extends in only a single direction (e.g., first direction or second direction) and is referred to as a one directional (1D) structure. In some embodiments, the first set of conductive structures includes a first and a second conductive structure. In some embodiments, the first conductive structure extends in the first direction, and the second conductive structure extends in the second direction, and the first and the second conductive structure are referred to as a two directional (2D) structure.

[0026] In some embodiments, the STI region is between the set of active regions and the first set of conductive structures. In some embodiments, a top surface of the set of conductive structures is below a top surface of the active region of the integrated circuit or a top surface of a fin of the integrated circuit. In some embodiments, the integrated circuit comprises a transmission gate.

[0027] In some embodiments, by utilizing the first set of conductive structures of the integrated circuit of the present disclosure the gate density of a standard cell is increased by up to 30% by offering more routing flexibility than other approaches. In some embodiments, the first set of conductive structures increases routing resources for standard cells compared to other approaches. In some embodiments, the current resistance (IR) and Electromigration (EM) of the present disclosure are improved compared to other approaches.

[0028] FIG. 1 is a circuit diagram of an integrated circuit 100, in accordance with some embodiments. In some embodiments, integrated circuit 100 corresponds to a transmission gate.

[0029] Integrated circuit 100 includes two P-type metal oxide semiconductor (PMOS) transistors P1 and P2, and two N-type metal oxide semiconductor (NMOS) transistors N1 and N2.

[0030] PMOS transistor P1 includes a gate terminal A1, a terminal T1 and a terminal T3.

[0031] NMOS transistor N1 includes a gate terminal B2, a terminal T2 and a terminal T4.

[0032] Terminal T1 of PMOS transistor P1 is electrically coupled to terminal T2 of NMOS transistor N1. Terminal T3 of PMOS transistor P1 is electrically coupled to terminal T4 of NMOS transistor N1. NMOS transistor N1 and PMOS transistor P1 are coupled to each other to form a first transistor pair.

[0033] PMOS transistor P2 includes a gate terminal B1, a terminal T5 and a terminal T7.

[0034] NMOS transistor N2 includes a gate terminal A2, a terminal T6 and a terminal T8.

[0035] Terminal T7 of PMOS transistor P2 is electrically coupled to terminal T8 of NMOS transistor N2. Terminal T5 of PMOS transistor P2 is electrically coupled to the terminal T6 of NMOS transistor N2. NMOS transistor N2 and PMOS transistor P2 are coupled to each other to form a second transistor pair.

[0036] The gate terminal A1 of PMOS transistor P1 is electrically coupled to the gate terminal A2 of NMOS transistor N2. The gate terminal B1 of PMOS transistor P2 is electrically coupled to the gate terminal B2 of NMOS

transistor N1. Terminal T3 of PMOS transistor P1 and terminal T4 of NMOS transistor N1 are electrically coupled to the terminal T6 of NMOS transistor N2 and the terminal T5 of PMOS transistor P2.

[0037] In some embodiments, at least one terminal of terminals T1, T2, T3, T4, T5, T6, T7 or T8 corresponds to a source terminal. In some embodiments, at least one terminal of terminals T1, T2, T3, T4, T5, T6, T7 or T8 corresponds to a drain terminal. Other configurations of terminals in integrated circuit 100 are within the scope of the present disclosure.

[0038] In some embodiments, at least integrated circuit 100, integrated circuit 400 (FIGS. 4A-4E), layout design 200 (FIG. 2), layout design 300 (FIG. 3), layout design 500A (FIG. 5A), layout design 600A (FIG. 6A), layout design 700A (FIG. 7A), layout design 800A (FIG. 8A) or layout design 900A (FIG. 9A) corresponds to one or more standard cells. In some embodiments, the standard cell is a transmission gate. In some embodiments, a standard cell is a logic gate cell. In some embodiments, a logic gate cell includes an AND, OR, NAND, NOR, XOR, INV, AND-OR-Invert (AOI), OR-AND-Invert (OAI), MUX, Flip-flop, BUFF, Latch, delay, clock cells, or the like. In some embodiments, a standard cell is a memory cell. In some embodiments, a memory cell includes a static random access memory (SRAM), a dynamic RAM (DRAM), a resistive RAM (RRAM), a magnetoresistive RAM (MRAM), a read only memory (ROM), or the like. In some embodiments, a standard cell includes one or more active or passive elements. Examples of active elements include, but are not limited to, transistors, diodes, or the like. Examples of transistors include, but are not limited to, metal oxide semiconductor field effect transistors (MOSFET), complementary metal oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJT), high voltage transistors, high frequency transistors, p-channel and/or n-channel field effect transistors (PFETs/NFETs), FinFETs, planar MOS transistors with raised source/drain, or the like. Examples of passive elements include, but are not limited to, capacitors, inductors, fuses, resistors or the like. In some embodiments, IC 100 includes other features not shown for ease of illustration.

[0039] FIG. 2 is a diagram of a layout design 200 of an integrated circuit, in accordance with some embodiments.

[0040] Layout design 200 is usable to manufacture integrated circuit 100 of FIG. 1, integrated circuit 400 of FIGS. 4A-4E, integrated circuit 500B of FIGS. 5B-5D, integrated circuit 600B of FIGS. 6B-6E, integrated circuit 700B of FIGS. 7B-7D, integrated circuit 800B of FIGS. 8B-8G or integrated circuit 900B of FIGS. 9B-9E. Components that are the same or similar to those in each of FIG. 2-3 or 5A-9E are given the same reference numbers, and detailed description thereof is thus omitted.

[0041] Layout design 200 includes active region layout patterns 202a, 202b (collectively referred to as a “set of active region layout patterns 202”) extending in a first direction X. Active region layout patterns 202a, 202b of the set of active region layout patterns 202 are separated from one another in a second direction Y different from the first direction X. The set of active region layout patterns 202 is usable to manufacture a corresponding set of active regions 402 (FIGS. 4A-4E) of integrated circuit 400. In some embodiments, the set of active region layout patterns 202 is located on a first layout level.

[0042] Layout design 200 further includes gate layout patterns 205a, 205b, 205c, 205d (collectively referred to as a “set of gate layout patterns 205”) and gate layout patterns 206a, 206b, 206c, 206d (collectively referred to as a “set of gate layout patterns 206”) each extending in the second direction Y. Each of the layout patterns of the set of gate layout patterns 205 is separated from an adjacent layout pattern of the set of gate layout patterns 205 in the first direction X by a first pitch. Each of the layout patterns of the set of gate layout patterns 206 is separated from an adjacent layout pattern of the set of gate layout patterns 206 in the first direction X by the first pitch. Set of gate layout patterns 205 and 206 are separated from each other in the second direction Y. In some embodiments, the set of gate layout patterns 205 and 206 are not separated from each other in the second direction Y forming a single set of gate layout patterns 204 (as shown in FIGS. 3 and 5A-9E). The set of gate layout patterns 204, 205 or 206 is usable to manufacture a corresponding set of gates 404 (FIGS. 4A-4E) of integrated circuit 400. The set of gate layout patterns 204, 205 and 206 are positioned on a second layout level (POLY) different from the first layout level. The set of active region layout patterns 202 is below the set of gate layout patterns 204, 205 or 206.

[0043] Gate layout pattern 206b is usable to manufacture the gate terminal A1 of PMOS transistor P1 of FIG. 1. Gate layout pattern 206c is usable to manufacture the gate terminal B1 of PMOS transistor P2 of FIG. 1. Gate layout pattern 205b is usable to manufacture the gate terminal B2 of NMOS transistor N1 of FIG. 1. Gate layout pattern 205c is usable to manufacture the gate terminal A2 of NMOS transistor N2 of FIG. 1.

[0044] Layout design 200 further includes metal over diffusion layout patterns 210a, 210b, 210c (collectively referred to as a “set of metal over diffusion layout patterns 210”) extending in the second direction Y. Each of the layout patterns of the set of metal over diffusion layout patterns 210 is separated from an adjacent layout pattern of the set of metal over diffusion layout patterns 210 in the first direction X. The set of metal over diffusion layout patterns is located on the second layout level. The set of metal over diffusion layout patterns 210 is usable to manufacture a corresponding set of contacts 410 (FIGS. 4A-4E) of integrated circuit 400.

[0045] In some embodiments, metal over diffusion layout pattern 210a is usable to manufacture terminal T1 of PMOS transistor P1 of FIG. 1, and terminal T2 of NMOS transistor N1 of FIG. 1. In some embodiments, metal over diffusion layout pattern 210a is usable to manufacture terminal T7 of PMOS transistor P2 of FIG. 1, and terminal T8 of NMOS transistor N2 of FIG. 1. In some embodiments, metal over diffusion layout pattern 210b is usable to manufacture terminal T3 of PMOS transistor P1 of FIG. 1, terminal T4 of NMOS transistor N1 of FIG. 1, terminal T5 of PMOS transistor P2 of FIG. 1, and terminal T6 of NMOS transistor N2 of FIG. 1. In some embodiments, metal over diffusion layout pattern 210c is usable to manufacture terminal T7 of PMOS transistor P2 of FIG. 1, and terminal T8 of NMOS transistor N2 of FIG. 1. In some embodiments, metal over diffusion layout pattern 210c is usable to manufacture terminal T1 of PMOS transistor P1 of FIG. 1, and terminal T2 of NMOS transistor N1 of FIG. 1.

[0046] Layout design 200 further includes conductive feature layout patterns 212a, 212b (collectively referred to as a “set of conductive feature layout patterns 212”) extend-

ing in at least the first direction X or the second direction Y. In some embodiments, the set of conductive feature layout patterns 212 is located on at least the first layout level, the second layout level or a third layout level. In some embodiments, the third layout level is different from the first layout level and the second layout level. The set of conductive feature layout patterns 212 is usable to manufacture a corresponding set of conductive structures (e.g., set of conductive structures 420 or set of conductive structures 430 (FIGS. 4A-4E)) of integrated circuit 400. In some embodiments, the set of conductive feature layout patterns 212 is usable to manufacture a corresponding set of conductive structures 420' of integrated circuit 500B, set of conductive structures 620' of integrated circuit 600B, set of conductive structures 720' of integrated circuit 700B, set of conductive structures 820' of integrated circuit 800B or set of conductive structures 920' of integrated circuit 900B. The conductive feature layout pattern 212a corresponds to fabricating a conductive structure (e.g., conductive structure 420a or conductive structures 430a, 430b, 430c) used to electrically couple the gate terminal A1 of PMOS transistor P1 to the gate terminal A2 of NMOS transistor N2 of FIG. 1. The conductive feature layout pattern 212b corresponds to fabricating a conductive structure (e.g., conductive structure 420a or conductive structures 430a, 430b, 430c) used to electrically couple the gate terminal B1 of PMOS transistor P2 to the gate terminal B2 of NMOS transistor N1 of FIG. 1. In some embodiments, the set of conductive feature layout patterns 212 is not included in layout design 200.

[0047] FIG. 3 is a diagram of a layout design 300 of an integrated circuit, in accordance with some embodiments. Layout design 300 is usable to manufacture integrated circuit 100 (FIG. 1) or integrated circuit 400 (FIGS. 4A-4E).

[0048] Layout design 300 is a variation of layout design 200 (FIG. 2). Layout design 300 is usable to manufacture an integrated circuit similar to integrated circuit 100 (FIG. 1) or integrated circuit 400 (FIGS. 4A-4E).

[0049] In comparison with layout design 200 of FIG. 2, layout design 300 further includes one or more of cut feature layout patterns 304a and 304b, fin layout patterns 316a and 316b, conductive feature layout patterns 320a, 330a, 330b, 330c and 340a, and via layout patterns 322a, 322b, 332a, 332b, 342a and 342b.

[0050] In comparison with layout design 200 of FIG. 2, set of gate layout patterns 204 of layout design 300 replaces set of gate layout patterns 205 and 206. Set of gate layout patterns 204 corresponds to the set of gate layout patterns 205 and 206 not being separated from each other in the second direction Y thus forming a single set of gate layout patterns (e.g., set of gate layout patterns 204). The set of gate layout patterns 204 is similar to the set of gate layout patterns 205 and 206, and similar detailed description is therefore omitted.

[0051] Cut feature layout patterns 304a, 304b (collectively referred to as a “set of cut feature layout patterns 304”) extend in the first direction X. The set of cut feature layout patterns 304 overlaps at least a first gate layout pattern of the set of gate layout pattern 204. Cut feature layout patterns 304a, 304b overlap corresponding gate layout patterns 204b, 204c of the set of gate layout patterns 204. Each layout pattern of the set of cut feature layout patterns 304 has a pattern width in the second direction Y, and a pattern length in the first direction X.

[0052] In some embodiments, cut feature layout pattern 304a is usable to identify a location of gates 404b1 and 404b2 (FIGS. 4A-4E) of integrated circuit 400 that are removed during operation 1006 of method 1000 (FIG. 10). In some embodiments, cut feature layout pattern 304b is usable to identify a location of gates 404c1 and 404c2 (FIGS. 4A-4E) of integrated circuit 400 that are removed during operation 1006 of method 1000 (FIG. 10). The removed gate portion of gates 404b1 and 404b2 and gates 404c1 and 404c2 is also referred to as a cut region. In some embodiments, the set of gate layout patterns 204 and the cut feature layout pattern 304 are located on the second layout level. In some embodiments, the second layout level corresponds to a poly layout level (POLY). Other configurations or quantities of patterns in the set of cut feature layout patterns 304 are within the scope of the present disclosure.

[0053] Fin layout patterns 316a, 316b (collectively referred to as “set of fin layout patterns 316”) each extend in the first direction X. Fin layout patterns 316a, 316b are over corresponding active region layout pattern 202a, 202b. The set of fin layout patterns 316 are below the set of gate layout patterns 204. Each of the layout patterns of the set of fin layout patterns 316 is separated from an adjacent layout pattern of the set of fin layout patterns 316 in the second direction Y by a fin pitch. The set of fin layout patterns 316 is usable to manufacture a corresponding set of fins of integrated circuit 400. Other configurations or quantities of fins in the set of fin layout patterns 316 are within the scope of the present disclosure.

[0054] Conductive feature layout pattern 320a (hereinafter referred to as a “set of conductive feature layout patterns 320”) extends in at least the first direction X or the second direction Y. In some embodiments, the set of conductive feature layout patterns 320 extends in a single direction (e.g., first direction X or second direction Y). In some embodiments, the set of conductive feature layout patterns 320 extends in two directions (e.g., first direction X and second direction Y (FIG. 5A)). The set of conductive feature layout patterns 320 includes one or more conductive feature layout patterns. The set of conductive feature layout patterns 320 is located on the first layout level. In some embodiments, the first layout level of layout design 300 is the active region. In some embodiments, the first layout level of layout design 300 is divided into a metal under (MU) portion and a via under (VU) portion. In some embodiments, the MU portion is below the VU portion. In some embodiments, the set of conductive feature layout patterns 320 is located on the MU portion of the first layout level. In some embodiments, the MU portion is positioned beneath at least the M0 level, the POLY level or the MD level of layout design 300.

[0055] The set of conductive feature layout patterns 320 is between the set of active region layout patterns 202. Conductive feature layout pattern 320a is positioned between active region layout pattern 202a and active region layout pattern 202b. The set of conductive feature layout patterns 320 is usable to manufacture a corresponding set of conductive structures 420 (FIGS. 4A-4E) of integrated circuit 400. The set of conductive feature layout patterns 320 is below the set of gate layout patterns 204 and the set of metal over diffusion layout patterns 210. In some embodiments, the set of metal over diffusion layout patterns 210 overlaps the set of conductive feature layout patterns 320. The set of gate layout patterns 204 overlaps at least the set of conductive feature layout patterns 320. Other configurations or

quantities of patterns in the set of conductive feature layout patterns 320 are within the scope of the present disclosure.

[0056] Via layout patterns 322a, 322b (collectively referred to as a “set of via layout patterns 322”) are between the set of conductive feature layout patterns 320 and the set of gate layout patterns 204. Via layout patterns 322a, 322b of the set of via layout patterns 322 are between corresponding gate layout pattern 204c, 204b of the set of gate layout patterns 204 and the set of conductive feature layout patterns 320. Via layout patterns 322a, 322b are positioned at a VU portion of the first layout level of layout design 300. In some embodiments, the VU portion is above the MU portion. In some embodiments, the VU portion is positioned beneath at least the V0 level, the VG level or the VD level of layout design 300.

[0057] The set of via layout patterns 322 is usable to manufacture a corresponding set of vias 422 (FIGS. 4A-4E). The set of vias 422 couple the set of conductive structures 420 to the set of gates 404.

[0058] In some embodiments, a via layout pattern 322a, 322b of the set of via layout patterns 322 is located where a corresponding gate layout pattern 204c, 204b of the set of gate layout patterns 204 overlaps a corresponding layout pattern of the set of conductive feature layout patterns 320. In some embodiments, a center of via layout pattern 322a, 322b of the set of via layout patterns 322 is below a center of a corresponding gate layout pattern 204c, 204b of the set of gate layout patterns 204. In some embodiments, a center of via layout pattern 322a, 322b of the set of via layout patterns 322 is aligned in at least the first direction X or the second direction Y with a center of a corresponding gate layout pattern 204c, 204b of the set of gate layout patterns 204. Other configurations of via layout patterns 322a, 322b of the set of via layout patterns 322 are within the scope of the present disclosure.

[0059] Conductive feature layout patterns 330a, 330b, 330c (collectively referred to as a “set of conductive feature layout patterns 330”) extend in the first direction X. The set of conductive feature layout patterns 330 is positioned between the set of active region layout patterns 202. Each of conductive feature layout patterns 330a, 330b, 330c is positioned between active region layout pattern 202a and active region layout pattern 202b. In some embodiments, each of the layout patterns 330a, 330b, 330c of the set of conductive feature layout patterns 330 is separated from an adjacent layout pattern of the set of conductive feature layout patterns 330 in the second direction Y. The set of conductive feature layout patterns 330 is usable to manufacture a corresponding set of conductive structures 430 (FIGS. 4A-4E) of integrated circuit 400.

[0060] The set of conductive feature layout patterns 330 is located on a third layout level different from the first layout level and the second layout level. In some embodiments, the third layout level corresponds to the metal zero (M0) layout level. In some embodiments, the third layout level corresponds to an upper metal layer including the metal one (M1) layout level, metal two (M2) layout level, metal three (M3) layout level, or the like. The set of conductive feature layout patterns 330 overlaps at least the set of gate layout patterns 204 or the set of metal over diffusion layout patterns 210. The set of conductive feature layout patterns 330 is above the set of conductive feature layout patterns 320 and the set of active region layout patterns 202.

[0061] Via layout patterns 332a, 332b (collectively referred to as a “set of via layout patterns 332”) are between the set of conductive feature layout patterns 330 and the set of gate layout patterns 204. Via layout patterns 332a, 332b are positioned at a via over gate (VG) level of layout design 300. Via layout patterns 332a, 332b of the set of via layout patterns 332 are between a corresponding conductive feature layout pattern 330c, 330a of the set of conductive feature layout patterns 330 and a corresponding gate layout pattern 204b, 204c of the set of gate layout patterns 204. The set of via layout patterns 332 is usable to manufacture a corresponding set of vias 432 (FIGS. 4A-4E). The set of vias 432 couples the set of conductive structures 430 to the set of gates 404.

[0062] In some embodiments, a via layout pattern 332a, 332b of the set of via layout patterns 332 is located where a corresponding layout pattern 330c, 330a of the set of conductive feature layout patterns 330 overlaps a corresponding gate layout pattern 204b, 204c of the set of gate layout patterns 204. In some embodiments, a center of via layout pattern 332a, 332b of the set of via layout patterns 332 is over a center of a corresponding gate layout pattern 204b, 204c of the set of gate layout patterns 204. In some embodiments, a center of via layout pattern 332a, 332b of the set of via layout patterns 332 is below a center of a corresponding conductive feature layout pattern 330c, 330a of the set of conductive feature layout patterns 330. In some embodiments, a center of via layout pattern 332a, 332b of the set of via layout patterns 332 is aligned in at least the first direction X or the second direction Y with a center of a corresponding gate layout pattern 204b, 204c of the set of gate layout patterns 204. In some embodiments, a center of via layout pattern 332a, 332b of the set of via layout patterns 332 is aligned in at least the first direction X or the second direction Y with a center of a corresponding conductive feature layout pattern 330c, 330a of the set of conductive feature layout patterns 330. Other configurations of via layout patterns 332a, 332b of the set of via layout patterns 332 are within the scope of the present disclosure.

[0063] Conductive feature layout pattern 340a (hereinafter referred to as a “set of conductive feature layout patterns 340”) extends in the second direction Y. The set of conductive feature layout patterns 340 includes one or more conductive feature layout patterns. The set of conductive feature layout patterns 340 is between the set of gate layout patterns 204. In some embodiments, conductive feature layout pattern 340a is positioned between gate layout pattern 204c and gate layout pattern 204d. In some embodiments, each of the layout patterns of the set of conductive feature layout patterns 340 is separated from an adjacent layout pattern of the set of conductive feature layout patterns 340 in the first direction X. The set of conductive feature layout patterns 340 is usable to manufacture a corresponding set of conductive structures 440 (FIGS. 4A-4E) of integrated circuit 400.

[0064] The set of conductive feature layout patterns 340 overlaps at least the set of conductive feature layout patterns 320, the set of conductive feature layout patterns 330, the set of metal over diffusion layout patterns 210 or the set of active region layout patterns 202. The set of conductive feature layout patterns 340 is located on a fourth layout level different from the first layout level, the second layout level and the third layout level. In some embodiments, the fourth layout level corresponds to a metal one (M1) level of layout

design 300. Other configurations or quantities of patterns in the set of conductive feature layout patterns 340 are within the scope of the present disclosure.

[0065] Via layout patterns 342a, 342b (collectively referred to as a “set of via layout patterns 342”) are between the set of conductive feature layout patterns 330 and the set of conductive feature layout patterns 340. Via layout patterns 342a, 342b of the set of via layout patterns 342 are between corresponding conductive feature layout pattern 330a, 330c of the set of conductive feature layout patterns 330 and the set of conductive feature layout patterns 340. Via layout patterns 342a, 342b are positioned at a via zero (V0) level of layout design 300. The set of via layout patterns 342 is usable to manufacture a corresponding set of vias 442 (FIGS. 4A-4E). The set of vias 442 couples the set of conductive structures 430 to the set of conductive structures 440.

[0066] In some embodiments, a via layout pattern 342a, 342b of the set of via layout patterns 342 is located where a corresponding conductive feature layout pattern 330a, 330c of the set of conductive feature layout patterns 330 is overlapped by a layout pattern of the set of conductive feature layout patterns 340. In some embodiments, a center of via layout pattern 342a, 342b of the set of via layout patterns 342 is above a center of a corresponding conductive feature layout pattern 330a, 330c of the set of conductive feature layout patterns 330. In some embodiments, a center of via layout pattern 342a, 342b of the set of via layout patterns 342 is aligned in at least the first direction X or the second direction Y with a center of a corresponding conductive feature layout pattern 330a, 330c of the set of conductive feature layout patterns 330. Other configurations of via layout patterns 342a, 342b of the set of via layout patterns 342 are within the scope of the present disclosure.

[0067] In some embodiments, gate terminal A1, 404b1 (FIG. 4C) (manufactured by gate layout pattern 204b) of layout design 300 is electrically coupled to gate terminal A2, 404c2 (FIG. 4B) (manufactured by gate layout pattern 204c). For example, in some embodiments, gate terminal A1, 404b1 (manufactured by gate layout pattern 204b) is electrically connected through via 422b (FIG. 4C) (manufactured by via layout pattern 322b) to conductive structure 420a (manufactured by conductive feature layout pattern 320a), and conductive structure 420a is electrically coupled through via 422a (FIG. 4B) (manufactured by via layout pattern 322a) to gate terminal A2, 404c2 (manufactured by gate layout pattern 204c).

[0068] In some embodiments, gate terminal B1, 404c1 (manufactured by gate layout pattern 204c) of layout design 300 is electrically coupled to gate terminal B2, 404b2 (manufactured by gate layout pattern 204b). For example, in some embodiments, gate terminal B1, 404c1 (FIG. 4B) (manufactured by gate layout pattern 204c) is electrically connected through via 432b (manufactured by via layout pattern 332b) to a conductive structure 430a (manufactured by conductive feature layout pattern 330a). In some embodiments, conductive structure 430a is electrically coupled through via 442a, V0 (manufactured by via layout pattern 342a) to a conductive structure 440a, M1 (manufactured by conductive feature layout pattern 340a). In some embodiments, conductive structure 440a is electrically coupled through via 442b, V0 (manufactured by via layout pattern 342b) to conductive structure 430c (manufactured by conductive feature layout pattern 330c). In some embodiments,

conductive structure **430c** is electrically coupled through via **432a** (manufactured by via layout pattern **332a**) to gate terminal **B2**, **404b2** (manufactured by gate layout pattern **204b**).

[0069] In some embodiments, the set of conductive feature layout patterns **320** are located below at least the set of gate layout patterns **304**, the metal-0 (M0) level or the MD level of layout design **300**. In some embodiments, by using the set of conductive feature layout patterns **320**, at least one upper metal layer track is reduced in the set of conductive feature layout patterns **330** compared to other approaches. In some embodiments, an upper metal layer track includes one or more of M0, M1 or M2, or the like. In some embodiments, the set of conductive feature layout patterns **330** includes an upper metal layer track layout pattern located on the M1 level, M2 level, or the like. In some embodiments, by using one less upper metal layer track in the set of conductive feature layout patterns **330**, a standard cell with a smaller height and smaller area results compared with other approaches. In some embodiments, since one upper metal layer track in the set of conductive feature layout patterns **330** is reduced, the width of one or more layout patterns in the set of conductive feature layout patterns **330** is increased, yielding a standard cell with better current resistance (IR) and Electromigration (EM) compared with other approaches. In some embodiments, by utilizing the set of conductive feature layout patterns **320** of the present disclosure, the gate density of the set of gate layout patterns of a standard cell is increased by up to 30% compared with other approaches. In some embodiments, by utilizing the set of conductive feature layout patterns **320** of the present disclosure, layout design **200** offers more routing flexibility and increases routing resources compared to other approaches.

[0070] FIGS. 4A, 4B, 4C, 4D and 4E are cross-sectional views of an integrated circuit **400**, in accordance with some embodiments. FIG. 4A is a cross-sectional view of an integrated circuit **400** corresponding to layout design **300** as intersected by plane A-A', FIG. 4B is a cross-sectional view of an integrated circuit **400** corresponding to layout design **300** as intersected by plane B-B', FIG. 4C is a cross-sectional view of an integrated circuit **400** corresponding to layout design **300** as intersected by plane C-C', FIG. 4D is a cross-sectional view of an integrated circuit **400** corresponding to layout design **300** as intersected by plane D-D', and FIG. 4E is a cross-sectional view of an integrated circuit **400** corresponding to layout design **300** as intersected by plane E-E', in accordance with some embodiments. Integrated circuit **400** is manufactured by layout design **300**.

[0071] Integrated circuit **400** is manufactured by layout design **300**. Structural relationships including alignment, lengths and widths, as well as configurations of integrated circuit **400** are similar to the structural relationships and configurations of layout design **200** of FIG. 2 and layout design **300** of FIG. 3, and will not be described in FIGS. 4A-4E for brevity.

[0072] Integrated circuit **400** includes active regions **402a**, **402b** (collectively referred to as "set of active regions **402**") in a substrate **480**. The set of active regions **402** extends in the first direction X and is located on a first level of integrated circuit **400**. Each active region of the set of active regions **402** is separated from each other in the second direction Y. In some embodiments, set of active regions **402** of integrated circuit **400** is referred to as an oxide definition

(OD) region which defines the source or drain diffusion regions of integrated circuit **400**. In some embodiments, the first level of integrated circuit **400** is referred to as the Active/Fin level. In some embodiments, active region **402a**, **402b** includes a set of fins (not shown) extending in the first direction and being below the set of gates **404**. In some embodiments, each of the fins (not shown) is separated from an adjacent fin of the set of fins (not shown) in the second direction Y by a fin pitch (not shown). Other quantities or configurations of the set of active region **402** are within the scope of the present disclosure.

[0073] Integrated circuit **400** further includes a conductive structure **420a** (collectively referred to as a "set of conductive structures **420**") extending in at least the first direction X or the second direction Y. The set of conductive structures **420** is located on the first level of integrated circuit **400**. The set of conductive structures **420** is between the set of active regions **402**. In some embodiments, conductive structure **420a** is positioned between active regions **402a**, **402b**. In some embodiments, a top surface of conductive structure **420a** of the set of conductive structures **420** is below a top surface of active region **402a**, **402b** of the set of active regions **402** or a top surface of a fin (not shown) of the set of fins (not shown).

[0074] In some embodiments, the first level of integrated circuit **400** is the active region of integrated circuit **400**. In some embodiments, the first level of integrated circuit **400** is divided into a metal under (MU) region and a via under (VU) region. In some embodiments, the MU region is below the VU region. In some embodiments, the set of conductive structures **420** is in the MU region of the first level of integrated circuit **400**. In some embodiments, the conductive structures of the set of conductive structures **420** are buried inside the substrate **480**, and positioned beneath at least the M0 layer, the POLY layer, the MD layer of integrated circuit **400**. In some embodiments, each conductive structure **420a** in the set of conductive structures **420** has a width (not shown) in the second direction Y ranging from about 1-50 nm. In some embodiments, each conductive structure **420a** in the set of conductive structures **420** has a length (not shown) in the first direction X ranging from about 1-10 μm . Other quantities or configurations of the set of conductive structures **404** are within the scope of the present disclosure.

[0075] Integrated circuit **400** further includes a shallow trench isolation (STI) region **490** extending in at least the first direction X or the second direction Y. STI region **490** is located on at least the first level of integrated circuit **400**. In some embodiments, STI region **490** is positioned between the set of active regions **402** and the set of conductive structures **420**. In some embodiments, conductive structure **420a** is separated from each of active regions **402a**, **402b** by STI region **490**. Other quantities or configurations of the STI region **490** are within the scope of the present disclosure.

[0076] Integrated circuit **400** further includes gates **404b1**, **404b2**, **404c1**, **404c2** (collectively referred to as "set of gates **404**") extending in the second direction Y, overlapping at least the set of conductive structures **420** and being located on a second level of integrated circuit **400**. In some embodiments, the second level of integrated circuit **400** is different from the first level. Each of the gates of the set of gates **404** is separated from an adjacent gate of the set of gates **404** in the first direction X by a first pitch. In some embodiments, the second level of integrated circuit **400** is referred to as the

Poly level. Other quantities or configurations of the set of gates **404** are within the scope of the present disclosure.

[0077] Integrated circuit **400** further includes contacts **410a**, **410b**, **410c** (collectively referred to as a “set of contacts **410**”) extending in the second direction Y, overlapping the set of conductive structures **420**, and being located on the second level of integrated circuit **400**. Each of the contacts **410a**, **410b**, **410c** of the set of contacts **410** is separated from an adjacent contact of the set of contacts **410** in the first direction X. In some embodiments, the set of contacts **410** is located on the MD level of integrated circuit **400**. In some embodiments, at least a contact of the set of contacts **410** is electrically coupled to at least a source or a drain of the active regions **402a**, **402b**. Other quantities or configurations of the set of contacts **410** are within the scope of the present disclosure.

[0078] Integrated circuit **400** further includes vias **422a**, **422b** (collectively referred to as a “set of vias **422**”) coupling the set of conductive structures **420** to the set of gates **404**. The set of vias **422** is between the set of conductive structures **420** and the set of gates **404**. Via **422b** is located where gates **404b1**, **404b2** overlap conductive structure **420a**. Via **422a** is located where gates **404c1**, **404c2** overlap conductive structure **420a**. In some embodiments, set of vias **422** is in the VU level of integrated circuit **400**. The VU level of integrated circuit **400** is between the first level and the second level. Other quantities or configurations of the set of vias **422** are within the scope of the present disclosure.

[0079] Integrated circuit **400** further includes conductive structures **430a**, **430b**, **430c** (collectively referred to as a “set of conductive structures **430**”) extending in the first direction X, being between the set of active regions **402**, overlapping at least the set of gates **404**, and being located on a third level of integrated circuit **400**. In some embodiments, the third level of integrated circuit **400** is different from the first level and the second level. In some embodiments, the third level of integrated circuit **400** is referred to as the M0 level. Each of the conductive structures **430a**, **430b**, **430c** of the set of conductive structures **430** is separated from an adjacent conductive structure of the set of conductive structures **430** in the second direction Y. Other quantities or configurations of the set of conductive structures **430** are within the scope of the present disclosure.

[0080] Integrated circuit **400** further includes vias **432a**, **432b** (collectively referred to as a “set of vias **432**”) coupling the set of conductive structures **430** to the set of gates **404**. The set of vias **432** is between the set of conductive structures **430** and the set of gates **404**. Via **432a** is located where conductive structure **430c** overlaps gates **404b1**, **404b2**. Via **432b** is located where conductive structure **430a** overlaps gates **404c1**, **404c2**. In some embodiments, set of vias **432** is in the VG level of integrated circuit **400**. The VG level of integrated circuit **400** is between the second level and the third level. Other quantities or configurations of the set of vias **432** are within the scope of the present disclosure.

[0081] Integrated circuit **400** further includes conductive structure **440a** (collectively referred to as “set of conductive structures **440**”) extending in the second direction Y, being between the set of gates **404**, overlapping at least the set of conductive structures **420**, and being located on a fourth level of integrated circuit **400**. In some embodiments, the fourth level of integrated circuit **400** is different from the first level, the second level and the third level. In some embodiments, the fourth level of integrated circuit **400** is

referred to as the M1 level. In some embodiments, each of the structures of the set of conductive structures **440** is separated from an adjacent structure of the set of conductive structures **440** in the first direction X. Other quantities or configurations of the set of conductive structures **440** are within the scope of the present disclosure.

[0082] Integrated circuit **400** further includes vias **442a**, **442b** (collectively referred to as “set of vias **442**”) coupling the set of conductive structures **440** to the set of conductive structures **430**. The set of vias **442** is between the set of conductive structures **440** and the set of conductive structures **430**. Via **442a** is located where conductive structure **440a** overlaps conductive structure **430a**. Via **442b** is located where conductive structure **440a** overlaps conductive structure **430c**. In some embodiments, set of vias **442** are in the V0 level of integrated circuit **400**. The V0 level of integrated circuit **400** is between the third level and the fourth level. Other quantities or configurations of the set of vias **442** are within the scope of the present disclosure.

[0083] In some embodiments, one or more conductive structure of the set of conductive structures **420**, **430** or **440** includes a metal, a metal compound or a doped semiconductor. In some embodiments, one or more vias of the set of vias **422**, **432**, **442** includes a metal, a metal compound or a doped semiconductor. In some embodiments, a metal includes at least Cu (Copper), Co, W, Ru, Al, or the like. In some embodiments, a metal compound includes at least AlCu, W—TiN, TiSix, NiSix, TiN, TaN, or the like. In some embodiments, a doped semiconductor includes at least doped silicon, or the like.

[0084] In some embodiments, gate terminal A1, **404b1** is electrically coupled to gate terminal A2, **404c2**. For example, gate terminal A1, **404b1** is electrically connected through via **422b** to conductive structure **420a**, and conductive structure **420a** is electrically coupled through via **422a** to gate terminal A2, **404c2**.

[0085] In some embodiments, gate terminal B1, **404c1** is electrically coupled to gate terminal B2, **404b2**. For example, in some embodiments, gate terminal B1, **404c1** is electrically connected through via **432b** to conductive structure **430a**. In some embodiments, conductive structure **430a** is electrically coupled through via **442a**, V0 to conductive structure **440a**, M1. In some embodiments, conductive structure **440a** is electrically coupled through via **442b**, V0 to conductive structure **430c**. In some embodiments, conductive structure **430c** is electrically coupled through via **432a** to gate terminal B2, **404b2**.

[0086] In some embodiments, the set of conductive structures **420** is located below at least the set of gates **404**, the metal 0 (M0) layer or the MD layer of integrated circuit **400**. In some embodiments, the set of conductive structures **420** is referred to as a buried metal track (BMT), as the conductive structures of the set of conductive structures **420** are buried inside the substrate **480**, and positioned beneath at least the M0 layer, the POLY layer, the MD layer of integrated circuit **400**.

[0087] In some embodiments, by using the set of conductive structures **420**, one upper metal layer track (e.g., MD track, M0 track, M1 track, or the like) is reduced in the set of conductive structures **430** yielding a standard cell with a smaller height and smaller area compared with other approaches. In some embodiments, by using the set of conductive structures **420**, one upper metal layer track (e.g., MD track, M0 track, M1 track, or the like) is reduced in the

set of conductive structures **430**, and the space occupied by the removed upper layer metal track can be utilized by increasing a width of at least one conductive structure in the set of conductive structures **430**, yielding a standard cell with better current resistance (IR) and EM compared with other approaches.

[0088] In some embodiments, by reducing one upper metal layer track of the present disclosure, the cell height can be reduced resulting in an increase in the gate density of a standard cell by up to 30% compared with other approaches. In some embodiments, by reducing one upper metal layer track of the present disclosure, integrated circuit **400** offers more routing flexibility and increases routing resources compared to other approaches.

[0089] FIG. 5A is a diagram of a layout design **500A** of an integrated circuit, in accordance with some embodiments.

[0090] FIGS. 5B, 5C and 5D are cross-sectional views of integrated circuit **500B**, in accordance with some embodiments.

[0091] Layout design **500A** is a variation of layout design **300** (FIG. 3). For example, layout design **500A** illustrates an example of where the MU layer (e.g., set of conductive feature layout patterns **520**) extends in two directions.

[0092] Layout design **500A** is usable to manufacture an integrated circuit similar to integrated circuit **100** (FIG. 1), integrated circuit **400** (FIGS. 4A-4E) or **500B**.

500A

[0093] In comparison with layout design **300** of FIG. 3, conductive feature layout patterns **520a**, **520b**, and **520c** of layout design **500A** replace conductive feature layout pattern **320a**.

[0094] Conductive feature layout patterns **520a**, **520b**, **520c** (hereinafter referred to as a “set of conductive feature layout patterns **520**”) are located on the first layout level. The set of conductive feature layout patterns **520** is usable to manufacture a corresponding set of two-dimensional (2D) conductive structures **420** or **420'** of integrated circuit **100**, **400** or **500B**.

[0095] In some embodiments, the set of conductive feature layout patterns **520** is located on the MU portion of the first layout level. The set of conductive feature layout patterns **520** extends in two directions (e.g., first direction X and second direction Y). Conductive feature layout pattern **520a** and **520c** each extend in the first direction X. Conductive feature layout pattern **520b** extends in the second direction Y.

[0096] The set of conductive feature layout patterns **520** is between the set of active region layout patterns **202**. Conductive feature layout pattern **520a** is positioned between conductive feature layout pattern **330a** and conductive feature layout pattern **330b**. Conductive feature layout pattern **520c** is positioned between conductive feature layout pattern **330b** and conductive feature layout pattern **330c**.

[0097] Conductive feature layout pattern **520b** is positioned between gate layout patterns **204b** and **204c**. In some embodiments, conductive feature layout pattern **520b** is positioned between conductive feature layout pattern **330a** and conductive feature layout pattern **330c**. In some embodiments, metal over diffusion layout pattern **210a**, **210b**, **210c** overlaps a corresponding conductive feature layout pattern **520a**, **520b**, **520c**.

[0098] The set of gate layout patterns **204** overlaps the set of conductive feature layout patterns **520**. In some embodi-

ments, gate layout patterns **204a**, **204b** overlap conductive feature layout pattern **520a**. In some embodiments, gate layout patterns **204c**, **204d** overlap conductive feature layout pattern **520c**. Other configurations or quantities of patterns in the set of conductive feature layout patterns **520** are within the scope of the present disclosure.

[0099] In comparison with layout design **300** of FIG. 3, via layout patterns **522a**, **522b** of layout design **500A** replace corresponding via layout patterns **322a**, **322b**. The set of via layout patterns **522** is similar to the set of via layout patterns **322**, and similar detailed description is therefore omitted. In some embodiments, the set of via layout patterns **522** is usable to manufacture a corresponding set of vias **422'** (FIGS. 5A-5D) coupled between the set of gates **404** and the set of conductive structures **420'** manufactured by the set of conductive feature layout patterns **520**.

[0100] FIG. 5B is a cross-sectional view of integrated circuit **500B** corresponding to layout design **500A** as intersected by plane F-F', FIG. 5C is a cross-sectional view of integrated circuit **500B** corresponding to layout design **500A** as intersected by plane G-G', and FIG. 5D is a cross-sectional view of integrated circuit **500B** corresponding to layout design **500A** as intersected by plane H-H' in accordance with some embodiments. Structural relationships including alignment, lengths and widths, as well as configurations of integrated circuit **500B** are similar to the structural relationships and configurations of layout design **500A** of FIG. 5A, and will not be described in FIGS. 5B-5D for brevity.

[0101] Integrated circuit **500B** is manufactured by layout design **500A**. In some embodiments, integrated circuit **500B** is similar to integrated circuit **400**, and similar detailed description is therefore omitted for brevity.

[0102] In comparison with integrated circuit **400** of FIGS. 4A-4E, set of conductive structures **420'** of integrated circuit **500B** replaces set of conductive structures **420**, and set of vias **422'** of integrated circuit **500B** replaces set of vias **422**.

[0103] In some embodiments, the set of conductive structures **420'** includes one or more of conductive structures **420a'**, **420b'** and **420c'**. In some embodiments, the set of vias **422'** includes one or more of vias **422a'** and **422b'**.

[0104] Conductive structures **420a'** and **420c'** extend in the first direction X, and are separated from each other in the second direction Y. Conductive structure **420b'** extends in the second direction Y. Conductive structure **420b'** directly contacts conductive structures **420a'** and **420c'**. In some embodiments, conductive structures **420a'**, **420b'** and **420c'** are part of a same continuous conductive structure. In some embodiments, set of conductive structures **420'** is referred to as a 2D conductive structure.

[0105] In some embodiments, gate terminal A1 or gate **404b1** (manufactured by gate layout pattern **204b**) of integrated circuit **500B** is electrically coupled to gate terminal A2 or gate **404c2** (manufactured by gate layout pattern **204c**). For example, in some embodiments, gate terminal A1 or gate **404b1** (manufactured by gate layout pattern **204b**) is electrically connected through via **422b'** (manufactured by via layout pattern **522b**) to the two dimensional (2-D) conductive structure **420a'**, **420b'** and **420c'** (manufactured by conductive feature layout patterns **520a**, **520b** **520c**), and the 2-D conductive structure **420a'**, **420b'** and **420c'** is electrically coupled through via **422a'** (manufactured by via layout pattern **522a**) to gate terminal A2 or gate **404c2** (manufactured by gate layout pattern **204c**).

[0106] In some embodiments, gate terminal B1 or gate 404c1 (manufactured by gate layout pattern 204c) of integrated circuit 500B is electrically coupled to gate terminal B2 or gate 404b2 (manufactured by gate layout pattern 204b) similar to integrated circuit 400 (FIGS. 4A-4E), and similar detailed description is therefore omitted.

[0107] FIG. 6A is a diagram of a layout design 600A of an integrated circuit 600B, in accordance with some embodiments.

[0108] FIGS. 6B, 6C, 6D and 6E are cross-sectional views of integrated circuit 600B, in accordance with some embodiments.

[0109] Layout design 600A is a variation of layout design 300 (FIG. 3). For example, layout design 600A illustrates an example of where the gate layout pattern (e.g., gate layout patterns 204b, 204c) is continuous, and the MU layer (e.g., a set of conductive feature layout patterns 620) extends in a single direction.

[0110] Layout design 600A is usable to manufacture an integrated circuit similar to integrated circuit 100 (FIG. 1), integrated circuit 400 (FIGS. 4A-4E) or 600B (FIGS. 6B-6E).

[0111] In comparison with layout design 300 of FIG. 3, layout design 600A does not include set of cut feature layout patterns 304. By not including the set of cut feature layout patterns 304, layout design 600A is usable to manufacture integrated circuit 600B having gate terminal A1 directly coupled to gate terminal A2, and gate terminal B1 directly coupled to gate terminal B2. For ease of illustration, each of layout design 600A of FIG. 6A, layout design 700A of FIG. 7A, layout design 800A of FIG. 8A and layout design 900A of FIG. 9A do not show set of fin layout patterns 316 of FIG. 3.

[0112] In comparison with layout design 300 of FIG. 3, a set of metal over diffusion layout patterns 610 of layout design 600A replaces set of metal over diffusion layout patterns 210, and similar detailed description is therefore omitted. The set of metal over diffusion layout patterns 610 is usable to manufacture a corresponding set of contacts 610' (FIGS. 6B-6E) of integrated circuit 600B.

[0113] Set of metal over diffusion layout patterns 610 includes one or more of metal over diffusion layout patterns 610a, 610b, 610c, 612a or 612c. Metal over diffusion layout patterns 610a, 610b, 610c, 612a and 612c are usable to manufacture corresponding contacts 610a', 610b', 610c', 612a' and 612c' (FIGS. 6B-6E) of integrated circuit 600B.

[0114] In comparison with layout design 300 of FIG. 3, metal over diffusion layout patterns 610a, 612a of layout design 600A replace metal over diffusion layout pattern 210a, metal over diffusion layout pattern 610b replaces metal over diffusion layout pattern 210b, and metal over diffusion layout patterns 610c, 612c replace metal over diffusion layout pattern 210c, and similar detailed description is therefore omitted.

[0115] Metal over diffusion layout patterns 610a, 612a are similar to metal over diffusion layout patterns 210a. Metal over diffusion layout pattern 610a is separated or discontinuous from metal over diffusion layout patterns 612a. In some embodiments, metal over diffusion layout pattern 610a extends in the second direction Y from an edge of active region layout pattern 202b to an edge of conductive feature layout pattern 330a. In some embodiments, metal over diffusion layout pattern 612a extends in the second direction

Y from an edge of active region layout pattern 202a to an edge of a conductive feature layout pattern 630b.

[0116] Metal over diffusion layout pattern 610b is similar to metal over diffusion layout pattern 210b with the reference numeral increased by 400, and similar detailed description is therefore omitted.

[0117] Metal over diffusion layout patterns 610c, 612c are similar to metal over diffusion layout patterns 210c. Metal over diffusion layout pattern 610c is separated or discontinuous from metal over diffusion layout patterns 612c. In some embodiments, metal over diffusion layout pattern 610c extends in the second direction Y from an edge of active region layout pattern 202b to an edge of conductive feature layout pattern 630b'. In some embodiments, metal over diffusion layout pattern 612c extends in the second direction Y from an edge of active region layout pattern 202a to an edge of conductive feature layout pattern 330c or 630b'.

[0118] In comparison with layout design 300 of FIG. 3, a set of via layout patterns 622, 632 or 642 of layout design 600A replaces corresponding set of via layout patterns 322, 332 or 342, and similar detailed description is therefore omitted. The set of via layout patterns 622, 632 or 642 is usable to manufacture a corresponding set of vias 622', 632' or 642' (FIGS. 6B-6E) of integrated circuit 600B.

[0119] Set of via layout patterns 622 includes one or more of via layout patterns 622a or 622b. Set of via layout patterns 632 includes one or more of via layout patterns 632a or 632b. Set of via layout patterns 642 includes one or more of via layout patterns 642a or 642b.

[0120] Via layout patterns 622a, 622b, 632a, 632b, 642a and 642b are usable to manufacture a corresponding via 622a', 622b', 632a', 632b', 642a' and 642b' (FIGS. 6B-6E) of integrated circuit 600B.

[0121] In comparison with layout design 300 of FIG. 3, via layout patterns 622a, 622b of layout design 600A replace corresponding via layout patterns 322a, 322b, via layout patterns 632a, 632b replace corresponding via layout patterns 332a, 332b, and via layout patterns 642a, 642b replace corresponding via layout patterns 342a, 342b, and similar detailed description is therefore omitted.

[0122] Via layout patterns 622a, 622b are similar to corresponding via layout patterns 322a, 322b. Via layout patterns 622a, 622b (collectively referred to as a "set of via layout patterns 622") are between conductive feature layout pattern 620a and corresponding metal over diffusion layout patterns 612a, 610c. The set of via layout patterns 622 is usable to manufacture a corresponding set of vias 622' (VU) that couple the set of conductive structures 620' to the set of contacts 610'.

[0123] In some embodiments, a via layout pattern 622a, 622b of the set of via layout patterns 622 is located where a corresponding layout pattern 612a, 610c of the set of metal over diffusion layout patterns 610 overlaps conductive feature layout pattern 620a of the set of conductive feature layout patterns 620. In some embodiments, a center of via layout patterns 622a, 622b of the set of via layout patterns 622 is below a center of a corresponding metal over diffusion layout pattern 612a, 610c of the set of metal over diffusion layout patterns 610. In some embodiments, a center of via layout patterns 622a, 622b of the set of via layout patterns 622 is aligned in at least the first direction X or the second direction Y with a center of a corresponding metal over diffusion layout pattern 612a, 610c of the set of metal over diffusion layout patterns 610. Other configura-

tions of via layout patterns **622a**, **622b** of the set of via layout patterns **622** are within the scope of the present disclosure.

[0124] Via layout patterns **632a**, **632b** (collectively referred to as a “set of via layout patterns **632**”) are similar to corresponding via layout patterns **332a**, **332b**. The set of via layout patterns **632** are between the set of conductive feature layout patterns **630** and the set of gate layout patterns **204**. Via layout pattern **632a**, **632b** of the set of via layout patterns **632** is between a corresponding conductive feature layout pattern **630b**, **630b'** and a corresponding gate layout pattern **204b**, **204c** of the set of gate layout patterns **204**. The set of via layout patterns **632** is usable to manufacture a corresponding set of vias **632'** (FIGS. 6B-6E). As shown in FIGS. 6B-6E, vias **632a'**, **632b'** couple the corresponding conductive structures **660b1'**, **660b2'** to corresponding gates **604b**, **604c**.

[0125] In some embodiments, a via layout pattern **632a**, **632b** of the set of via layout patterns **632** is located where a corresponding layout pattern **630b**, **630b'** overlaps a corresponding gate layout pattern **204b**, **204c** of the set of gate layout patterns **204**. In some embodiments, a center of via layout pattern **632a**, **632b** of the set of via layout patterns **632** is over a center of a corresponding gate layout pattern **204b**, **204c** of the set of gate layout patterns **204**. In some embodiments, a center of via layout pattern **632a**, **632b** of the set of via layout patterns **632** is below a center of a corresponding conductive feature layout pattern **630b**, **630b'**. In some embodiments, a center of via layout pattern **632a**, **632b** of the set of via layout patterns **632** is aligned in at least the first direction X or the second direction Y with a center of a corresponding gate layout pattern **204b**, **204c** of the set of gate layout patterns **204**. In some embodiments, a center of via layout pattern **632a**, **632b** of the set of via layout patterns **632** is aligned in at least the first direction X or the second direction Y with a center of a corresponding conductive feature layout pattern **630b**, **630b'**. Other configurations of via layout patterns **632a**, **632b** of the set of via layout patterns **632** are within the scope of the present disclosure.

[0126] Via layout patterns **642a**, **642b** are similar to corresponding via layout patterns **342a**, **342b**. Via layout pattern **642a**, **642b** (collectively referred to as “set of via layout patterns **642**”) is between conductive feature layout pattern **640a** and corresponding conductive feature layout patterns **330a**, **330c**. Via layout patterns **642a**, **642b** are usable to manufacture corresponding vias **642a'**, **642b'**. As shown in FIGS. 6B-6E, vias **642a'**, **642b'** couple conductive structure **640a'** to corresponding conductive structures **430a**, **430c**.

[0127] In some embodiments, a via layout pattern **642a**, **642b** of the set of via layout patterns **642** is located where layout pattern **640a** of the set of conductive feature layout patterns **640** overlaps a corresponding conductive feature layout pattern **330a**, **330c**. In some embodiments, a center of via layout patterns **642a**, **642b** of the set of via layout patterns **642** is below a center of conductive feature layout pattern **640a**. In some embodiments, a center of via layout patterns **642a**, **642b** of the set of via layout patterns **642** is above a center of a corresponding conductive feature layout pattern **330a**, **330c**. In some embodiments, a center of via layout patterns **642a**, **642b** of the set of via layout patterns **642** is aligned in at least the first direction X or the second direction Y with a center of a corresponding conductive feature layout pattern **330a**, **330c**. Other configurations of

via layout patterns **642a**, **642b** of the set of via layout patterns **642** are within the scope of the present disclosure.

[0128] In comparison with layout design **300** of FIG. 3, layout design **600A** further includes via layout patterns **636a**, **636b** (collectively referred to as a “set of via layout patterns **636**”).

[0129] Via layout patterns **636a**, **636b** (collectively referred to as a “set of via layout patterns **636**”) are between the set of conductive feature layout patterns **630** and the set of metal over diffusion layout patterns **610**. Via layout pattern **636a**, **636b** of the set of via layout patterns **636** is between a corresponding conductive feature layout pattern **330a**, **330c** and a corresponding metal over diffusion layout pattern **610a**, **612c**. The set of via layout patterns **636** is usable to manufacture a corresponding set of vias **636'** (e.g., VD). As shown in FIGS. 6B-6E, vias **636a'**, **636b'** couple corresponding conductive structures **430a**, **430c** to corresponding contacts **610a'**, **612c'**.

[0130] In some embodiments, a via layout pattern **636a**, **636b** of the set of via layout patterns **636** is located where a corresponding layout pattern **330a**, **330c** overlaps a corresponding metal over diffusion layout pattern **610a**, **612c**. In some embodiments, a center of via layout pattern **636a**, **636b** of the set of via layout patterns **636** is over a center of a corresponding metal over diffusion layout pattern **610a**, **612c**. In some embodiments, a center of via layout pattern **636a**, **636b** of the set of via layout patterns **636** is below a center of a corresponding conductive feature layout pattern **330a**, **330c**. In some embodiments, a center of via layout pattern **636a**, **636b** of the set of via layout patterns **636** is aligned in at least the first direction X or the second direction Y with a center of a corresponding metal over diffusion layout pattern **610a**, **612c**. In some embodiments, a center of via layout pattern **636a**, **636b** of the set of via layout patterns **636** is aligned in at least the first direction X or the second direction Y with a center of a corresponding conductive feature layout pattern **330a**, **330c**. Other configurations of via layout patterns **636a**, **636b** of the set of via layout patterns **636** are within the scope of the present disclosure.

[0131] In comparison with layout design **300** of FIG. 3, conductive feature layout pattern **620a** of layout design **600A** replaces conductive feature layout pattern **320a**, conductive feature layout patterns **630b**, **630b'** replace conductive feature layout pattern **330b**, and conductive feature layout pattern **640a** replaces conductive feature layout pattern **340a**, and similar detailed description is therefore omitted.

[0132] Conductive feature layout pattern **620a** is similar to conductive feature layout pattern **320a** with the reference numerals increased by 300. Conductive feature layout pattern **620a** is usable to manufacture a corresponding conductive structure **620a'** (FIGS. 6B-6E). Other configurations of conductive feature layout pattern **620a** of the set of conductive feature layout patterns **620** are within the scope of the present disclosure.

[0133] Conductive feature layout patterns **630b**, **630b'** are similar to conductive feature layout pattern **330b** with the reference numerals increased by 300. Conductive feature layout patterns **630b**, **630b'** are usable to manufacture corresponding conductive structures **660b1'**, **660b2'** (FIGS. 6B-6E). Conductive feature layout pattern **630b** is separated or discontinuous from conductive feature layout pattern **630b'**. In some embodiments, conductive feature layout pattern **630b** extends in the first direction X to an edge of

metal over diffusion layout pattern **610b**. In some embodiments, conductive feature layout pattern **630b'** extends in the first direction X to another edge of metal over diffusion layout pattern **610b**. Other configurations of conductive feature layout patterns **630b**, **630b'** of the set of conductive feature layout patterns **630** are within the scope of the present disclosure.

[0134] Conductive feature layout pattern **640a** is similar to conductive feature layout pattern **340a** with the reference numerals increased by 300. Conductive feature layout pattern **640a** is usable to manufacture corresponding conductive structure **440a** (FIGS. 6B-6E). Conductive feature layout pattern **640a** is positioned between gate layout patterns **204b** and **204c**. Other configurations of conductive feature layout pattern **640a** of a set of conductive feature layout patterns **640** are within the scope of the present disclosure.

[0135] FIG. 6B is a cross-sectional view of integrated circuit **600B** corresponding to layout design **600A** as intersected by plane I-I', FIG. 6C is a cross-sectional view of integrated circuit **600B** corresponding to layout design **600A** as intersected by plane J-J', FIG. 6D is a cross-sectional view of integrated circuit **600B** corresponding to layout design **600A** as intersected by plane K-K', and FIG. 6E is a cross-sectional view of integrated circuit **600B** corresponding to layout design **600A** as intersected by plane L-L', in accordance with some embodiments. Structural relationships including alignment, lengths and widths, as well as configurations of integrated circuit **600B** are similar to the structural relationships and configurations of layout design **600A** of FIG. 6A, and will not be described in FIGS. 6B-6E for brevity.

[0136] Integrated circuit **600B** is manufactured by layout design **600A**. In some embodiments, integrated circuit **600B** is similar to integrated circuit **400**, and similar detailed description is therefore omitted for brevity.

[0137] In comparison with integrated circuit **400** of FIGS. 4A-4E, set of contacts **610'** of integrated circuit **600B** replaces set of contacts **410**, set of vias **622'**, **632'**, **642'** of integrated circuit **600B** replace corresponding set of vias **422**, **432**, **442**, conductive structure **620a'** of integrated circuit **600B** replaces conductive structure **420a**, conductive structure **640a'** of integrated circuit **600B** replaces conductive structure **440a**, conductive structures **660b1'**, **660b2'** of integrated circuit **600B** replaces conductive structure **430b**, gate **604b** of integrated circuit **600B** replaces gates **404b1** and **404b2**, and gate **604c** of integrated circuit **600B** replaces gates **404c1** and **404c2**, and similar detailed description is therefore omitted for brevity.

[0138] In some embodiments, the set of contacts **610'** includes one or more of contacts **610a'**, **610b'**, **610c'**, **612a'** and **612c'**. In some embodiments, the set of vias **622'** includes one or more of vias **622a'** or **622b'**. In some embodiments, the set of vias **632'** includes one or more of vias **632a'** or **632b'**. In some embodiments, the set of vias **642'** includes one or more of vias **642a'** or **642b'**.

[0139] In comparison with contact **410a** of integrated circuit **400** of FIGS. 4A-4E, contacts **610a'** and **612a'** are not a single, continuous structure. Contacts **610a'** and **612a'** are separated from each other in the second direction Y.

[0140] In comparison with contact **410c** of integrated circuit **400** of FIGS. 4A-4E, contacts **610c'** and **612c'** are not a single, continuous structure. Contacts **610c'** and **612c'** are separated from each other in the second direction Y.

[0141] In comparison with conductive structure **430b** of integrated circuit **400** of FIGS. 4A-4E, conductive structures **660b1'**, **660b2'** are not a single, continuous structure. In other words, conductive structures **660b1'** and **660b2'** are separated from each other in the first direction X.

[0142] In comparison with integrated circuit **400** of FIGS. 4A-4E, integrated circuit **600B** further includes a set of vias **636'**. In some embodiments, the set of vias **636'** includes one or more of vias **636a'** or **636b'**.

[0143] Set of vias **636'** are similar to set of vias **432**, except the set of vias **636'** are positioned between the set of contacts **610'** and one or more of conductive structures **430a**, **430c**, **660b1'** or **660b2'**, and similar detailed description is therefore omitted for brevity. In other words, the set of vias **636'** (e.g., VD level) are above the set of contacts **610'** (e.g., MD level), and are configured to couple the set of contacts **610'** to upper metal layers (e.g., M0 level). For example, vias **636a'**, **636b'** couple corresponding conductive structures **430a**, **430c** to corresponding contacts **610a'**, **612c'**. Other configurations of set of vias **636'** are within the scope of the present disclosure.

[0144] In comparison with gates **404b1** and **404b2** of integrated circuit **400** of FIGS. 4A-4E, gate **604b** is a continuous structure. In some embodiments, gate terminal A1 (manufactured by gate layout pattern **204b**) of integrated circuit **600B** is directly coupled to gate terminal A2 (manufactured by gate layout pattern **204b**) by gate **604b**.

[0145] In comparison with gates **404c1** and **404c2** of integrated circuit **400** of FIGS. 4A-4E, gate **604c** is a continuous structure. In some embodiments, gate terminal B1 (manufactured by gate layout pattern **204c**) of integrated circuit **600B** is directly coupled to gate terminal B2 (manufactured by gate layout pattern **204c**) by gate **604c**.

[0146] In some embodiments, terminal A1D corresponds to contact **610a'**, and terminal B2D corresponds to contact **612c'**. In these embodiments, terminal A1D or contact **610a'** (manufactured by metal over diffusion layout pattern **610a**) of integrated circuit **600B** is electrically coupled to terminal B2D or contact **612c'** (manufactured by metal over diffusion layout pattern **612c**). For example, in some embodiments, contact **610a'** (manufactured by metal over diffusion layout pattern **610a**) is electrically connected through via **636a'** (manufactured by via layout pattern **636a**) to conductive structure **430a** (manufactured by conductive feature layout pattern **330a**). In some embodiments, conductive structure **430a** is electrically coupled through via **642a'** (manufactured by via layout pattern **642a**) to conductive structure **640a'** (manufactured by conductive feature layout pattern **640a**). In some embodiments, conductive structure **640a'** is electrically coupled through via **642b'** (manufactured by via layout pattern **642b**) to conductive structure **430c** (manufactured by conductive feature layout pattern **330c**). In some embodiments, conductive structure **430c** is electrically coupled through via **636b'** (manufactured by via layout pattern **636b**) to contact **612c'** (manufactured by metal over diffusion layout pattern **612c**). Thus, terminal A1D or contact **610a'** is electrically coupled to terminal B2D or contact **612c'**.

[0147] In some embodiments, terminal BID corresponds to contact **610c'**, and terminal A2D corresponds to contact **612a'**. In these embodiments, terminal BID or contact **610c'** (manufactured by metal over diffusion layout pattern **610c**) of integrated circuit **600B** is electrically coupled to terminal A2D or contact **612a'** (manufactured by metal over diffusion

layout pattern 612a). For example, in some embodiments, contact 610c' (manufactured by metal over diffusion layout pattern 610c) is electrically connected through via 622b' (manufactured by via layout pattern 622b) to conductive structure 620a' (manufactured by conductive feature layout pattern 620a). In some embodiments, conductive structure 620a' is electrically coupled through via 622a' (manufactured by via layout pattern 622a) to contact 612a' (manufactured by metal over diffusion layout pattern 612a). Thus, terminal B1D or contact 610c' of integrated circuit 600B is electrically coupled to terminal A2D or contact 612a'.

[0148] Other configurations of terminals A1D, A2D, B1D or B2D are within the scope of the present disclosure. For example, in some embodiments, terminal A1D corresponds to contact 610c', and terminal B2D corresponds to contact 612a'. For example, in some embodiments, terminal A2D corresponds to contact 610a', and terminal B1D corresponds to contact 612c'.

[0149] FIG. 7A is a diagram of a layout design 700A of an integrated circuit, in accordance with some embodiments.

[0150] FIG. 7B, 7C or 7D are cross-sectional views of integrated circuit 700B, in accordance with some embodiments.

[0151] Layout design 700A is a variation of layout design 600A (FIG. 6A). For example, layout design 700A illustrates an example of where the MU layer (e.g., a set of conductive feature layout patterns 720) extends in two directions.

[0152] Layout design 700A is usable to manufacture an integrated circuit similar to integrated circuit 100 (FIG. 1), integrated circuit 400 (FIGS. 4A-4E) or 700B (FIGS. 7B-7D).

[0153] In comparison with layout design 600A of FIG. 6A, conductive feature layout patterns 720a, 720b, and 720c of layout design 700A replace conductive feature layout pattern 620a, and similar detailed description is therefore omitted.

[0154] Conductive feature layout pattern 720a, 720b, 720c (hereinafter referred to as a "set of conductive feature layout patterns 720") is similar to corresponding conductive feature layout patterns 520a, 520b, 520c of FIG. 5A.

[0155] The set of conductive feature layout patterns 720 is usable to manufacture a corresponding set of two-dimensional (2D) conductive structures 720' of integrated circuit 100 or 700B (FIGS. 7B-7D).

[0156] In some embodiments, the set of conductive feature layout patterns 720 is located on the MU portion of the first layout level. The set of conductive feature layout patterns 720 extends in two directions (e.g., first direction X and second direction Y). Conductive feature layout pattern 720a and 720c each extend in the first direction X. Conductive feature layout pattern 720b extends in the second direction Y.

[0157] The set of conductive feature layout patterns 720 is between the set of active region layout patterns 202. Conductive feature layout pattern 720a is positioned between conductive feature layout pattern 330c and conductive feature layout pattern 630b. Conductive feature layout pattern 720c is positioned between conductive feature layout pattern 330a and conductive feature layout pattern 630b'.

[0158] Conductive feature layout pattern 720b is positioned between gate layout patterns 204b and 204c. In some embodiments, conductive feature layout pattern 720b is positioned between conductive feature layout pattern 330a and conductive feature layout pattern 330c. In some embodi-

ments, metal over diffusion layout pattern 612a, 610b, 610c overlaps a corresponding conductive feature layout pattern 720a, 720b, 720c.

[0159] The set of gate layout patterns 204 overlaps the set of conductive feature layout patterns 720. In some embodiments, gate layout patterns 204a, 204b overlap conductive feature layout pattern 720a. In some embodiments, gate layout patterns 204c, 204d overlap conductive feature layout pattern 720c. Other configurations or quantities of patterns in the set of conductive feature layout patterns 720 are within the scope of the present disclosure.

[0160] FIG. 7B is a cross-sectional view of integrated circuit 700B corresponding to layout design 700A as intersected by plane M-M', FIG. 7C is a cross-sectional view of integrated circuit 700B corresponding to layout design 700A as intersected by plane N-N', and FIG. 7D is a cross-sectional view of integrated circuit 700B corresponding to layout design 700A as intersected by plane O-O' in accordance with some embodiments. Structural relationships including alignment, lengths and widths, as well as configurations of integrated circuit 700B are similar to the structural relationships and configurations of layout design 700A of FIG. 7A, and will not be described in FIGS. 7B-7D for brevity.

[0161] Integrated circuit 700B is manufactured by layout design 700A. In some embodiments, integrated circuit 700B is similar to integrated circuit 400 or 600A, and similar detailed description is therefore omitted for brevity.

[0162] In comparison with integrated circuit 600B of FIGS. 6B-6E, set of conductive structures 720' of integrated circuit 700B replaces set of conductive structures 620'.

[0163] In some embodiments, the set of conductive structures 720' includes one or more of conductive structures 720a', 720b' and 720c'.

[0164] Conductive structures 720a' and 720c' extend in the first direction X, and are separated from each other in the second direction Y. Conductive structure 720b' extends in the second direction Y. Conductive structure 720b' directly contacts conductive structures 720a' and 720c'. In some embodiments, conductive structures 720a', 720b' and 720c' are part of a same continuous conductive structure. In some embodiments, set of conductive structures 720' is referred to as a 2D conductive structure.

[0165] In some embodiments, gate terminal A1 (manufactured by gate layout pattern 204b) of integrated circuit 700B is directly coupled to gate terminal A2 (manufactured by gate layout pattern 204b) by gate 604b (manufactured by gate layout pattern 204b) similar to integrated circuit 600B FIGS. 6B-6E, and detailed description is therefore omitted.

[0166] In some embodiments, gate terminal B1 (manufactured by gate layout pattern 204c) of integrated circuit 700B is directly coupled to gate terminal B2 (manufactured by gate layout pattern 204c) by gate 604c (manufactured by gate layout pattern 204c) similar to integrated circuit 600B of FIGS. 6B-6E, and detailed description is therefore omitted.

[0167] In some embodiments, terminal A1D (manufactured by metal over diffusion layout pattern 610a) of integrated circuit 700B is electrically coupled to terminal B2D (manufactured by metal over diffusion layout pattern 612c) similar to integrated circuit 600B of FIGS. 6B-6E, and detailed description is therefore omitted.

[0168] In some embodiments, terminal B1D corresponds to contact 610c', and terminal A2D corresponds to contact

612a'. In these embodiments, terminal BID or contact **610c'** (manufactured by metal over diffusion layout pattern **610c**) of integrated circuit **700B** is electrically coupled to terminal A2D or contact **612a'** (manufactured by metal over diffusion layout pattern **612a**). For example, in some embodiments, contact **610c'** (manufactured by metal over diffusion layout pattern **610c**) is electrically connected through via **622b'** (manufactured by via layout pattern **622b**) to conductive structure **720c'** (manufactured by conductive feature layout pattern **720c**) (manufactured by conductive feature layout pattern **720c**) is electrically connected to conductive structure **720a'** (manufactured by conductive feature layout pattern **720a**) by conductive structure **720b'** (manufactured by conductive feature layout pattern **720b**). In some embodiments, conductive structure **720a'** (manufactured by conductive feature layout pattern **720a**) is electrically coupled through via **622a'** (manufactured by via layout pattern **622a**) to contact **612a** (manufactured by metal over diffusion layout pattern **612a**). Thus, terminal BID or contact **610c'** of integrated circuit **700B** is electrically coupled to terminal A2D or contact **612a'**.

[0169] FIG. 8A is a diagram of a layout design **800A** of an integrated circuit, in accordance with some embodiments.

[0170] FIGS. 8B, 8C, 8D, 8E, 8F and 8G are cross-sectional views of integrated circuit **800B**, in accordance with some embodiments.

[0171] Layout design **800A** is a variation of layout design **300** (FIG. 3) and layout design **700A** (FIG. 7A). For example, layout design **800A** illustrates an example layout design having a 4 poly pitch width in the first direction X, and the MU layer (e.g., a set of conductive feature layout patterns **820**) extends in a single direction, but includes two portions (e.g., conductive feature layout patterns **820a** and **820b**).

[0172] Layout design **800A** is usable to manufacture an integrated circuit similar to integrated circuit **100** (FIG. 1), integrated circuit **400** (FIGS. 4A-4E) or integrated circuit **800B** (FIGS. 8B-8G).

[0173] In comparison with layout design **300** of FIG. 3, layout design **800A** does not include set of cut feature layout patterns **304**. By not including the set of cut feature layout patterns **304**, layout design **800A** is usable to manufacture an integrated circuit having gate terminal A1 directly coupled to gate terminal A2.

[0174] In comparison with layout design **200**, **300**, **500A**, **600A** or **700A**, layout design **800A** further includes an additional gate layout pattern **804e** extending in the second direction Y, resulting in a wider layout design (e.g., layout design **800A**). For example, layout design **800A** has a 4 poly pitch width in the first direction X. Layout design **800A** has features of layout design **300** and **600A**. For example, layout design incorporates at least metal diffusion layout patterns **610a**, **610c**, **612a** and **612c** from layout design **600A**. Similarly, layout design **800A** incorporates at least metal diffusion layout patterns **210a**, and **210c** from layout design **300**.

[0175] Layout design **800A** includes set of active region layout patterns **202**, gate layout patterns **204a**, **204b**, **204c**, **204d**, **804e** (collectively referred to as a “set of gate layout patterns **804**”), metal over diffusion layout patterns **210a**, **210c**, **610a**, **610c**, **612a**, **612c** (collectively referred to as a “set of metal over diffusion layout patterns **810**”), set of conductive feature layout patterns **330**, conductive feature

layout pattern **820a**, **820b** (collectively referred to as a “set of conductive feature layout patterns **820**”), via layout patterns **822a**, **822b**, **822c**, **822d** (collectively referred to as a “set of via layout patterns **822**”), via layout patterns **832a**, **832b**, **832c** (collectively referred to as a “set of via layout patterns **832**”) and via layout patterns **836a**, **836b** (collectively referred to as a “set of via layout patterns **836**”).

[0176] In comparison with layout design **200**, **300**, **500A**, **600A** or **700A**, set of active region layout patterns **202** and set of conductive feature layout patterns **330** extend across at least gate layout patterns **204a** and **804e**.

[0177] Gate layout pattern **804e** is similar to one or more of gate layout patterns **204a**, **204b**, **204c**, **204d** of the set of gate layout patterns **204**, and detailed description is therefore omitted. Gate layout pattern **804e** is separated from gate layout pattern **204d** in the first direction X by a Poly pitch. The set of gate layout patterns **804** is usable to manufacture a corresponding set of gates **804'** (FIGS. 8B-8G) of integrated circuit **800B**. Gate layout pattern **804e** is usable to manufacture a corresponding gate **804e'** (FIGS. 8B-8G) of integrated circuit **800B**.

[0178] Set of metal over diffusion layout patterns **810** is similar to set of metal over diffusion layout patterns **210** of FIG. 2 and set of metal over diffusion layout patterns **610** of FIGS. 6A & 7A, and detailed description is therefore omitted. The set of metal over diffusion layout patterns **810** is usable to manufacture a corresponding set of contacts **810'** (FIGS. 8B-8G) of integrated circuit **800B**. Metal over diffusion layout patterns **210a**, **210c**, **610a**, **610c**, **612a** and **612c** are usable to manufacture corresponding contacts **410a**, **410c**, **610a'**, **610c'**, **612a'** and **612c'** (FIGS. 8B-8G) of integrated circuit **800B**.

[0179] Metal over diffusion layout pattern **210a** is positioned between gate layout patterns **204a** and **204b**. Metal over diffusion layout pattern **210c** is positioned between gate layout patterns **204d** and **804e**. Each of metal over diffusion layout patterns **610a** and **612a** are positioned between gate layout patterns **204b** and **204c**. Each of metal over diffusion layout patterns **610c** and **612c** are positioned between gate layout patterns **204c** and **204d**.

[0180] Set of conductive feature layout patterns **820** is similar to set of conductive feature layout patterns **320**, **420**, **520**, **620**, **720**, and detailed description is therefore omitted. The set of conductive feature layout patterns **820** is usable to manufacture a corresponding set of conductive structures **820'** (FIGS. 8B-8G) of integrated circuit **800B**. Conductive feature layout patterns **820a** and **820b** are similar to corresponding conductive feature layout patterns **720a** and **720c**, and detailed description is therefore omitted. Conductive feature layout patterns **820a** and **820b** are usable to manufacture corresponding conductive structures **820a'** and **820b'** (FIGS. 8B-8G) of integrated circuit **800B**.

[0181] Conductive feature layout pattern **820a** extends in the first direction X, and is positioned between conductive feature layout pattern **330b** and conductive feature layout pattern **330c**. Conductive feature layout pattern **820a** is positioned between gate layout pattern **204a** and gate layout pattern **204c**. Conductive feature layout pattern **820a** is below at least metal over diffusion layout pattern **210a**, **610a** or **612a**. In some embodiments, metal over diffusion layout pattern **210a** or metal over diffusion layout pattern **612a** overlaps conductive feature layout pattern **820a**.

[0182] Conductive feature layout pattern **820b** extends in the first direction X, and is positioned between conductive

feature layout pattern 330a and conductive feature layout pattern 330b. Conductive feature layout pattern 820b is positioned between gate layout pattern 204c and gate layout pattern 804e. Conductive feature layout pattern 820b is below at least metal over diffusion layout pattern 210c, 610c or 612c. In some embodiments, metal over diffusion layout pattern 210c or metal over diffusion layout pattern 610c overlaps conductive feature layout pattern 820b.

[0183] Set of via layout patterns 822 is similar to set of via layout patterns 322, 522, 622, and detailed description is therefore omitted. The set of via layout patterns 822 is usable to manufacture a corresponding set of vias 822' (FIGS. 8B-8G) of integrated circuit 800B.

[0184] Via layout patterns 822a, 822b, 822c and 822d are usable to manufacture corresponding vias 822a', 822b', 822c' and 822d' (FIGS. 8B-8G) of integrated circuit 800B. Via layout patterns 822a, 822c is similar to via layout pattern 622a, and detailed description is therefore omitted. Via layout pattern 822c, 822a is between conductive feature layout pattern 820a and corresponding metal over diffusion layout patterns 210a, 612a. In some embodiments, via layout pattern 822c, 822a of the set of via layout patterns 822 is located where a corresponding layout pattern 210a, 612a of the set of metal over diffusion layout patterns 810 overlaps conductive feature layout pattern 820a of the set of conductive feature layout patterns 820. In some embodiments, a center of via layout patterns 822c, 822a of the set of via layout patterns 822 is below a center of a corresponding metal over diffusion layout pattern 210a, 612a of the set of metal over diffusion layout patterns 810. In some embodiments, a center of via layout patterns 822c, 822a of the set of via layout patterns 822 is aligned in at least the first direction X or the second direction Y with a center of a corresponding metal over diffusion layout pattern 210a, 612a of the set of metal over diffusion layout patterns 810.

[0185] Via layout patterns 822b, 822d is similar to via layout pattern 622b, and detailed description is therefore omitted. Via layout pattern 822b, 822d is between conductive feature layout pattern 820b and corresponding metal over diffusion layout patterns 610c, 210c. In some embodiments, a via layout pattern 822b, 822d of the set of via layout patterns 822 is located where a corresponding layout pattern 610c, 210c of the set of metal over diffusion layout patterns 810 overlaps conductive feature layout pattern 820b of the set of conductive feature layout patterns 820. In some embodiments, a center of via layout patterns 822b, 822d of the set of via layout patterns 822 is below a center of a corresponding metal over diffusion layout pattern 610c, 210c of the set of metal over diffusion layout patterns 810. In some embodiments, a center of via layout patterns 822b, 822d of the set of via layout patterns 822 is aligned in at least the first direction X or the second direction Y with a center of a corresponding metal over diffusion layout pattern 610c, 210c of the set of metal over diffusion layout patterns 810. Other configurations of via layout patterns 822a, 822b, 822c, 822d of the set of via layout patterns 822 are within the scope of the present disclosure.

[0186] Set of via layout patterns 832 is similar to set of via layout patterns 332, 632, and detailed description is therefore omitted. The set of via layout patterns 832 is usable to manufacture a corresponding set of vias 832' (FIGS. 8B-8G) of integrated circuit 800B. The set of via layout patterns 832 are between the set of conductive feature layout patterns 330 and the set of gate layout patterns 804. Via layout patterns

832a, 832b and 832c are usable to manufacture corresponding vias 832a', 832b' and 832c' (FIGS. 8B-8G) of integrated circuit 800B. Via layout patterns 832a, 832c of the set of via layout patterns 832 are between conductive feature layout pattern 330c and a corresponding gate layout pattern 204b, 204d of the set of gate layout patterns 804. Via layout pattern 832b of the set of via layout patterns 832 is between conductive feature layout pattern 330a and gate layout pattern 204c of the set of gate layout patterns 804.

[0187] In some embodiments, a via layout pattern 832a, 832c of the set of via layout patterns 832 is located where layout pattern 330c overlaps a corresponding gate layout pattern 204b, 204d of the set of gate layout patterns 804. In some embodiments, via layout pattern 832b of the set of via layout patterns 832 is located where layout pattern 330a overlaps gate layout pattern 204c of the set of gate layout patterns 804. In some embodiments, a center of via layout pattern 832a, 832c of the set of via layout patterns 832 is over a center of a corresponding gate layout pattern 204b, 204d of the set of gate layout patterns 804. In some embodiments, a center of via layout pattern 832b of the set of via layout patterns 832 is over a center of gate layout pattern 204c of the set of gate layout patterns 804. In some embodiments, a center of via layout pattern 832a, 832c of the set of via layout patterns 832 is below a center of conductive feature layout pattern 330c. In some embodiments, a center of via layout pattern 832b of the set of via layout patterns 832 is below a center of conductive feature layout pattern 330a. Other configurations of via layout patterns 832a, 832b, 832c of the set of via layout patterns 832 are within the scope of the present disclosure.

[0188] Set of via layout patterns 836 is similar to set of via layout patterns 636, and detailed description is therefore omitted. The set of via layout patterns 836 is usable to manufacture a corresponding set of vias 836' (FIGS. 8B-8G) of integrated circuit 800B. Set of via layout patterns 836 are between the set of conductive feature layout patterns 330 and the set of metal over diffusion layout patterns 810. Via layout patterns 836a and 836b are usable to manufacture corresponding vias 836a' and 836b' (FIGS. 8B-8G) of integrated circuit 800B. Via layout patterns 836a, 836b of the set of via layout patterns 836 are between conductive feature layout pattern 330b and a corresponding metal over diffusion layout pattern 610a, 612c.

[0189] In some embodiments, a via layout pattern 836a, 836b of the set of via layout patterns 836 is located where layout pattern 330b at least partially overlaps a corresponding metal over diffusion layout pattern 610a, 612c. Via layout pattern 836a, 836b of the set of via layout patterns 836 is over a corresponding metal over diffusion layout pattern 610a, 612c. In some embodiments, a center of via layout pattern 836a, 836b of the set of via layout patterns 836 is over a center of a corresponding metal over diffusion layout pattern 610a, 612c. In some embodiments, a center of via layout pattern 836a, 836b of the set of via layout patterns 836 is below a center of conductive feature layout pattern 330b. In some embodiments, a center of via layout pattern 836a, 836b of the set of via layout patterns 836 is aligned in at least the first direction X or the second direction Y with a center of a corresponding metal over diffusion layout pattern 610a, 612c. In some embodiments, a center of via layout pattern 836a, 836b of the set of via layout patterns 836 is aligned in at least the first direction X or the second direction Y with a center of conductive feature layout pattern

330b. Other configurations of via layout patterns **836a**, **836b** of the set of via layout patterns **836** are within the scope of the present disclosure.

[0190] FIG. 8B is a cross-sectional view of integrated circuit **800B** corresponding to layout design **800A** as intersected by plane P-P', FIG. 8C is a cross-sectional view of integrated circuit **800B** corresponding to layout design **800A** as intersected by plane Q-Q', FIG. 8D is a cross-sectional view of integrated circuit **800B** corresponding to layout design **800A** as intersected by plane R-R', FIG. 8E is a cross-sectional view of integrated circuit **800B** corresponding to layout design **800A** as intersected by plane S-S', FIG. 8F is a cross-sectional view of integrated circuit **800B** corresponding to layout design **800A** as intersected by plane T-T', and FIG. 8G is a cross-sectional view of integrated circuit **800B** corresponding to layout design **800A** as intersected by plane U-U', in accordance with some embodiments. Structural relationships including alignment, lengths and widths, as well as configurations of integrated circuit **800B** are similar to the structural relationships and configurations of layout design **800A** of FIG. 8A, and will not be described in FIGS. 8B-8G for brevity.

[0191] Integrated circuit **800B** is manufactured by layout design **800A**. In some embodiments, integrated circuit **800B** is similar to integrated circuit **400**, **600B** or **700B**, and similar detailed description is therefore omitted for brevity.

[0192] Integrated circuit **800B** has a 4 poly pitch width in the first direction X. Integrated circuit **800B** has features of integrated circuit **400** and **600B**. For example, integrated circuit **800B** incorporates at least contacts **610a'**, **610c'**, **612a'** and **612c'** from integrated circuit **600B**. Similarly, integrated circuit **800B** incorporates at least contacts **410a** and **410c** from integrated circuit **400**.

[0193] Integrated circuit **800B** includes set of active regions **402**, gates **404a**, **404d**, **604b**, **604c** and **804e'** (collectively referred to as a set of gates **804'**), contacts **410a**, **410c**, **610a'**, **610c'**, **612a'**, **612c'** (collectively referred to as a set of contacts **810'**), set of conductive structures **430**, conductive structures **820a'**, **820b'** (collectively referred to as a set of conductive structures **820'**), vias **822a'**, **822b'**, **822c'**, **822d'** (collectively referred to as a set of vias **822'**), vias **832a'**, **832b'**, **832c'** (collectively referred to as a set of vias **832'**) and vias **836a'**, **836b'** (collectively referred to as a set of vias **836'**).

[0194] In comparison with integrated circuit **600** (FIGS. 6B-6E) or **700** (FIGS. 7B-7D), conductive structures **820a'**, **820b'** replace conductive structures **720a'**, **720c'** of integrated circuit **700B**, vias **822a'**, **822b'**, **822c'**, **822d'** replace vias **622a'**, **622b'** of integrated circuit **700B**, vias **832a'**, **832b'**, **832c'** replace vias **632a'**, **632b'** of integrated circuit **600B**, vias **836a'**, **836b'** replace vias **636a'**, **636b'** of integrated circuit **600B**, and similar detailed description is therefore omitted for brevity.

[0195] In comparison with integrated circuit **600B** (FIGS. 6B-6E) or **700B** (FIGS. 7B-7D), gate **804e'** is similar to gate **404a**, **404d**, **604b** or **604c**, and similar detailed description is therefore omitted for brevity.

[0196] In comparison with conductive structures **720a'** and **720c'** of integrated circuit **700B** of FIGS. 7B-7D, conductive structures **820a'** and **820b'** are not coupled together. In other words, conductive structures **820a'** and **820b'** are not coupled together through another conductive structure (e.g., conductive structure **720b'**).

[0197] The set of vias **822'** electrically couple the set of conductive structures **820'** to the set of contacts **810'**. The set of vias **832'** electrically couple the set of conductive structures **430** to the set of gates **804'**. The set of vias **836'** electrically couple the set of conductive structures **430** to the set of contacts **810'**.

[0198] In some embodiments, terminal **A1** or **A2** corresponds to gate **604c**. Gate **604c** is a continuous structure. In some embodiments, gate terminal **A1** is directly coupled to gate terminal **A2** by gate **604c** (manufactured by gate layout pattern **204c** of layout design **800A**).

[0199] In some embodiments, terminal **B1** corresponds to gate **404d**, and terminal **B2** corresponds to gate **604b**. In these embodiments, gate **404d** (manufactured by gate layout pattern **204d** of layout design **800A**) of integrated circuit **800B** is electrically coupled to gate **604b** (manufactured by gate layout pattern **204b** of layout design **800A**). For example, in some embodiments, gate **404d** (manufactured by gate layout pattern **204d** of layout design **800A**) is electrically connected through via **832c'** (manufactured by via layout pattern **832c**) to conductive structure **430c** (manufactured by conductive feature layout pattern **330c**), and conductive structure **430c** is electrically coupled through via **832a'** (manufactured by via layout pattern **832a**) to gate **604b** (manufactured by gate layout pattern **204b** of layout design **800A**).

[0200] In some embodiments, terminal **B2D** corresponds to contact **410a**, and terminal **B2D'** corresponds to contact **612a'**. In these embodiments, terminal **B2D** or contact **410a** (manufactured by metal over diffusion layout pattern **210a**) of integrated circuit **800B** is electrically coupled to terminal **B2D'** or contact **612a'** (manufactured by metal over diffusion layout pattern **612a**). For example, in some embodiments, contact **410a** (manufactured by metal over diffusion layout pattern **210a**) is electrically connected through via **822c'** (manufactured by via layout pattern **822c**) to conductive structure **820a'** (manufactured by conductive feature layout pattern **820a**). In some embodiments, conductive structure **820a'** is electrically coupled through via **822a'** (manufactured by via layout pattern **822a**) to contact **612a'** (manufactured by metal over diffusion layout pattern **612a**). Thus, terminal **B2D** or contact **410a** is electrically coupled to terminal **B2D'** or contact **612a'**.

[0201] In some embodiments, terminal **B1D** corresponds to contact **410c**, and terminal **B1D'** corresponds to contact **610c'**. In these embodiments, terminal **B1D** or contact **410c** (manufactured by metal over diffusion layout pattern **210c**) of integrated circuit **800B** is electrically coupled to terminal **B1D'** or contact **610c'** (manufactured by metal over diffusion layout pattern **610c**). For example, in some embodiments, contact **410c** (manufactured by metal over diffusion layout pattern **210c**) is electrically connected through via **822d'** (manufactured by via layout pattern **822d**) to conductive structure **820b'** (manufactured by conductive feature layout pattern **820b**). In some embodiments, conductive structure **820b** is electrically coupled through via **822b'** (manufactured by via layout pattern **822b**) to contact **610c'** (manufactured by metal over diffusion layout pattern **610c**). Thus, terminal **B1D** or contact **410c** is electrically coupled to terminal **B1D'** or contact **610c'**.

[0202] FIG. 9A is a diagram of a layout design **900A** of an integrated circuit **900B**, in accordance with some embodiments.

[0203] FIGS. 9B, 9C, 9D and 9E are cross-sectional views of integrated circuit 900B, in accordance with some embodiments.

[0204] Layout design 900A is a variation of layout design 800A (FIG. 8A). For example, layout design 900A illustrates an example where the gate (e.g., gate 904b1' and 904b2') manufactured by gate layout patterns 204b is not continuous, and the MU layer (e.g., a set of conductive feature layout patterns 920) extends in a single direction, but includes two portions (e.g., conductive feature layout patterns 920a and 920b).

[0205] Layout design 900A is usable to manufacture an integrated circuit similar to integrated circuit 100 (FIG. 1) or integrated circuit 400 (FIGS. 4A-4E).

[0206] In comparison with layout design 800A of FIG. 8A, layout design 900A further includes a set of cut feature layout patterns 904 and via layout pattern 932a. By including the set of cut feature layout patterns 904, layout design 900A is usable to manufacture an integrated circuit 900B (FIGS. 9B-9E) having gate terminal A1 not being coupled to gate terminal B2.

[0207] Set of cut feature layout patterns 904 is usable to cut gate 204b to form gates 904b1' and 904b2' of integrated circuit 900B (FIGS. 9B-9E). Set of cut feature patterns 904 is similar to the set of cut feature patterns 304, and detailed description is therefore omitted.

[0208] Via layout pattern 932a is similar to one or more layout patterns of set of via layout patterns 332, 632 and 832, and detailed description is therefore omitted. The via layout pattern 932a is usable to manufacture via 932a' in the set of vias 932' (FIGS. 9B-9E).

[0209] Via layout pattern 932a of layout design 900A is similar to via layout pattern 832a, and detailed description is therefore omitted. Via layout pattern 932a is between the set of conductive feature layout patterns 330 and the set of gate layout patterns 804. Via layout pattern 932a is between conductive feature layout pattern 330a and gate layout pattern 204b of the set of gate layout patterns 804. In some embodiments, via layout pattern 932a is located where layout pattern 330a overlaps gate layout pattern 204b of the set of gate layout patterns 804. In some embodiments, a center of via layout pattern 932a is over a center of gate layout pattern 204b of the set of gate layout patterns 804. In some embodiments, a center of via layout pattern 932a is below a center of conductive feature layout pattern 330a. Other configurations of via layout pattern 932a is within the scope of the present disclosure.

[0210] Layout design 900A is a variation of layout design 800A (FIG. 8A). In comparison with layout design 800A of FIG. 8A, layout design 900A does not include via layout pattern 836b. In comparison with layout design 800A of FIG. 8A, set of metal over diffusion layout patterns 910 of layout design 900A replaces set of metal over diffusion layout patterns 810, conductive feature layout patterns 920a, 920b (collectively referred to as "set of conductive feature layout patterns 920") replace a corresponding conductive feature layout pattern 820a, 820b, and via layout patterns 922a, 922c, 922d (collectively referred to as "set of via layout patterns 922") replace a corresponding via layout pattern 822a, 822c, 822d, and similar detailed description is therefore omitted.

[0211] The set of metal over diffusion layout patterns 910 is usable to manufacture a corresponding set of contacts 910' (FIGS. 9B-9E) of integrated circuit 900B. Set of metal over

diffusion layout patterns 910 includes one or more of metal over diffusion layout patterns 210a, 910b, 610c, 612c, 910c and 912c. Metal over diffusion layout patterns 210a, 910b, 610c, 612c, 910c and 912c are usable to manufacture corresponding contacts 410a, 910b', 610c', 612c', 910c' and 912c' (FIGS. 9B-9E) of integrated circuit 900B. Metal over diffusion layout pattern 910b replaces metal over diffusion layout pattern 610a and metal over diffusion layout pattern 612a of layout design 800A, and similar detailed description is therefore omitted. Metal over diffusion layout patterns 910c and 912c replace metal over diffusion layout pattern 210c of layout design 800A, and similar detailed description is therefore omitted.

[0212] In some embodiments, metal over diffusion layout pattern 910b extends in the second direction Y from an edge of active region layout pattern 202b to an edge of active region layout pattern 202a. In some embodiments, metal over diffusion layout pattern 910c extends in the second direction Y from an edge of active region layout pattern 202b to an edge of conductive feature layout pattern 330b. In some embodiments, metal over diffusion layout pattern 912c extends in the second direction Y from an edge of active region layout pattern 202a to another edge of conductive feature layout pattern 330b. Other configurations of metal over diffusion layout patterns 210a, 910b, 610c, 612c, 910c and 912c of the set of metal over diffusion layout patterns 910 are within the scope of the present disclosure.

[0213] Set of conductive feature layout patterns 920 is similar to set of conductive feature layout patterns 320, 420, 520, 620, 720 and 820, and detailed description is therefore omitted. The set of conductive feature layout patterns 920 is usable to manufacture a corresponding set of conductive structures 920' (FIGS. 9B-9E) of integrated circuit 900B. Conductive feature layout patterns 920a and 920b are similar to corresponding conductive feature layout patterns 820a and 820c, and detailed description is therefore omitted. Conductive feature layout patterns 920a and 920b are usable to manufacture corresponding conductive structures 920a' and 920b' (FIGS. 9B-9E) of integrated circuit 900B.

[0214] Conductive feature layout pattern 920a extends in the first direction X, and is positioned between conductive feature layout pattern 330b and conductive feature layout pattern 330c. Conductive feature layout pattern 920a is positioned between gate layout pattern 204c and gate layout pattern 804e. Conductive feature layout pattern 920a is below at least metal over diffusion layout patterns 612c or 912c. In some embodiments, metal over diffusion layout pattern 612c or metal over diffusion layout pattern 912c overlaps conductive feature layout pattern 920a.

[0215] Conductive feature layout pattern 920b extends in the first direction X, and is positioned between conductive feature layout pattern 330a and conductive feature layout pattern 330b. Conductive feature layout pattern 920b is positioned between gate layout pattern 204b and gate layout pattern 204d. Conductive feature layout pattern 920b is below at least metal over diffusion layout pattern 910b or 610c. In some embodiments, metal over diffusion layout pattern 910b or metal over diffusion layout pattern 610c overlaps conductive feature layout pattern 920b. Other configurations of conductive feature layout patterns 920a, 920b of the set of conductive feature layout patterns 920 are within the scope of the present disclosure.

[0216] Set of via layout patterns 922 includes via layout patterns 922a, 822b 922c, 922d. Set of via layout patterns

922 is similar to set of via layout patterns **322**, **522**, **622** and **822** and detailed description is therefore omitted. The set of via layout patterns **922** is usable to manufacture a corresponding set of vias **922'** (FIGS. 9B-9E) of integrated circuit **900B** that couple the set of conductive structures **920'** to the set of contacts **910'**.

[0217] Via layout patterns **922a**, **822b**, **922c** and **922d** are usable to manufacture corresponding vias **922a'**, **822b'**, **922c'** and **922d'** (FIGS. 9B-9E) of integrated circuit **900B**. Via layout patterns **922a**, **922c** is similar to corresponding via layout patterns **822a**, **822c**, and detailed description is therefore omitted. Via layout pattern **922c**, **922a** is between conductive feature layout pattern **920a** and corresponding metal over diffusion layout patterns **612c**, **912c**. In some embodiments, via layout patterns **922c**, **922a** of the set of via layout patterns **922** is located where a corresponding layout pattern **612c**, **912c** of the set of metal over diffusion layout patterns **910** overlaps conductive feature layout pattern **920a** of the set of conductive feature layout patterns **920**. In some embodiments, a center of via layout patterns **922c**, **922a** of the set of via layout patterns **922** is below a center of a corresponding metal over diffusion layout pattern **612c**, **912c** of the set of metal over diffusion layout patterns **910**. In some embodiments, a center of via layout patterns **922c**, **922a** of the set of via layout patterns **922** is aligned in at least the first direction X or the second direction Y with a center of a corresponding metal over diffusion layout pattern **612c**, **912c** of the set of metal over diffusion layout patterns **910**.

[0218] Via layout patterns **922d** is similar to via layout pattern **622b** or **822d**, and detailed description is therefore omitted. Via layout pattern **922d** is between conductive feature layout pattern **920b** and metal over diffusion layout pattern **910b**. In some embodiments, via layout pattern **922d** of the set of via layout patterns **922** is located where layout pattern **910b** of the set of metal over diffusion layout patterns **910** overlaps conductive feature layout pattern **920b** of the set of conductive feature layout patterns **920**. In some embodiments, a center of via layout pattern **922d** of the set of via layout patterns **922** is below a center of metal over diffusion layout pattern **910b** of the set of metal over diffusion layout patterns **910**. In some embodiments, a center of via layout pattern **922d** of the set of via layout patterns **922** is aligned in at least the first direction X or the second direction Y with a center of metal over diffusion layout pattern **910b** of the set of metal over diffusion layout patterns **910**. Other configurations of via layout patterns **922a**, **922c**, **922d** of the set of via layout patterns **922** are within the scope of the present disclosure.

[0219] FIG. 9B is a cross-sectional view of integrated circuit **900B** corresponding to layout design **900A** as intersected by plane V-V', FIG. 9C is a cross-sectional view of integrated circuit **900B** corresponding to layout design **900A** as intersected by plane W-W', FIG. 9D is a cross-sectional view of integrated circuit **900B** corresponding to layout design **900A** as intersected by plane X-X', and FIG. 9E is a cross-sectional view of integrated circuit **900B** corresponding to layout design **900A** as intersected by plane Y-Y', in accordance with some embodiments. Structural relationships including alignment, lengths and widths, as well as configurations of integrated circuit **900B** are similar to the structural relationships and configurations of layout design **900A** of FIG. 9A, and will not be described in FIGS. 9B-9E for brevity.

[0220] Integrated circuit **900B** is manufactured by layout design **900A**. In some embodiments, integrated circuit **900B** is similar to integrated circuit **400**, **600B**, **700B** or **800B**, and similar detailed description is therefore omitted for brevity.

[0221] Integrated circuit **900B** has a 4 poly pitch width in the first direction X.

[0222] Integrated circuit **900B** includes set of active regions **402**, gates **404a**, **404d**, **904b1'**, **904b2'**, **604c** and **804e'** (collectively referred to as a set of gates **904'**), contacts **410a**, **910b'**, **610c'**, **612c'**, **910c'** and **912c'** (collectively referred to as a set of contacts **910'**), set of conductive structures **430**, conductive structures **920a'**, **920b'** (collectively referred to as a set of conductive structures **920'**), vias **922a'**, **822b'**, **922c'**, **922d'** (collectively referred to as a set of vias **922'**), vias **832a'**, **832b'**, **832c'**, **932a'** (collectively referred to as a set of vias **932'**) and via **836a'**.

[0223] In comparison with integrated circuit **800B** (FIGS. 8B-8G), gates **904b1'** and **904b2'** replace gate **604b**, contacts **910b'** replaces contacts **610a'** and **612a'**, contacts **910c'** and **912c'** replace contact **410c**, conductive structures **920a'** and **920b'** replace corresponding conductive structures **820a'** and **820b'**, vias **922a'**, **922c'** and **922d'** replace corresponding vias **822a'**, **822c'** and **822d'**, and similar detailed description is therefore omitted for brevity.

[0224] In comparison with integrated circuit **800B** (FIGS. 8B-8G), integrated circuit **900B** further includes via **932a'**. Via **932a'** is similar to via **832a'**, and similar detailed description is therefore omitted for brevity. Via **932a'** electrically couples conductive structures **430a** to gate **904b1'**.

[0225] In comparison with integrated circuit **800B** (FIGS. 8B-8G), integrated circuit **900B** does not include via **836b'**.

[0226] The set of vias **932'** electrically couple the set of conductive structures **430** to the set of gates **904'**. The set of vias **922'** electrically couple the set of conductive structures **920'** to the set of contacts **910'**.

[0227] In some embodiments, terminal A1 corresponds to gate **904b1'**, terminal B2 corresponds to gate **904b2'** and terminal A2 corresponds to gate **604c**. In these embodiments, gate **904b1'** (manufactured by gate layout pattern **204b** of layout design **900A**) is not directly coupled to gate **904b2'** (manufactured by gate layout pattern **204b** of layout design **900A**) because of cut feature layout pattern **904** of layout design **900A**. In other words, in some embodiments, cut feature layout pattern **904** of layout design **900A** is used to identify a removed portion of gate **604b** (shown in FIGS. 8F-8G) thereby forming discontinuous gates **904b1'** and **904b2'**.

[0228] In some embodiments, terminal A1 corresponds to gate **904b1'** and terminal A2 corresponds to gate **604c**. In these embodiments, gate **904b1'** (manufactured by gate layout pattern **204b** of layout design **900A**) of integrated circuit **900B** is electrically coupled to gate **604c** (manufactured by gate layout pattern **204c** of layout design **900A**). For example, in some embodiments, gate **904b1'** (manufactured by gate layout pattern **204b** of layout design **900A**) is electrically connected through via **932a'** (manufactured by via layout pattern **932a**) to conductive structure **430a** (manufactured by conductive feature layout pattern **330a**), and conductive structure **430a** is electrically coupled through via **832b'** (manufactured by via layout pattern **832b**) to gate **604c** (manufactured by gate layout pattern **204c** of layout design **900A**).

[0229] In some embodiments, terminal B1 corresponds to gate **404d** and terminal B2 corresponds to gate **904b2'**. In

these embodiments, gate **404d** (manufactured by gate layout pattern **204d** of layout design **900A**) is electrically coupled to gate **904b2'** (manufactured by gate layout pattern **204b** of layout design **900A**). For example, in some embodiments, gate **404d** (manufactured by gate layout pattern **204d** of layout design **900A**) is electrically connected through via **832c'** (manufactured by via layout pattern **832c**) to conductive structure **430c** (manufactured by conductive feature layout pattern **330c**), and conductive structure **430c** is electrically coupled through via **832a'** (manufactured by via layout pattern **832a**) to gate **904b2'** (manufactured by gate layout pattern **204b** of layout design **900A**).

[0230] In some embodiments, terminal **A2D** corresponds to contact **612c'**, and terminal **A2D'** corresponds to contact **912c'**. In these embodiments, terminal **A2D** or contact **612c'** (manufactured by metal over diffusion layout pattern **612c** of layout design **900A**) is electrically coupled to terminal **A2D'** or contact **912c'** (manufactured by metal over diffusion layout pattern **912c** of layout design **900A**). For example, in some embodiments, contact **612c'** (manufactured by metal over diffusion layout pattern **612c**) is electrically connected through via **922c'** (manufactured by via layout pattern **922c**) to conductive structure **920a'** (manufactured by conductive feature layout pattern **920a**). In some embodiments, conductive structure **920a'** (manufactured by conductive feature layout pattern **920a**) is electrically coupled through via **922a'** (manufactured by via layout pattern **922a**) to contact **912c'** (manufactured by metal over diffusion layout pattern **912c** of layout design **900A**). Thus, terminal **A2D** or contact **612c'** is electrically coupled to terminal **A2D'** or contact **912c'**.

[0231] In some embodiments, terminal **A1D** corresponds to contact **910b'**, and terminal **A1D'** corresponds to contact **610c'**. In these embodiments, terminal **A1D** or contact **910b'** (manufactured by metal over diffusion layout pattern **910b** of layout design **900A**) is electrically coupled to terminal **A1D'** or contact **610c'** (manufactured by metal over diffusion layout pattern **610c** of layout design **900A**). For example, in some embodiments, contact **910b'** (manufactured by metal over diffusion layout pattern **910b**) is electrically connected through via **922d'** (manufactured by via layout pattern **922d**) to conductive structure **920b'** (manufactured by conductive feature layout pattern **920b** of layout design **900A**). In some embodiments, conductive structure **920b'** is electrically coupled through via **822b'** (manufactured by via layout pattern **822b**) to contact **610c'** (manufactured by metal over diffusion layout pattern **610c**). Thus, terminal **A1D** or contact **910b'** is electrically coupled to terminal **A1D'** or contact **610c'**.

[0232] In some embodiments, the set of conductive feature layout patterns **320**, **520**, **620**, **720**, **820** and **920** are located below the set of gate layout patterns **304** and the metal-0 (M0) level of layout design **300** & **500A-900A**.

[0233] In some embodiments, by using the set of conductive feature layout patterns **320**, **520**, **620**, **720**, **820** and **920**, one M0 track is reduced in the set of conductive feature layout patterns **330** of FIGS. 3 & 5A-9A, yielding a standard cell with a smaller height and smaller area compared with other approaches. In some embodiments, by using the set of conductive feature layout patterns **320**, **520**, **620**, **720**, **820** and **920**, the width of one or more layout patterns in the set of conductive feature layout patterns **330** of FIGS. 3 & 5A-9A is increased since one M0 track is reduced, yielding a standard cell with better current resistance (IR) and Electromigration (EM) compared with other approaches.

[0234] In some embodiments, by utilizing the set of conductive feature layout patterns **320**, **520**, **620**, **720**, **820** and **920** of the present disclosure, the gate density of the set of gate layout patterns of a standard cell is increased by up to 30% compared with other approaches. In some embodiments, by utilizing the set of conductive feature layout patterns **320**, **520**, **620**, **720**, **820** and **920** of the present disclosure, layout design **200-300** and **500A-900A** and integrated circuits **400** and **500B-900B** of FIGS. 2-3 & 5A-9E offer more routing flexibility and increases routing resources compared to other approaches.

[0235] FIG. 10 is a flowchart of a method **1000** of forming an integrated circuit in accordance with some embodiments. It is understood that additional operations may be performed before, during, and/or after the method **1000** depicted in FIG. 10, and that some other processes may only be briefly described herein. In some embodiments, the method **1000** is usable to form integrated circuits, such as integrated circuit **100** (FIG. 1), integrated circuit **400** (FIGS. 4A-4E), integrated circuit **500B**, integrated circuit **600B**, integrated circuit **700B**, integrated circuit **800B** or integrated circuit **900B**.

[0236] In operation **1002** of method **1000**, a layout design **300** (FIG. 3) of an integrated circuit (e.g., integrated circuit **100** (FIG. 1) or integrated circuit **400** (FIGS. 4A-4E)) is generated. Operation **1002** is performed by a processing device (e.g., processor **1202** (FIG. 12)) configured to execute instructions for generating a layout design (e.g., layout design **200** or **300**). In some embodiments, layout design **200** or **300** is a graphic database system (GDSII) file format. In some embodiments, operation **1002** of method **1000** is utilized to generate one or more of layout design **200** (FIG. 2), layout design **300** (FIG. 3), layout design **500A** (FIG. 5A), layout design **600A** (FIG. 6A), layout design **700A** (FIG. 7A), layout design **800A** (FIG. 8A) or layout design **900A** (FIG. 9A).

[0237] Method **1000** continues with operation **1004**, where the integrated circuit (e.g., integrated circuit **100** (FIG. 1) or integrated circuit **400** (FIGS. 4A-4E)) is manufactured based on the layout design **300** (FIG. 3). In some embodiments, operation **1004** includes manufacturing integrated circuit **500B**, **600B**, **700B**, **800B** or **900B** based on corresponding layout design **500A**, **600A**, **700A**, **800A** or **900A**. In some embodiments, integrated circuit **100** (FIG. 1) or integrated circuit **400** (FIGS. 4A-4E) includes gate **404b1**, **404b2**, **404c1** or **404c2**. In some embodiments, operation **1004** comprises manufacturing at least one mask based on one or more of layout design **200** (FIG. 2), layout design **300** (FIG. 3), layout design **500A** (FIG. 5A), layout design **600A** (FIG. 6A), layout design **700A** (FIG. 7A), layout design **800A** (FIG. 8A) or layout design **900A** (FIG. 9A), and manufacturing the integrated circuit (e.g., integrated circuit **100** (FIG. 1), integrated circuit **400** (FIGS. 4A-4E), or integrated circuit **500B**, **600B**, **700B**, **800B** or **900B**) based on the at least one mask. In some embodiments, operation **1004** of method **1000** is utilized to manufacture integrated circuit (e.g., integrated circuit **100** (FIG. 1), integrated circuit **400** (FIGS. 4A-4E), or integrated circuit **500B**, **600B**, **700B**, **800B** or **900B**) based on one or more of layout design **200** (FIG. 2), layout design **300** (FIG. 3), layout design **500A** (FIG. 5A), layout design **600A** (FIG. 6A), layout design **700A** (FIG. 7A), layout design **800A** (FIG. 8A) or layout design **900A** (FIG. 9A).

[0238] Method 1000 continues with operation 1006, where a portion of a gate (e.g., gate A1, A2, B1 or B2) is removed thereby forming a first gate structure (e.g., gate 404b1 or 404c1) and a second gate structure (e.g., gate 404b2 or 404c2) of integrated circuit 100, 400, 500B, 600B, 700B, 800B or 900B. In some embodiments, the portion of the gate (e.g., gate A1, A2, B1 or B2) that is removed in operation 1006 is identified in layout designs 300, 500A and 900A by cut feature layout pattern 304a or 304b (FIGS. 3 and 5A) or cut feature layout pattern 904 (FIG. 9A). In some embodiments, operation 1006 of method 1000 is referred to as a cut-poly (CPO) process. In some embodiments, operation 1006 results in the formation of integrated circuit 100 (FIG. 1), 400 (FIGS. 4A-4E), 500B, 600B, 700B, 800B or 900B. In some embodiments, the removed gate portion of gates 404b1 and 404b2 and gates 404c1 and 404c2 is referred to as a cut region.

[0239] In some embodiments, operation 1006 is performed by a removal process. In some embodiments, the removal process includes one or more etching processes suitable to remove a portion of gate (e.g., gate A1, A2, B1 or B2). In some embodiments, the etching process of operation 1006 includes identifying a portion of the gate (e.g., gate A1, A2, B1 or B2) that is to be removed, and etching the portion of the gate (e.g., gate A1, A2, B1 or B2) that is to be removed. In some embodiments, a mask is used to specify portions of the gate (e.g., gate A1, A2, B1 or B2) that are to be cut or removed. In some embodiments the mask is a hard mask. In some embodiments, the mask is a soft mask. In some embodiments, etching corresponds to plasma etching, reactive ion etching, chemical etching, dry etching, wet etching, other suitable processes, any combination thereof, or the like.

[0240] In some embodiments, method 1000 is utilized with layout designs other than layout design 200 or 300. For example, in some embodiments, layout design 200 (FIG. 2) or 300 (FIG. 3) of operation 1004 or 1006 is replaced with one or more of layout design 500A (FIG. 5A), layout design 600A (FIG. 6A), layout design 700A (FIG. 7A), layout design 800A (FIG. 8A) or layout design 900A (FIG. 9A).

[0241] In some embodiments, operation 1006 is not performed. For example, in some embodiments, operation 1006 is not performed to manufacture integrated circuit 100 or 400. In some embodiments, method 1000 is performed to manufacture an integrated circuit, 600B, 700B or 800B based on one or more of layout designs 600A, 700A or 800A, and operation 1006 is not performed on the manufactured integrated circuit.

[0242] FIGS. 11A-11B is a flowchart of a method 1100 of generating a layout design of an integrated circuit in accordance with some embodiments. It is understood that additional operations may be performed before, during, and/or after the method 1100 depicted in FIGS. 11A-11B, and that some other processes may only be briefly described herein. In some embodiments, the method 1100 is usable to generate one or more layout designs, such as layout design 200 (FIG. 2), layout design 300 (FIG. 3), layout design 500A (FIG. 5A), layout design 600A (FIG. 6A), layout design 700A (FIG. 7A), layout design 800A (FIG. 8A) or layout design 900A (FIG. 9A) of an integrated circuit, such as integrated circuit 100 (FIG. 1), integrated circuit 400 (FIGS. 4A-4E) or integrated circuit 500B, 600B, 700B, 800B or 900B. In some embodiments, the method 1100 is usable to form integrated circuits having similar structural relationships as one or

more of layout designs 200-300 or 500A-900A (FIG. 2-3 or 5A-9A). In some embodiments, method 1100 is performed by a processing device (e.g., processor 1202 (FIG. 12)) configured to execute instructions for generating layout design 200 (FIG. 2), layout design 300 (FIG. 3), layout design 500A (FIG. 5A), layout design 600A (FIG. 6A), layout design 700A (FIG. 7A), layout design 800A (FIG. 8A) or layout design 900A (FIG. 9A).

[0243] In operation 1102 of method 1100, a set of active region layout patterns is generated. In some embodiments, the set of active region layout patterns of method 1100 includes one or more patterns of the set of active region layout patterns 202 of FIGS. 2-3 and 5A-9A, and detailed description of these layout patterns is therefore omitted.

[0244] In operation 1104 of method 1100, the set of active region layout patterns is placed on a first layout level of layout design 200, 300 or 500A-900A. In some embodiments, the first layout level corresponds to the OD layout level.

[0245] In operation 1106 of method 1100, a set of fin layout patterns is generated. In some embodiments, the set of fin layout patterns of method 1100 includes one or more patterns of the set of fin layout patterns 316 of FIG. 3, and detailed description of these layout patterns is therefore omitted.

[0246] In operation 1108 of method 1100, the set of fin layout patterns is placed on a second layout level of layout design 200, 300 or 500A-900A. In some embodiments, the second layout level corresponds to the FIN layout level. In some embodiments, the second layout level is the same as the first layout level. In some embodiments, the second layout level is different from the first layout level. In some embodiments, the second layout level is above the first layout level. In some embodiments, the second layout level is below the first layout level.

[0247] In operation 1110 of method 1100, a first set of conductive feature layout patterns is generated. In some embodiments, the first set of conductive feature layout patterns of method 1100 includes one or more patterns of the set of conductive feature layout patterns 320, 520, 620, 720, 820 or 920 of FIGS. 3 & 5A-9A, and detailed description of these layout patterns is therefore omitted.

[0248] In operation 1112 of method 1100, the first set of conductive feature layout patterns is placed on a first portion of the first layout level of layout design 200, 300 or 500A-900A. In some embodiments, the first portion of the first layout level corresponds to the MU layout level.

[0249] In operation 1114 of method 1100, a set of gate layout patterns is generated. In some embodiments, the set of gate layout patterns of method 1100 includes one or more patterns of the set of gate layout patterns 205 of FIG. 2, set of gate layout patterns 206 of FIG. 2, set of gate layout patterns 204 of FIGS. 3 & 5A-7A, and set of gate layout patterns 804 of FIGS. 8A-9A, and detailed description of these layout patterns is therefore omitted.

[0250] In operation 1116 of method 1100, the set of gate layout patterns is placed on a third layout level of layout design 200, 300 or 500A-900A. In some embodiments, the third layout level corresponds to the Poly layout level or MID layout level. In some embodiments, the third layout level is different from the first layout level and the second layout level. In some embodiments, the third layout level is above one or more of the first or second layout level. In some

embodiments, the third layout level is below one or more of the first or second layout level.

[0251] In operation 1118 of method 1100, a first set of via layout patterns is generated. In some embodiments, the first set of via layout patterns of method 1100 includes one or more patterns of the set of via layout patterns 322, 522, 622, 822, 922 of FIGS. 3 & 5A-9A, and detailed description of these layout patterns is therefore omitted.

[0252] In operation 1120 of method 1100, the first set of via layout patterns is placed between the first layout level and the third layout level of layout design 200, 300 or 500A-900A. In some embodiments, the first set of via layout patterns is placed on the VU layout level of layout design 200, 300 or 500A-900A. In some embodiments, the first set of via layout patterns is placed between the set of gate layout patterns and the first set of conductive feature layout patterns.

[0253] In operation 1122 of method 1100, a set of metal over diffusion layout patterns is generated. In some embodiments, the set of metal over diffusion layout patterns of method 1100 includes one or more layout patterns of the set of metal over diffusion layout patterns 210, 610, 612, 810 or 910 of FIGS. 2-3 & 5A-9A, and detailed description of these layout patterns is therefore omitted.

[0254] In operation 1124 of method 1100, the set of metal over diffusion layout patterns is placed on the third layout level of layout design 200, 300 or 500A-900A. In some embodiments, the third layout level corresponds to the MD layout level.

[0255] In operation 1126 of method 1100, a second set of via layout patterns is generated. In some embodiments, the second set of via layout patterns of method 1100 includes one or more patterns of the set of via layout patterns 322, 522, 622, 822, 922 of FIGS. 3 & 5A-9A, and detailed description of these layout patterns is therefore omitted.

[0256] In operation 1128 of method 1100, the second set of via layout patterns is placed between the third layout level and the first layout level of layout design 200, 300 or 500A-900A. In some embodiments, the second set of via layout patterns is placed on the VU layout level of layout design 200, 300 or 500A-900A. In some embodiments, the second set of via layout patterns is placed between the set of metal over diffusion layout patterns and the first set of conductive feature layout patterns.

[0257] In operation 1130 of method 1100, a second set of conductive feature layout patterns is generated. In some embodiments, the second set of conductive feature layout patterns of method 1100 includes one or more patterns of the set of conductive feature layout patterns 330 of FIGS. 3 & 5A-9A, or conductive feature layout pattern 630b, 630b' of FIGS. 3 & 5A-9A, and detailed description of these layout patterns is therefore omitted.

[0258] In operation 1132 of method 1100, the second set of conductive feature layout patterns is placed on a fourth layout level of layout design 200, 300 or 500A-900A. In some embodiments, the fourth layout level is different from the first layout level, the second layout level and the third layout level. In some embodiments, the fourth layout level is above one or more of the first, second or third layout level. In some embodiments, the fourth layout level is below one or more of the first, second or third layout level. In some embodiments, the fourth layout level corresponds to the M0 layout level.

[0259] In operation 1134 of method 1100, a third set of via layout patterns is generated. In some embodiments, the third set of via layout patterns of method 1100 includes one or more patterns of the set of via layout patterns 332, 632, 832 or via layout pattern 932a of FIGS. 3 & 5A-9A, and detailed description of these layout patterns is therefore omitted.

[0260] In operation 1136 of method 1100, the third set of via layout patterns is placed between the third layout level and the fourth layout level of layout design 200, 300 or 500A-900A. In some embodiments, the third set of via layout patterns is placed on the VG layout level of layout design 200, 300 or 500A-900A. In some embodiments, the third set of via layout patterns is placed between the set of gate layout patterns and the second set of conductive feature layout patterns.

[0261] In operation 1138 of method 1100, a fourth set of via layout patterns is generated. In some embodiments, the fourth set of via layout patterns of method 1100 includes one or more patterns of the set of via layout patterns 636, 836 of FIGS. 6A-9A, and detailed description of these layout patterns is therefore omitted.

[0262] In operation 1140 of method 1100, the fourth set of via layout patterns is placed between the third layout level and the fourth layout level of layout design 200, 300 or 500A-900A. In some embodiments, the fourth set of via layout patterns is placed on the VD layout level of layout design 200, 300 or 500A-900A. In some embodiments, the fourth set of via layout patterns is placed between the set of metal over diffusion layout patterns and the second set of conductive feature layout patterns.

[0263] In operation 1142 of method 1100, a third set of conductive feature layout patterns is generated. In some embodiments, the third set of conductive feature layout patterns of method 1100 includes one or more patterns of the set of conductive feature layout patterns 340, 640 of FIGS. 3 & 5A-7A, and detailed description of these layout patterns is therefore omitted.

[0264] In operation 1144 of method 1100, the third set of conductive feature layout patterns is placed on a fifth layout level of layout design 200, 300 or 500A-700A. In some embodiments, the fifth layout level is different from the first layout level, the second layout level, the third layout level, and the fourth layout level. In some embodiments, the fifth layout level is above one or more of the first, second, third or fourth layout level. In some embodiments, the fifth layout level is below one or more of the first, second, third or fourth layout level. In some embodiments, the fifth layout level corresponds to the M1 layout level.

[0265] In operation 1146 of method 1100, a fifth set of via layout patterns is generated. In some embodiments, the fifth set of via layout patterns of method 1100 includes one or more patterns of the set of via layout patterns 342, 642 of FIGS. 3 & 5A-7A, and detailed description of these layout patterns is therefore omitted.

[0266] In operation 1148 of method 1100, the fifth set of via layout patterns is placed between the fourth layout level and the fifth layout level of layout design 200, 300 or 500A-700A. In some embodiments, the fifth set of via layout patterns is placed on the V0 layout level of layout design 200, 300 or 500A-700A. In some embodiments, the fifth set of via layout patterns is placed between the first set of conductive layout patterns and the second set of conductive feature layout patterns.

[0267] In operation 1150 of method 1100, a cut feature layout pattern is generated. In some embodiments, the cut feature layout pattern of method 1100 includes cut feature layout pattern 304a or 304b (FIGS. 3 and 5A) or cut feature layout pattern 904 (FIG. 9A), and detailed description of these layout patterns is therefore omitted.

[0268] In operation 1152 of method 1100, the cut feature layout pattern is placed on one or more of the third, fourth or fifth layout level of layout design 200, 300 or 500A-900A.

[0269] In some embodiments, one or more of the operations of method 1100 is not performed. One or more of the operations of methods 1000-1100 is performed by a processing device configured to execute instructions for manufacturing an integrated circuit, such as integrated circuit 100, 400, 500B, 600B, 700B, 800B or 900B. In some embodiments, one or more operations of methods 1000-1100 is performed using a same processing device as that used in a different one or more operations of methods 1000-1100. In some embodiments, a different processing device is used to perform one or more operations of methods 1000-1100 from that used to perform a different one or more operations of methods 1000-1100.

[0270] FIG. 12 is a schematic view of a system 1200 for designing an IC layout design in accordance with some embodiments. In some embodiments, system 1200 generates or places one or more IC layout designs described herein. System 1200 includes a hardware processor 1202 and a non-transitory, computer readable storage medium 1204 encoded, i.e., storing, with the computer program code 1206, i.e., a set of executable instructions. Computer readable storage medium 1204 is configured for interfacing with manufacturing machines for producing the integrated circuit. The processor 1202 is electrically coupled to the computer readable storage medium 1204 via a bus 1208. The processor 1202 is also electrically coupled to an I/O interface 1210 by bus 1208. A network interface 1212 is also electrically connected to the processor 1202 via bus 1208. Network interface 1212 is connected to a network 1214, so that processor 1202 and computer readable storage medium 1204 are capable of connecting to external elements via network 1214. The processor 1202 is configured to execute the computer program code 1206 encoded in the computer readable storage medium 1204 in order to cause system 1200 to be usable for performing a portion or all of the operations as described in method 1000 or 1100.

[0271] In some embodiments, the processor 1202 is a central processing unit (CPU), a multi-processor, a distributed processing system, an application specific integrated circuit (ASIC), and/or a suitable processing unit.

[0272] In some embodiments, the computer readable storage medium 1204 is an electronic, magnetic, optical, electromagnetic, infrared, and/or a semiconductor system (or apparatus or device). For example, the computer readable storage medium 1204 includes a semiconductor or solid-state memory, a magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and/or an optical disk. In some embodiments using optical disks, the computer readable storage medium 1204 includes a compact disk-read only memory (CD-ROM), a compact disk-read/write (CD-R/W), and/or a digital video disc (DVD).

[0273] In some embodiments, the storage medium 1204 stores the computer program code 1206 configured to cause system 1200 to perform method 1000 or 1100. In some

embodiments, the storage medium 1204 also stores information needed for performing method 1000 or 1100 as well as information generated during performing method 1000 or 1100, such as layout design 1216 and user interface 1218, and/or a set of executable instructions to perform the operation of method 1000 or 1100. In some embodiments, layout design 1216 comprises one or more of layout designs 200-300 or 500A-900A. In some embodiments, layout design 1216 comprises one or more layout patterns of layout designs 200-300 or 500A-900A.

[0274] In some embodiments, the storage medium 1204 stores instructions (e.g., computer program code 1206) for interfacing with manufacturing machines. The instructions (e.g., computer program code 1206) enable processor 1202 to generate manufacturing instructions readable by the manufacturing machines to effectively implement method 1000 or 1100 during a manufacturing process.

[0275] System 1200 includes I/O interface 1210. I/O interface 1210 is coupled to external circuitry. In some embodiments, I/O interface 1210 includes a keyboard, keypad, mouse, trackball, trackpad, and/or cursor direction keys for communicating information and commands to processor 1202.

[0276] System 1200 also includes network interface 1212 coupled to the processor 1202. Network interface 1212 allows system 1200 to communicate with network 1214, to which one or more other computer systems are connected. Network interface 1212 includes wireless network interfaces such as BLUETOOTH, WIFI, WIMAX, GPRS, or WCDMA; or wired network interface such as ETHERNET, USB, or IEEE-1394. In some embodiments, method 1000 or 1100 is implemented in two or more systems 1200, and information such as layout design, and user interface are exchanged between different systems 1200 by network 1214.

[0277] System 1200 is configured to receive information related to a layout design through I/O interface 1210 or network interface 1212. The information is transferred to processor 1202 by bus 1208 to determine a layout design for producing an integrated circuit, e.g., integrated circuit 100, 400, 500B, 600B, 700B, 800B or 900B. The layout design is then stored in computer readable medium 1204 as layout design 1216. System 1200 is configured to receive information related to a user interface through I/O interface 1210 or network interface 1212. The information is stored in computer readable medium 1204 as user interface 1218.

[0278] In some embodiments, method 1000 or 1100 is implemented as a standalone software application for execution by a processor. In some embodiments, method 1000 or 1100 is implemented as a software application that is a part of an additional software application. In some embodiments, method 1000 or 1100 is implemented as a plug-in to a software application. In some embodiments, method 1000 or 1100 is implemented as a software application that is a portion of an EDA tool. In some embodiments, method 1000 or 1100 is implemented as a software application that is used by an EDA tool. In some embodiments, the EDA tool is used to generate a layout of the integrated circuit device. In some embodiments, the layout is stored on a non-transitory computer readable medium. In some embodiments, the layout is generated using a tool such as VIRTUOSO® available from CADENCE DESIGN SYSTEMS, Inc., or another suitable layout generating tool. In some embodiments, the layout is generated based on a netlist which is created based on the

schematic design. In some embodiments, method **1000** or **1100** is implemented by a manufacturing device to manufacture an integrated circuit (e.g., integrated circuit **100**, **400**, **500B**, **600B**, **700B**, **800B** or **900B**) using a set of masks manufactured based on one or more layout designs (e.g., layout design **200-300** or **500A-900A**) generated by system **1200**. System **1200** of FIG. **12** generates layout designs (e.g., layout design **200-300** or **500A-900A**) of an integrated circuit (e.g., integrated circuit **100**, **400**, **500B**, **600B**, **700B**, **800B** or **900B**) that are smaller than other approaches.

[0279] One aspect of this description relates to an integrated circuit. In some embodiments, the integrated circuit includes a first active region and a second active region in a substrate, the first active region and the second active region extending in a first direction, being located on a first level, and being separated from one another in a second direction different from the first direction. In some embodiments, the integrated circuit further includes a first conductor extending in the first direction and the second direction, being located on a lower portion of the first level, and being between the first active region and the second active region. In some embodiments, the integrated circuit further includes a first contact extending in the second direction, overlapping the first conductor and being located on a second level different from the first level. In some embodiments, the integrated circuit further includes a second contact extending in the second direction, overlapping the first conductor, being located on the second level, and being separated from the first contact in at least the first direction. In some embodiments, the integrated circuit further includes a set of gates extending in the second direction, overlapping the first conductor, and being located on the second level, each gate of the set of gates being separated from an adjacent gate of the set of gates in the first direction. In some embodiments, the first conductor electrically couples the first contact and the second contact together.

[0280] Another aspect of this description relates to an integrated circuit. In some embodiments, the integrated circuit includes a first active region and a second active region in a substrate, the first active region and the second active region extending in a first direction, being located on a first level, and being separated from one another in a second direction different from the first direction. In some embodiments, the integrated circuit further includes a first conductor extending in the first direction and the second direction, being located on a lower portion of the first level, and being between the first active region and the second active region. In some embodiments, the integrated circuit further includes a first gate extending in the second direction, overlapping the first conductor and being located on a second level different from the first level. In some embodiments, the integrated circuit further includes a second gate extending in the second direction, overlapping the first conductor, being located on the second level, and being separated from the first gate in at least the first direction. In some embodiments, the integrated circuit further includes a set of contacts extending in the second direction, overlapping the first conductor, and being located on the second level, each of the contacts of the set of contacts being separated from an adjacent contact of the set of contacts in the first direction. In some embodiments, the first conductor electrically couples the first gate and the second gate together.

[0281] Still another aspect of this description relates to an integrated circuit. In some embodiments, the integrated circuit includes a first active region and a second active region in a substrate, the first active region and the second active region extending in a first direction, being located on a first level, and being separated from one another in a second direction different from the first direction. In some embodiments, the integrated circuit further includes a first conductive structure extending in the first direction, being located on the first level, and being between the first active region and the second active region. In some embodiments, the integrated circuit further includes a set of gates extending in the second direction, overlapping at least the first conductive structure and being located on a second level different from the first level, each gate of the set of gates being separated from an adjacent gate of the set of gates in the first direction by a first pitch. In some embodiments, the integrated circuit further includes a set of contacts extending in the second direction, overlapping the first conductive structure, and being located on the second level, each contact of the set of contacts being separated from an adjacent contact of the set of contacts in the first direction, and a first contact of the set of contacts being directly coupled to a first source/drain of the first active region. In some embodiments, the integrated circuit further includes a first set of vias coupling the first conductive structure to the set of contacts, the first set of vias being between the first conductive structure and the set of contacts, and a first via of the first set of vias being located where the first contact of the set of contacts overlaps the first conductive structure.

[0282] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. An integrated circuit, comprising:

- a first active region and a second active region in a substrate, the first active region and the second active region extending in a first direction, being located on a first level, and being separated from one another in a second direction different from the first direction;
- a first conductor extending in the first direction and the second direction, being located on a lower portion of the first level, and being between the first active region and the second active region;
- a first contact extending in the second direction, overlapping the first conductor and being located on a second level different from the first level;
- a second contact extending in the second direction, overlapping the first conductor, being located on the second level, and being separated from the first contact in at least the first direction; and
- a set of gates extending in the second direction, overlapping the first conductor, and being located on the

- second level, each gate of the set of gates being separated from an adjacent gate of the set of gates in the first direction,
- wherein the first conductor electrically couples the first contact and the second contact together.
2. The integrated circuit of claim 1, wherein the first conductor comprises:
- a first portion extending in the first direction, being located on the lower portion of the first level, and being electrically coupled to the first contact;
 - a second portion extending in the first direction, being located on the lower portion of the first level, being electrically coupled to the second contact, and being separated from the first portion in the second direction; and
 - a third portion extending in the second direction, being located on the lower portion of the first level, being coupled to the first portion and the second portion, and being between the first portion and the second portion.
3. The integrated circuit of claim 2, further comprising:
- a first via coupling the first portion to the first contact, and being between the first portion and the first contact, wherein the first contact overlaps the first portion.
4. The integrated circuit of claim 3, further comprising:
- a second via coupling the second portion to the second contact, and being between the second portion and the second contact, wherein the second contact overlaps the second portion.
5. The integrated circuit of claim 4, further comprising:
- a third contact extending in the second direction, being separated from the first contact in the second direction, being separated from the second contact in the first direction, and being located on the second level; and
 - a fourth contact extending in the second direction, being separated from the first contact in the first direction, being separated from the second contact in the second direction, and being located on the second level.
6. The integrated circuit of claim 5, further comprising:
- a second conductor extending in the first direction, being between the first active region and the second active region, overlapping the first contact and the fourth contact, and being located on a third level different from the first level and the second level;
 - a third conductor extending in the first direction, being between the first active region and the second active region, overlapping the second contact and the third contact, being located on the third level, and being separated from the second conductor in the second direction; and
 - a fourth conductor extending in the first direction, being between the third conductor and the fourth conductor, overlapping each of the first conductor, a first gate of the set of gates and a second gate of the set of gates, being located on the third level, and being separated from the second conductor and the third conductor in the second direction.
7. The integrated circuit of claim 6, further comprising:
- a third via coupling the second conductor to the fourth contact, and being between the second conductor and the fourth contact; and
 - a fourth via coupling the third conductor to the third contact, and being between the third conductor and the third contact.
8. The integrated circuit of claim 7, further comprising:
- a fifth via coupling the fourth conductor to the first gate of the set of gates, and being between the fourth conductor and the first gate of the set of gates; and
 - a sixth via coupling the fourth conductor to the second gate of the set of gates, and being between the fourth conductor and the second gate of the set of gates.
9. An integrated circuit, comprising:
- a first active region and a second active region in a substrate, the first active region and the second active region extending in a first direction, being located on a first level, and being separated from one another in a second direction different from the first direction;
 - a first conductor extending in the first direction and the second direction, being located on a lower portion of the first level, and being between the first active region and the second active region;
 - a first gate extending in the second direction, overlapping the first conductor and being located on a second level different from the first level;
 - a second gate extending in the second direction, overlapping the first conductor, being located on the second level, and being separated from the first gate in at least the first direction; and
 - a set of contacts extending in the second direction, overlapping the first conductor, and being located on the second level, each of the contacts of the set of contacts being separated from an adjacent contact of the set of contacts in the first direction,
- wherein the first conductor electrically couples the first gate and the second gate together.
10. The integrated circuit of claim 9, wherein the first conductor comprises:
- a first portion extending in the first direction, being located on the lower portion of the first level, and being electrically coupled to the first gate;
 - a second portion extending in the first direction, being located on the lower portion of the first level, being electrically coupled to the second gate, and being separated from the first portion in the second direction; and
 - a third portion extending in the second direction, being located on the lower portion of the first level, being coupled to the first portion and the second portion, and being between the first portion and the second portion.
11. The integrated circuit of claim 10, further comprising:
- a first via coupling the first portion to the first gate, and being between the first portion and the first gate, wherein the first gate overlaps the first portion; and
 - a second via coupling the second portion to the second gate, and being between the second portion and the second gate, wherein the second gate overlaps the second portion.
12. The integrated circuit of claim 11, further comprising:
- a third gate extending in the second direction, overlapping the second active region, being separated from the first gate in the second direction, being separated from the second gate in the first direction, and being located on the second level; and
 - a fourth gate extending in the second direction, overlapping the first active region, being separated from the first gate in the first direction, being separated from the second gate in the second direction, and being located on the second level.

- 13.** The integrated circuit of claim **12**, further comprising:
- a second conductor extending in the first direction, being between the first active region and the second active region, overlapping the first gate and the fourth gate, and being located on a third level different from the first level and the second level;
 - a third conductor extending in the first direction, being between the first active region and the second active region, overlapping the second gate and the third gate, being located on the third level, and being separated from the third conductor in the second direction; and
 - a fourth conductor extending in the first direction, being between the second conductor and the third conductor, overlapping the first conductor, being located on the third level, and being separated from the second conductor and the third conductor in the second direction.
- 14.** The integrated circuit of claim **13**, further comprising:
- a third via coupling the third conductor to the third gate, and being between the third conductor and the third gate; and
 - a fourth via coupling the second conductor to the fourth gate, and being between the second conductor and the fourth gate.
- 15.** An integrated circuit, comprising:
- a first active region and a second active region in a substrate, the first active region and the second active region extending in a first direction, being located on a first level, and being separated from one another in a second direction different from the first direction;
 - a first conductive structure extending in the first direction, being located on the first level, and being between the first active region and the second active region;
 - a set of gates extending in the second direction, overlapping at least the first conductive structure and being located on a second level different from the first level, each gate of the set of gates being separated from an adjacent gate of the set of gates in the first direction by a first pitch;
 - a set of contacts extending in the second direction, overlapping the first conductive structure, and being located on the second level, each contact of the set of contacts being separated from an adjacent contact of the set of contacts in the first direction, and a first contact of the set of contacts being directly coupled to a first source/drain of the first active region; and
 - a first set of vias coupling the first conductive structure to the set of contacts, the first set of vias being between the first conductive structure and the set of contacts, and a first via of the first set of vias being located where the first contact of the set of contacts overlaps the first conductive structure.

- 16.** The integrated circuit of claim **15**, further comprising:
- a second conductive structure extending in the first direction, being located on the first level, being between the first active region and the second active region, and being separated from the first conductive structure in the second direction.
- 17.** The integrated circuit of claim **16**, further comprising:
- a second set of vias coupling the second conductive structure to the set of contacts, the second set of vias being between the second conductive structure and the set of contacts, and a first via of the second set of vias being located where a second contact of the set of contacts overlaps the second conductive structure.
- 18.** The integrated circuit of claim **17**, further comprising:
- a first set of conductive structures extending in the first direction, being between the first active region and the second active region, overlapping at least the set of gates, and being located on a third level different from the first level and the second level, and each conductive structure of the first set of conductive structures being separated from an adjacent conductive structure of the first set of conductive structures in the second direction.
- 19.** The integrated circuit of claim **18**, further comprising:
- a second set of vias coupling the first set of conductive structures to the set of contacts, the second set of vias being between the first set of conductive structures and the set of contacts, and a first via of the second set of vias being located where a first conductive structure of the first set of conductive structures overlaps a second contact of the set of contacts; and
 - a third set of vias coupling the first set of conductive structures to the set of gates, the third set of vias being between the first set of conductive structures and the set of gates, and a first via of the third set of vias being located where a second conductive structure of the first set of conductive structures overlaps a first gate of the set of gates.
- 20.** The integrated circuit of claim **19**, further comprising:
- a second conductive structure extending in the second direction, being between the set of gates, overlapping at least the first conductive structure, and being located on a fourth level different from the first level, the second level and the third level; and
 - a fourth set of vias coupling the second conductive structure to the first set of conductive structures, the fourth set of vias being between the second conductive structure and the first set of conductive structures, and a first via of the fourth set of vias being located where the second conductive structure overlaps the first conductive structure of the first set of conductive structures.

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